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TEST BOOK OF
Zoology

Animal Form & Function
A Comparative Perspective

Subjective & Objective

For Other Universities

Paper C For B.Sc/ AD

For Punjab University

Paper D For B.Sc / AD (IVth Years)

For B.S. (Four Years)

Semester IV (Paper V & VI)

C



Communication I Nerves: Neurons:

The Basic Functional Units of the Nervous System. Neuron Structure. Neuron Communication Introductory Aspects: Resting Membrane Potential, Action potential (Nerve Impulse), Transmission of the Action Potential between Cells. Invertebrate and Vertebrate Nervous Systems: The Spinal Cord, Spinal Nerves, The Brain, Cranial Nerves, The Autonomic Nervous System.

Communication II: Senses:

Sensory Reception. Invertebrate Sensory Receptors: Baroreceptors, Chemoreceptors, Georeceptors, Hygroreceptors, Phonoreceptors, Photoreceptors, Proprioceptors, Tactile Receptors, Thermoreceptors, Vertebrate Sensory Receptors: Lateral-Line System and Electrical Sensing, Lateral-Line System and Mechanoreception, Hearing and Equilibrium in Air, Hearing and Equilibrium in Water, Skin Sensors of Damaging Stimuli, Skin Sensors of Heat and Cold, Skin Sensors of Mechanical Stimuli, Sonar, Smell, Taste, Vision.

Communication III: The Endocrine System and Chemical Messengers:

Chemical Messengers. Introductory aspects of hormones and Their Feedback Systems: Biochemistry of Hormones, Feedback Control System of Hormone Secretion, Mechanisms of Hormone Action, Fixed-Membrane-Receptor Mechanism, Mobile-Receptor Mechanism. Some Hormones of Invertebrates: Porifera, Cnidarians, Platyhelminths, Nemertean, Nematodes, Molluscs, Annelids, Arthropods, Echinoderms, An Overview of the Vertebrate Endocrine System, Endocrine Systems of Vertebrates Other Than Birds or Mammals, Endocrine Systems of Birds and Mammals.

Circulation and Gas Exchange: Internal Transport and Circulatory Systems: Transport Systems in Invertebrates and Vertebrates. The Hearts and Circulatory Systems of Bony Fishes, Amphibians, Reptiles, Birds and Mammals, Gas Exchange, Respiratory Surfaces, Invertebrate Respiratory Systems, Vertebrate Respiratory Systems, Cutaneous Exchange, Gills, Lungs, Lung Ventilation, Human Respiratory System, Gas Transport.

Nutrition and Digestion:

Evolution of Nutrition. The Metabolic Fates of Nutrients in Heterotrophs. Animal Strategies for Getting and Using Food Continuous Versus Discontinuous Feeders. Diversity in Digestive Structures: Invertebrates and Vertebrates. The Mammalian Digestive System: The various segments and their roles.

Cell Biology:

Introduction to Cell Biology; Difference between prokaryotic and eukaryote cells. Cytoplasmic organelles: Membrane system, Ultrastructure of cell, Endoplasmic reticulum; their role in protein synthesis and drug metabolism. Golgi apparatus: Role in synthesis of glycoprotein. Mitochondria: Cell respiration, significance as semi-autonomous organelle. Lysosomes: Diverse role due to hydrolytic activity of enzymes. Peroxisomes: Metabolism of hydrogen peroxide. Glyoxysomes: Plasma membrane and its function: Chemical composition and structure of plasma membrane, cell permeability, active transport, Endocytosis, Phagocytosis, Cytoskeleton: Microfilaments, Microtubules, Intermediate filaments. Introduction to Replication, Transcription and Translation.

Communication I Nerves: Neurons:

Neurons: The Basic Functional Units of the Nervous System

The functional unit of the nervous system is a highly specialized cell called the neuron (from the Greek word "nerve"). Neurons are designed to produce signals that can be sent over short or long distances, from one part of an animal's body to another. Neurons have two key properties:

1. **Excitability:** The ability to respond to stimuli.
2. **Conductivity:** The ability to carry a signal.

Types of Neurons

There are three main types of neurons:

1. **Sensory Neurons:** Act as receptors of stimuli themselves or are activated by receptors. Changes in the internal or external environments stimulate sensory neurons, which respond by sending signals to the major integrating centers where information is processed.
2. **Interneurons:** Comprise the integrating centers and receive signals from the sensory neurons and transmit them to motor neurons.
3. **Motor Neurons:** Send the processed information via a signal to the body's effectors (e.g., muscles), causing them to contract, or to glands, causing them to secrete.

Neuron Structure: The Key to Function

Most neurons contain three main parts:

1. **Cell Body:** Contains a large, central nucleus.
2. **Dendrites:** Short, thread-like branches that are extensions of the cell body and conduct signals towards the cell body. The motor neuron has many dendrites (from the Greek word "dendron," meaning tree).
3. **Axon:** A relatively long, cylindrical process that conducts signals (information) away from the cell body.

Sheathing of Neurons

1. **Neurons without Sheath:** The neurons of hydras and sea anemones do not have a sheath covering the axon of the neuron.
2. **Sheathed Neurons:** Other invertebrates and all vertebrates have sheathed neurons. When present, the laminated lipid sheath is called myelin. In some neurons, a neuromelocyte (formerly known as a Schwann cell) wraps the myelin sheath in layers. In these neurons, gaps called neurofibril nodes (formerly nodes of Ranvier) segment the myelin sheath at regular intervals. The neuromelocyte also assists in the regeneration of injured myelinated neurons.

Function of the Nervous System

The nervous system has the following functions:

1. **Receiving Data (Input Stimulus):** In a given neuron, the dendrites are the receptors.
2. **Integration:** The cell body acts as the integrator.

3. **Effecting Change (Output Response):** The ends of the axon are the effectors, resulting in a change in the animal's physiology.

Neuron Communication

Neuron Signals

The signal of a neuron is the nerve impulse or action potential. The key to this nerve impulse is the neuron's plasma membrane and its properties. Changes in membrane permeability and the movement of ions produce a nerve impulse that travels along the plasma membrane of the dendrites, cell body, and axon of each neuron.

Resting Membrane Potential

Definition

A "resting" neuron is not sending a nerve impulse. The plasma membrane of a resting neuron is polarized; the fluid on the inner side of the membrane is negatively charged compared to the positively charged fluid outside the membrane (see figure 24.3).

Cause of Polarization

The difference in electrical charge between the inside and outside of the membrane at any given point is due to the active movement of positive and negative ions in the fluids on either side of the membrane, and to the permeability of the plasma membrane to these ions. The difference in charge is called the resting membrane potential. All cells have such a resting potential, but neurons and muscle cells are unique in that they can change and recycle it quickly.

Measurement

The resting potential is measured in millivolts (mV). A millivolt is 1/1,000 of a volt. Normally, the resting membrane potential is about -70 mV. Due to the unequal distribution of various electrically charged ions, sodium (Na^+) ions are more highly concentrated in the fluid outside the plasma membrane, and potassium (K^+) and large negative protein ions are more highly concentrated inside.

Ion Movement

The Na^+ and K^+ ions constantly diffuse through ion channels in the plasma membrane, moving from areas of higher concentrations to areas of lower concentrations.

Ion Concentrations and the Sodium-Potassium Pump

There are also larger Cl^- ions and large negative protein ions, which cannot move easily from the inside of the neuron to the outside. However, the concentrations of Na^+ and K^+ ions on the two sides of the membrane remain constant due to the action of the sodium-potassium ATPase pump, which is powered by ATP (figure 24.3). The pump actively moves Na^+ ions to the outside of the cell and K^+ ions to the inside of the cell. Because it moves three Na^+ ions out for each two K^+ ions that it moves in, the pump helps to establish the resting potential of about -70 mV. Both ions leak back across the membrane—down their concentration gradients. K^+ ions, however, move more easily back to the outside, adding to the positive charge there and contributing to the membrane potential of about -70 mV.

Mechanism of Neuron Action: Changing the Resting Membrane Potential into the Action Potential (Nerve Impulse)

Resting Membrane Potential:

- Resting electrical potential across the plasma membrane is crucial for nerve impulse creation and conduction.
- A stimulus reaching a threshold level initiates an impulse, causing depolarization of the membrane.

Depolarization:

- Threshold stimulus increases Na ion permeability, leading to an inflow of Na ions.
- Positive Na ions influx reduces membrane polarity, termed depolarization.

Action Potential Generation:

- Depolarization triggers voltage-gated Na channels to open, allowing more Na ions inside.
- Subsequently, Na gates close, and voltage-gated K channels open, enabling rapid K ion efflux.
- Outward movement of K ions restores positive charge outside, repolarizing the membrane.

Conduction of Action Potential:

- Series of membrane changes induce depolarization and repolarization, propagating a wave down the axon.
- Refractory period follows each action potential, during which hyperpolarization occurs before restoration of resting potential.

All-or-None Law:

- Minimum threshold stimulus initiates action potential; increased stimulus intensity doesn't affect its strength.
- Axon fires at full power or not at all, following the all-or-none law.

Speed of Impulse Conduction:

- Axon diameter and myelin sheath influence impulse conduction speed.
- Larger diameter and myelinated axons transmit impulses faster.
- Myelination insulates axons, allowing action potential "jumping" at neurofibril nodes, known as saltatory conduction.

Evolutionary Adaptations:

- In invertebrates like crayfish and earthworms, large-diameter axons facilitate rapid escapes.
- Myelination in vertebrates enhances conduction velocity, enabling efficient nervous system function in compact spaces.

Transmission of the Action Potential Between Cells

After an action potential (a nerve signal) travels along an axon, it reaches the end, called the end bulb. The synapse is the connection point between the axon of one neuron and the dendrite of another neuron or an effector cell (like a muscle or gland cell). The space between the end bulb and the next neuron's dendrite is called the synaptic cleft.

Synapse Structure

- **Presynaptic Neuron:** The neuron sending the signal towards the synapse.
- **Postsynaptic Neuron:** The neuron receiving the signal after the synapse. This can also be a muscle or gland cell.

Types of Synapses

- **Electrical Synapse:** Nerve impulses transmit directly from one neuron to another through ions that move across the synapse, allowing rapid signal transmission in both directions. These are common in fish, helping them react quickly to predators.
- **Chemical Synapse:** Neurons communicate via a chemical called a neurotransmitter. The presynaptic neuron releases the neurotransmitter, which then binds to the receptor on the postsynaptic neuron, creating a new action potential in the postsynaptic neuron.

Chemical Synapse Mechanism

1. **Nerve Impulse Arrival:** When a nerve impulse reaches an end bulb, it causes vesicles containing neurotransmitters to fuse with the plasma membrane.
2. **Neurotransmitter Release:** The neurotransmitter is released into the synaptic cleft by exocytosis.
3. **Neurotransmitter Action:** The neurotransmitter (e.g., acetylcholine) binds to receptors on the postsynaptic membrane, causing depolarization and continuation of the nerve impulse.
4. **Neurotransmitter Inactivation:** Enzymes like acetylcholinesterase break down the neurotransmitter to prevent continuous stimulation, which can cause a diseased state.

Practical Example

Flea sprays often contain parathion, which prevents the breakdown of acetylcholine in fleas, causing continuous stimulation of their nerve cells and eventually immobilizing them. The dose is low enough not to harm pets or humans significantly.

Invertebrate Nervous Systems

Introduction to Nervous Systems

- **All cells can respond to stimuli** and send information both internally and externally. Even protozoa and sponges, which lack a true nervous system, show coordination and reaction to stimuli. For instance, protozoan cilia beat regularly, and flagellates respond to light changes, requiring internal coordination.
- **True nervous systems are present only in animals with tissue-level organization**, excluding protozoa and sponges.

Evolutionary Trends in Nervous System Development

- **Trend 1: Increased Complexity**
 - More complex animals have more detailed nervous systems.

- **Cnidarians** (e.g., hydras, jellyfish, sea anemones) possess the simplest nervous systems, consisting of a nerve net. This latticework of neurons conducts impulses bidirectionally. Cnidarians lack brains and clusters of neurons, with nerve impulses spreading throughout the body.
- **Trend 2: Cephalization**
 - As animals evolved to move forward, **sensory organs and nervous tissue became concentrated at the front (anterior end)**.
 - **Flatworms and Roundworms** exhibit this trend. Flatworms have ganglia (clusters of nerve cells) in the head region, functioning as a primitive brain. Lateral nerve cords carry sensory information to and from these ganglia.
- **Trend 3: Bilateral Symmetry**
 - **Bilateral symmetry** involves having symmetrical right and left body halves, leading to paired neurons, muscles, sensory structures, and brain centers.
 - This pairing allows coordinated movements, such as climbing, crawling, flying, or walking.
- **Trend 4: Increased Complexity in Nervous System Structure**
 - **Invertebrates like crustaceans, segmented worms, and arthropods** show advanced nervous system organization.
 - Axons join into nerve cords, and besides a small central brain, there are smaller peripheral ganglia coordinating different body regions. These ganglia occur in each body segment or near the organs they regulate.
- **Trend 5: Increase in Interneurons**
 - **More complex animals have more interneurons**, leading to more complex behavior patterns.
 - In **echinoderms** (e.g., sea stars), the nervous system is divided into parts: the ectoneural system for sensory and motor functions, a radial nerve for each arm, a deeper hyponeural system for motor functions, and an apical system with potential sensory functions.
 - The **brain, containing the highest number of neurons**, becomes more complex in animals with more intricate behaviors, like vertebrates.

Summary

- **Nervous systems evolved from simple nerve nets** in cnidarians to more complex arrangements with central brains and peripheral ganglia in advanced invertebrates and vertebrates.
- The increasing complexity of the nervous system in animals is characterized by **cephalization, bilateral symmetry, increased interneurons, and more specialized structures**, enabling more sophisticated behaviors and coordination.

Vertebrate Nervous System

Key Characteristics

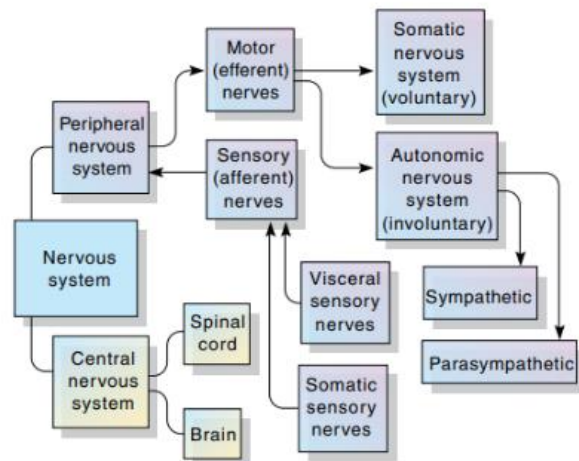
- **Bilateral Symmetry:** Vertebrates have symmetrical right and left halves.
- **Notochord:** A rod of tissue present in embryos, later replaced by the vertebral column in most species.
- **Tubular Nerve Cord:** Located above the notochord, it develops into the spinal cord and brain.

Notochord and Vertebral Column

- The notochord is **mesodermally derived** and provides structural support.
- **Vertebral Column:** Replaces the notochord during development, allowing for strong muscles and fast movement, facilitating predatory behavior.
- **Jaw Development:** Bones evolve into powerful jaws, enhancing predatory abilities.

Evolution of the Nerve Cord

- **Single, Tubular Nerve Cord:** Located above the notochord, it expands and specializes into the spinal cord and brain.
- **Brain Development:** The anterior end thickens and divides into the hindbrain, midbrain, and forebrain.
- **Sensory Receptors:** Paired structures like eyes and ears develop for better environmental awareness.



Divisions of the Nervous System

- **Central Nervous System (CNS):** Composed of the brain and spinal cord, responsible for information processing.
- **Peripheral Nervous System (PNS):** Consists of all nerves outside the brain and spinal cord, divided into sensory and motor nerves.

Sensory and Motor Nerves

- **Sensory (Afferent) Nerves:** Transmit information to the CNS.
- **Motor (Efferent) Nerves:** Carry commands away from the CNS.

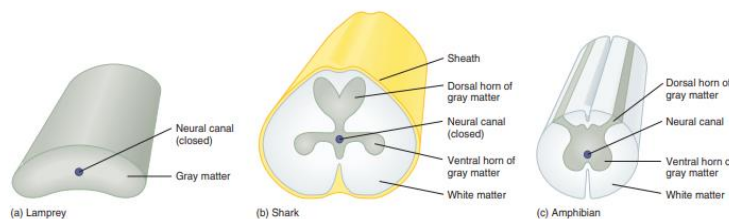
Motor Nerve Subdivisions

- **Voluntary (Somatic) Nervous System:** Controls skeletal muscles.
- **Involuntary (Autonomic) Nervous System:** Stimulates smooth and cardiac muscles, and glands.
 - **Sympathetic System:** Prepares the body for action.

- **Parasympathetic System:** Calms the body and conserves energy.

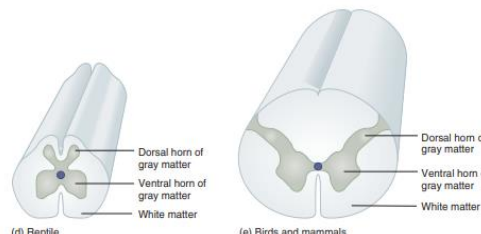
Nerve Fiber Bundles

- **CNS:** Bundles of nerve fibers are called **tracts**.
- **PNS:** Bundles of nerve fibers are called **nerves**.



Cell Body Clusters

- **CNS:** Clusters of cell bodies are called **nuclei**.
- **PNS:** Clusters of cell bodies are called **ganglia**.



The Spinal Cord

The spinal cord is essential for two main reasons: it links the brain with most of the body, and it manages spinal reflex actions.

Functions

- **Connecting Link:** Connects the brain to most of the body.
- **Reflex Actions:** Handles predictable, involuntary responses to stimuli, aiding both voluntary and involuntary limb movements and certain organ functions.

Structure

- **Extension:** Part of the central nervous system, running from the brain to near or into the tail.
- **Neural Canal:** Contains cerebrospinal fluid.
- **Gray Matter:** Made up of cell bodies and dendrites, mainly handling reflex connections at various levels.
- **Ventral and Dorsal Roots:** Extend from the spinal cord, containing motor and sensory fibers that form major spinal nerves.
- **White Matter:** Named for the whitish myelin covering the axons.

Protective Layers

- **Meninges:** Three layers of protective membranes surrounding the spinal cord and brain.
 - **Dura Mater:** Outer, tough, fibrous membrane.
 - **Arachnoid:** Middle, delicate layer connecting to the pia mater.
 - **Pia Mater:** Innermost layer with small blood vessels that nourish the spinal cord.

Spinal Nerves

Number of Spinal Nerves

- **Directly Related to Segments:** The number of spinal nerves corresponds to the number of segments in the trunk and tail of a vertebrate.

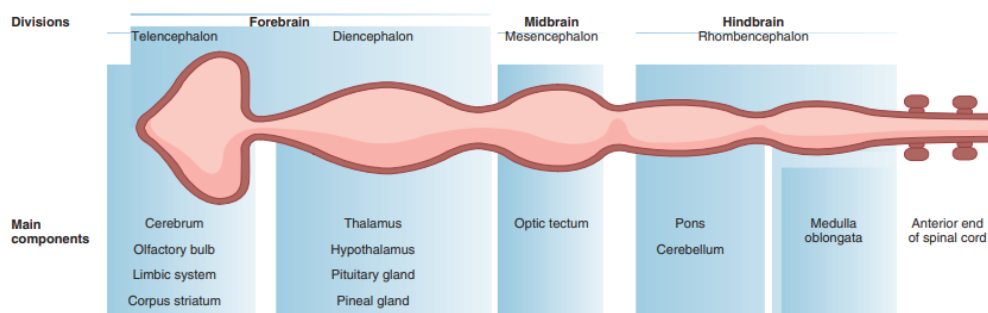
Examples

- **Frog:** Evolved strong hind legs for swimming or jumping, a reduced trunk, and no tail in the adult; has only 10 pairs of spinal nerves.
- **Snake:** Moves by lateral undulations of its long trunk and tail; has several hundred pairs of spinal nerves.

The Brain

Anatomical Development

- **Vertebrate Brain:** Develops at the anterior end of the spinal cord.
- **Embryonic Development:** Undergoes regional expansion, forming the hindbrain, midbrain, and forebrain.
- **Ventricles:** The central canal of the spinal cord extends into the brain, expanding into chambers called ventricles filled with cerebrospinal fluid.



Hindbrain

- **Components:** Includes the medulla oblongata, cerebellum, and pons.

Medulla Oblongata

- **Location:** Enlargement where the spinal cord enters the brain.
- **Functions:** Contains reflex centers for breathing, swallowing, cardiovascular function, and gastric secretion.
- **Development:** Well-developed in all jawed vertebrates, reflecting its role in controlling visceral functions and screening information entering or leaving the brain.

Cerebellum

- **Function:** Coordinates motor activity associated with limb movement, maintaining posture, and spatial orientation.
- **Structure:**
 - **Cartilaginous Fishes:** Distinct anterior and posterior lobes.

- **Teleosts:** Large in active swimmers, small in inactive fishes.
- **Amphibians:** Rudimentary cerebellum due to simple locomotor patterns.
- **Tetrapods:** Laterally expanded for locomotor control of limb muscles.
- **Birds and Mammals:** Much larger due to complex locomotor patterns and evolutionary history of limb development.

Pons

- **Function:** Acts as a bridge of transverse nerve tracts from the cerebrum of the forebrain to both sides of the cerebellum.
- **Connectivity:** Contains tracts connecting the forebrain and spinal cord in all vertebrates.

Midbrain

- **Original Function:** Coordinated reflex responses to visual input.
- **Evolved Functions:** Added roles in tactile (touch) and auditory (hearing) input without changing in size.
- **Optic Tectum:** The thickened region of gray matter in the roof of the midbrain that integrates visual and auditory signals.

Forebrain

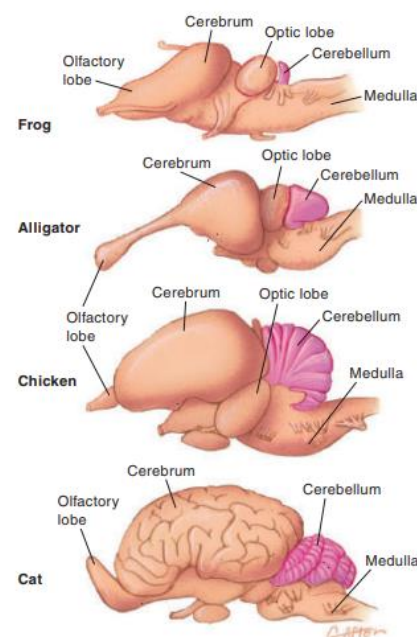
- **Evolutionary Changes:** The forebrain has significantly changed during vertebrate evolution.
- **Main Parts:** The forebrain consists of the diencephalon and telencephalon.

Diencephalon

- **Location:** Lies just in front of the midbrain.
- **Components:** Pineal gland, pituitary gland, hypothalamus, and thalamus.
- **Thalamus:** Relays all sensory information to higher brain centers.
- **Hypothalamus:** Regulates body temperature, sexual drive, carbohydrate metabolism, hunger, and thirst.
- **Pineal Gland:** Controls some body rhythms.
- **Pituitary Gland:** A major endocrine gland.

Telencephalon

- **Evolutionary Expansion:** Rapidly increased in size and complexity.



- **Corpus Striatum:** Important in reptiles and birds for complex behavior patterns.
- **Cerebrum:**
 - **Structure:** Divided into right and left cerebral hemispheres by a large groove.
 - **Function:** Related to sensory and motor integration.
- **Cerebral Cortex:**
 - **Expansion:** Progressively increased in size and complexity in mammals.
 - **Function:** Contains primary sensory and motor areas, involved in the perception of visual or auditory signals.
 - **Human Specificity:** Includes areas for language processing, both written and spoken.

TABLE 24.1
FUNCTIONS OF THE CRANIAL NERVES OF REPTILES, BIRDS, AND MAMMALS

NERVE	TYPE	INNERVATION AND FUNCTION
I Olfactory	Sensory	Smell
II Optic	Sensory	Vision
III Oculomotor	Primarily motor	Eyelids, eyes, adjustments of light entering eyes, lens focusing (motor)
IV Trochlear	Primarily motor	Condition of muscles (sensory) Eye muscles (motor)
V Trigeminal	Mixed	Condition of muscles (sensory) Eyes, tear glands, scalp, forehead, and upper eyelids (sensory) Upper teeth, upper gum, upper lip, lining of the palate, and skin of the face (sensory)
VI Abducens	Primarily motor	Scalp, skin of the jaw, lower teeth, lower gum, and lower lip (sensory) Jaws, floor of the mouth (motor) Eye muscles (motor)
VII Facial	Mixed	Condition of muscles (sensory) Taste receptors of the anterior tongue (sensory) Facial expression, tear glands, and salivary glands (motor)
VIII Vestibulocochlear	Sensory	Equilibrium; vestibule Hearing; cochlea
IX Glossopharyngeal	Mixed	Pharynx, tonsils, posterior tongue, and carotid arteries (sensory) Pharynx and salivary glands (motor)
X Vagus	Mixed	Speech and swallowing, heart, and visceral organs in the thorax and abdomen (motor) Pharynx, larynx, esophagus, and visceral organs of the thorax and abdomen (sensory)
XI Accessory	Motor	Soft palate, pharynx, and larynx
XII Hypoglossal	Motor	Neck and back Tongue muscles

Cranial Nerves

- **Cranial Nerves in Vertebrates:** The peripheral nervous system of vertebrates includes paired cranial nerves.
- **Number of Pairs:** Reptiles, birds, and mammals have 12 pairs, while fishes and amphibians have 10 pairs.
- **Types of Cranial Nerves:**
 - **Sensory Nerves:** Contain only sensory axons (e.g., optic nerve) that carry signals to the brain.
 - **Mixed Nerves:** Contain both sensory and motor axons (e.g., vagus nerve) with sensory axons leading to the brain and motor axons leading to the heart and smooth muscles of visceral organs in the thorax and abdomen.

The Autonomic Nervous System

- **Function:** Composed of two divisions that act in opposition to control involuntary muscles (smooth and cardiac) and glands.
- **Parasympathetic Nervous System:**
 - **Function:** Operates during relaxation.
 - **Nerve Origin:** Arises from the brain and sacral region of the spinal cord.
 - **Nerve Structure:** Long efferent nerve fibers synapse at ganglia near organs, and short efferent neurons extend from ganglia to organs.
- **Sympathetic Nervous System:**
 - **Function:** Responsible for the "fight-or-flight" response.
 - **Nerve Origin:** Arises from the thoracic and lumbar regions of the spinal cord.
 - **Nerve Structure:** Short efferent fibers extend to ganglia near the spine, and long efferent neurons extend from ganglia to each organ.
- **Dual Input to Organs:**
 - **Parasympathetic Input:** Stimulates salivary gland secretions, intestinal movements, contracts pupillary muscles, and relaxes sphincter muscles.
 - **Sympathetic Input:** Inhibits salivary gland secretions and intestinal movements, relaxes pupillary muscles, and contracts sphincters.

Communication II: Senses:

Sensory Reception

- **Aristotle's Five Human Senses:** About two thousand years ago, Aristotle identified five human senses—sight, hearing, smell, taste, and touch—commonly referred to as the “five senses.”
- **Additional Animal Senses:** Today, zoologists know that animals also have other senses. For example, invertebrates possess an impressive array of sensory receptors through which they receive information about their environment.
- **Examples of Invertebrate Sensory Receptors:**
 - Tactile receptors that sense touch
 - Georeceptors that sense the pull of gravity
 - Hygroreceptors that detect the water content of air
 - Proprioceptors that respond to mechanically induced changes caused by stretching, compression, bending, and tension
 - Phonoreceptors that are sensitive to sound
 - Baroreceptors that respond to pressure changes
 - Chemoreceptors that respond to air- and waterborne molecules
 - Photoreceptors that sense light
 - Thermoreceptors that are influenced by temperature changes
- **Vertebrate Sensory Abilities:** Most vertebrates have a sense of equilibrium (balance) and a sense of body movement, and they are also sensitive to fine touch, touch-pressure, heat, taste, vision, olfaction, audition, cold, pain, and various other tactile stimuli.
- **Circulatory and Digestive System Receptors:** Receptors in the circulatory system register changes in blood pressure and blood levels of carbon dioxide and hydrogen ions, and receptors in the digestive system are involved in the perception of hunger and thirst.
- **Importance of Senses:** Overall, an animal's senses limit and define its impression of the environment. In fact, all awareness depends on the reception and decoding of stimuli from the external environment and from within an animal's body.
- **Function of Sensory Receptors:** Sensory receptors consist of cells that can convert environmental information (stimuli) into nerve impulses.
- **Stimulus Definition:** A stimulus (pl., stimuli) is any form of energy an animal can detect with its receptors.
- **Transducer Function:** All receptors are transducers (“to change over”); that is, they convert one form of energy into another.
- **Generator Potential:** Different types of receptors convert different kinds of stimuli, such as light or heat, into a local electrical potential called the generator potential. If the generator

potential reaches the sensory neuron's threshold potential, it causes channels in the plasma membrane to open and creates an action potential.

- **Action Potential and Sensations:** All action potentials are alike. Furthermore, an action potential is an all-or-none phenomenon; it either occurs or it doesn't. In those animals that have brains, some nerve signals from specific receptors always end up in a specific part of the brain for interpretation.
- **Intensity of Stimulus:** When the stimulus strength increases, the number of action potentials per unit of time also increases. Thus, the brain can perceive the intensity and type of stimulus from the timing of the impulses and the "wiring" of neurons.

Basic Features of Sensory Receptors

1. **Sensitive Receptor Cells:** They contain sensitive receptor cells or finely branched peripheral endings of sensory neurons that respond to a stimulus by creating a generator potential.
2. **Specific Structure:** Their structure is designed to receive a specific stimulus.
3. **Synapse with Afferent Nerve Fibers:** Their receptor cells synapse with afferent nerve fibers that travel to the central nervous system along specific neural pathways.
4. **Translation into Recognizable Sensation:** In the central nervous system, the nerve impulse is translated into a recognizable sensation, such as sound.

Invertebrate Sensory Receptors

An animal's behavior is largely based on how it responds to information from its environment. Invertebrates have many different types of sensory receptors that help them gather this information. Let's look at a few examples of these receptors and understand their structure and function.

Baroreceptors

Function: Baroreceptors detect changes in pressure.

Details: While no specific baroreceptor structures have been identified in invertebrates, many ocean-dwelling invertebrates respond to pressure changes. Examples include:

- **Copepod crustaceans**
- **Ctenophores**
- **Jellyfish medusae**
- **Squids**

Some intertidal crustaceans may use pressure changes from water depth to coordinate their migratory activities with the tides.

Chemoreceptors

Function: Chemoreceptors respond to chemicals in the environment.

Details: Chemoreception is a direct sense where specific molecules stimulate a response. It is the oldest and most universal sense among animals. Examples of chemoreception include:

- **Protozoa** reacting to acids, alkalis, and salts with avoidance behavior.

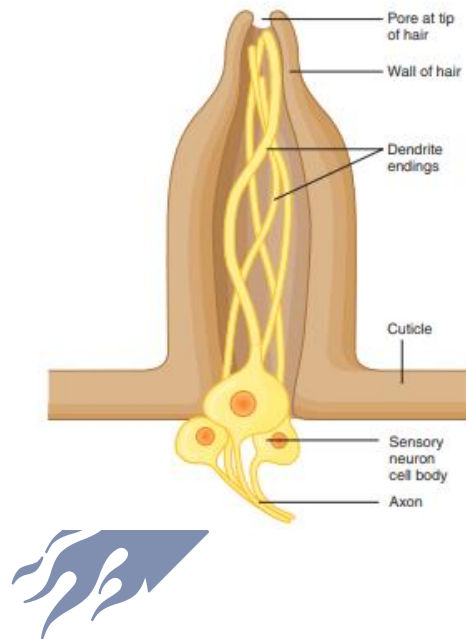
- **Predatory ciliates** being attracted to chemicals released by their prey.

In aquatic invertebrates, chemoreceptors are often located in pits or depressions that allow water carrying chemicals to circulate through. In arthropods, chemoreceptors are usually found on the antennae, mouthparts, and legs in the form of hollow hairs called sensilla. These sensilla contain chemosensory neurons.

Types of Information Gathered:

- **Humidity detection**
- **pH assessment**
- **Prey tracking**
- **Food recognition**
- **Mate location**

For example, male silkworm moths (*Bombyx mori*) can detect a single molecule of bombykol, a sex attractant released by female moths, among a trillion molecules of air. This ability helps males locate females from several miles away, which is a significant reproductive advantage for this species.

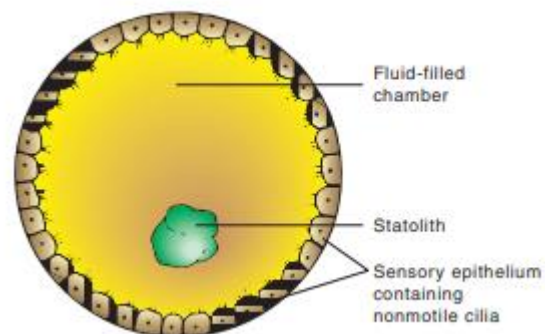


Georeceptors

Function: Georeceptors respond to the force of gravity, providing an animal with information about its orientation relative to "up" and "down."

Structure: Most georeceptors are statocysts, which consist of a fluid-filled chamber lined with cilia-bearing sensory epithelium. Inside the chamber is a solid granule called a statolith.

Mechanism: When an animal moves, the position of the statolith and the fluid within the chamber change, altering the information from the sensory epithelium. This provides information about the animal's linear and rotational acceleration.



Examples: Statocysts are found in various **gastropods, cephalopods, crustaceans, nemertines, polychaetes, and scyphozoans**. These animals use statocysts to orient themselves within their environment. For instance, **burrowing invertebrates** rely on georeceptors rather than photoreceptors for orientation within the substratum, while **planktonic animals** use statocysts to orient in their three-dimensional aquatic environment, especially at night and in deep water.

Special Adaptation: Some aquatic insects detect gravity using air bubbles trapped in passageways, such as tracheal tubes. These air bubbles act like the air bubble in a carpenter's level, moving according to gravity and stimulating sensory bristles that line the tubes.

Hygroreceptors

Function: Hygroreceptors detect the water content of air.

Examples: Some insects have hygroreceptors that can detect small changes in ambient relative humidity. This sense enables them to seek environments with specific humidity or to modify their physiology or behavior (e.g., controlling the opening or closing of spiracles).

Location: Hygrosensory structures are found on the antennae, palps, underside of the body, and near the spiracles of insects.

Unknown Mechanism: The exact process by which a hygroreceptor transduces humidity into an action potential is not yet known.

Phonoreceptors

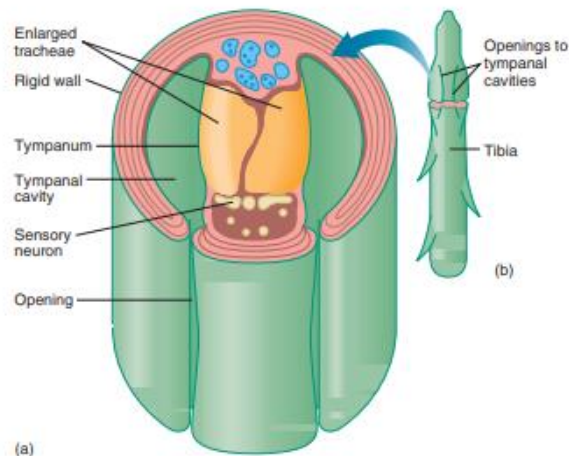
Function: Phonoreceptors respond to sound.

Presence: True phonoreceptors are found in **insects, arachnids, and centipedes**. Other invertebrates may respond to sound-induced vibrations of the substratum.

Examples in Insects: Crickets, grasshoppers, and cicadas possess phonoreceptors called **tympanic or tympanal organs**, which have a tough, flexible tympanum covering an internal sac. When sound waves strike the tympanum, it vibrates, stimulating sensory neurons attached to it and producing a generator potential.

Examples in Arachnids: Most arachnids possess phonoreceptors in their cuticle called **slit sense organs** that can sense sound-induced vibrations.

Examples in Centipedes: Centipedes have **organs of Tomosvary**, which may be sensitive to sound, though the physiology of both slit sense organs and organs of Tomosvary is poorly understood.



Photoreceptors

Function: Photoreceptors are sensitive to light.

Light-Sensitive Pigments: All photoreceptors have pigments (like carotenoids and rhodopsin) that absorb light and produce a generator potential.

Types and Examples:

- **Euglena:** These flagellated protozoa have photoreceptor granules called stigmata (red pigments). The actual photoreceptor is at the base of the flagellum, helping Euglena orient toward light for photosynthesis.
- **Earthworm (Lumbricus):** Simple unicellular photoreceptor cells scattered over the epidermis or in specific areas.
- **Ocelli:** Small cups lined with light-sensitive receptors and backed by light-absorbing pigment. Found in Annelida, Mollusca, and Arthropoda, ocelli provide information about light direction and intensity but do not form images.

- **Compound Eyes:** Made up of many units called ommatidia. Best developed in arthropods, these eyes detect movement and may form images. Each ommatidium has its own nerve tract, and the eye adapts to light changes and may provide color vision.
- **Complex Camera Eyes:** Found in squids and octopuses, these eyes are similar to vertebrate eyes. They have a cornea, lens, and retina. The receptor sites face the incoming light, unlike in vertebrates where they are inverted. Squid eyes focus by moving the lens, while vertebrate eyes focus by changing the lens shape.

Proprioceptors

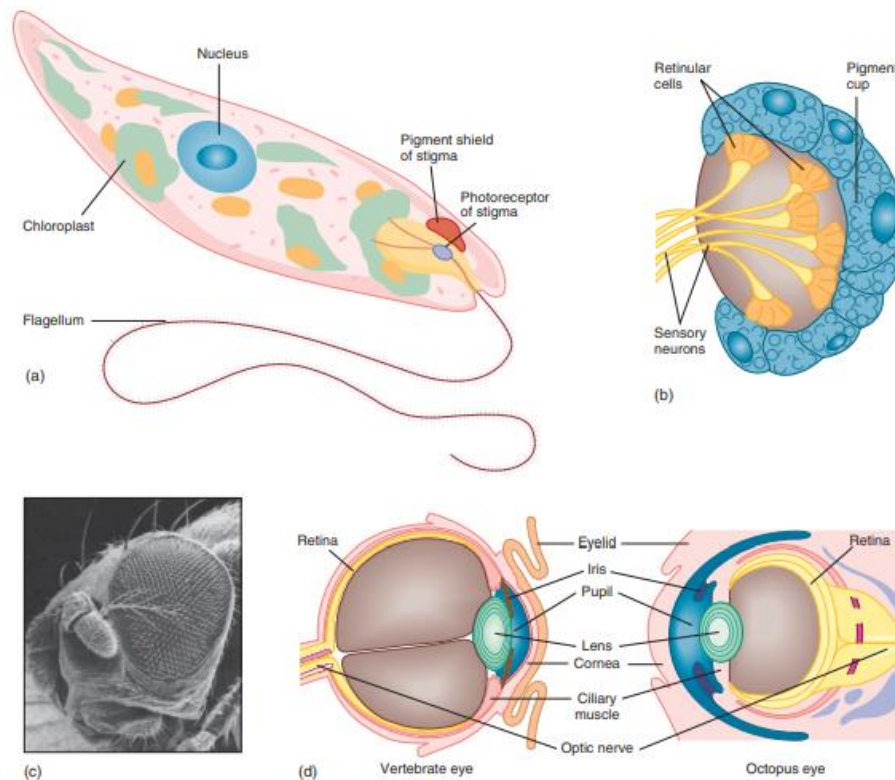
Function: Proprioceptors, also known as "stretch receptors," respond to mechanical changes like stretching, compression, bending, or tension.

Role: They provide information about the movement and position of an animal's body parts.

Examples in Arthropods: Proprioceptors are associated with appendage joints and body extensor muscles. Sensory neurons are attached to parts of the body that stretch, such as muscle cells, connective-tissue fibers, or membranes spanning joints. Changes in shape of these structures distort sensory nerve endings, initiating a generator potential.

Key Terms and Their Meanings

- **Georeceptors (earth receptor):** Respond to gravity.
- **Statocysts (standing bladder):** Fluid-filled chambers with sensory epithelium and a solid granule (statolith) that detect movement.
- **Hygroreceptors (moist receptor):** Detect humidity.
- **Phonoreceptors (voice receptor):** Respond to sound.
- **Photoreceptors (light receptor):** Sensitive to light, containing pigments like carotenoids and rhodopsin.
- **Stigma (eye-spot):** Mass of photoreceptor granules in Euglena.
- **Ocellus (small eye):** Simple light-detecting organ.
- **Ommatidia (little eye):** Units making up compound eyes.
- **Proprioceptors (self receptor):** Internal receptors for body part movement and position.



Tactile Receptors

Function: Tactile (touch) receptors detect physical contact with objects in the environment.

Structure: These receptors are generally derived from modifications of epithelial cells associated with sensory neurons. They often involve projections from the body surface, such as bristles, spines, setae, and tubercles.

Mechanism: When an animal contacts an object, these receptors are mechanically deformed. This deformation activates the receptor, which in turn activates underlying sensory neurons, initiating a generator potential.

Examples:

- **Tube-dwelling polychaetes:** Have receptors that allow them to retract quickly into their tubes in response to movements in their surroundings.
- **Web-building spiders:** Possess tactile receptors that can sense struggling prey through vibrations of the web threads.

Thermoreceptors

Function: Thermoreceptors (heat receptors) respond to temperature changes.

Sensing Mechanism: Some invertebrates can directly sense differences in environmental temperatures.

Examples:

Protozoan Paramecium: Collects in areas where water temperature is moderate and avoids temperature extremes.

Leeches and Ticks: Attracted to warm-blooded hosts through a heat-sensing mechanism.

Certain Insects, Some Crustaceans, and Horseshoe Crab (*Limulus*): Can sense thermal variations.

Unknown Structure: In all these cases, the specific receptor structures responsible for thermoreception have not been identified.

Summary

Main Points:

- **Tactile Receptors:** Respond to mechanical deformation and vibrations, essential for detecting contact and movement.
- **Thermoreceptors:** Detect temperature changes, important for environmental adaptation.
- **Environmental Adaptations:** Sensory receptors evolve based on environmental demands, highlighting the intricate relationship between sensory systems and habitats.
- **Vertebrate and Invertebrate Similarities:** Both groups have sensory receptors adapted to their environments, underscoring the universality of sensory adaptation in the animal kingdom.

Vertebrate Sensory Receptors

Adaptations to Environments: Vertebrate sensory receptors reflect adaptations to the sensory stimuli present in different external and internal environments. Each environment has specific chemical and physical characteristics that affect the types of energy and molecules that carry sensory information.

Examples of External Environments:

- **Human:** The earth and air.
- **Trout:** Cool, clear water of a mountain stream.
- **Turtle:** Turbid water of a swamp.
- **Salmon:** Salty water of the sea.

Media Characteristics:

- **Air:** Transmits light well, conducts sound efficiently, carries a limited number of small molecules detectable by smell, and passes little or no electrical energy.
- **Water:** Sound travels faster and farther, dissolves and carries a wide range of chemicals, conducts electricity well (especially seawater), but absorbs many wavelengths of light.

Evolutionary Adaptations: Vertebrate sensory receptors, like invertebrate sensory receptors, have evolved to function effectively in their environments. Despite different habitats, many underlying similarities unite vertebrate senses. Each sense involves:

1. **Environmental Information:** The type of sensory information present in the environment.
2. **Receptor Cell Adaptation:** How receptor cells evolved to detect this information.
3. **Information Processing:** How the central nervous system processes this information for the animal's use.

Lateral-Line System and Electrical Sensing

Function: Specialized organs for balance, gravity detection, hearing, and magnetoreception evolved from the lateral-line system of fishes. This system is used for electrical sensing in most fishes, some amphibians, and the platypus.

Structure:

- **Sensory Pores:** Located in the skin's surface, leading to canals connected to electroreceptors called **ampullary organs**.
- **Ampullary Organs:** These electroreceptors can sense electrical currents in the surrounding water.

Benefits: The ability to detect weak electrical fields generated by living organisms helps fish find mates, capture prey, and avoid predators. This sense is especially valuable in deep, turbulent, or murky water where vision is limited. Some fishes generate electrical fields and use their electroreceptors for **electrocommunication**, detecting how surrounding objects distort the field to navigate in murky or turbulent waters.

Vertebrate Sensory Receptors

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Lateral-Line System and Mechanoreception

Mechanoreceptor: A receptor excited by mechanical pressures or distortions (e.g., sound, touch, and muscular contractions).

Lateral-Line System: This system is present in cyclostomes, sharks, some advanced fishes, and aquatic amphibians. It includes various hair-cell mechanoreceptors called **neuromasts**, which are found in pits along the body, but not in the head region.

Function of Neuromasts: These are responsive to local water displacement or disturbance. When water moves near the lateral line, it moves the water in the pits and distorts the hair cells, causing a generator potential in the associated sensory neurons. This allows the animal to detect the direction and force of water currents and the movement of other animals or prey in the water.

Hearing and Equilibrium in Air

Hearing: Initially important to vertebrates as a mechanism to alert them to potentially dangerous activity, and later became important for finding food, mates, and communication.

Ear Functions:

1. **Auditory Apparatus:** Concerned with hearing.
2. **Vestibular Apparatus:** Concerned with posture and equilibrium.

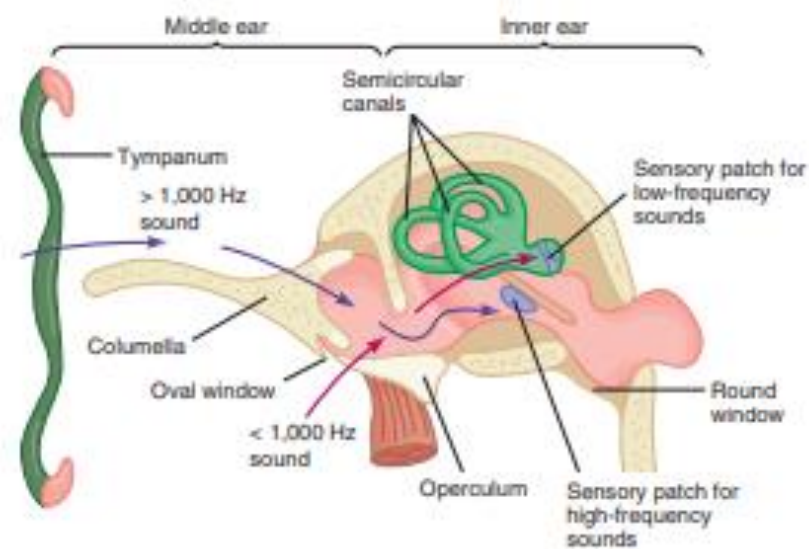
Sound Transmission: Sound results from pressure waves transmitting energy through a medium, such as air or water. In air, sound produces less than 0.1% of the pressure it produces in water, requiring adaptation for effective hearing.

Evolution of Tympanum: Adaptation to hearing in air resulted in the evolution of an acoustic transformer incorporating a thin, stretched membrane (eardrum or tympanic membrane). This structure first evolved in amphibians.

Anuran (Frog) Ear Structure:

- **Tympanum:** Modified integument stretched over a cartilaginous ring that vibrates in response to sounds.
- **Middle Ear:** A chamber behind the tympanum with a small bone called the **columella** (or stapes) that touches the oval window.
- **Inner Ear:** Contains fluid-filled semicircular canals where pressure waves stimulate receptor cells.

High-Frequency Sounds: Transmitted through the middle ear via the columella, causing pressure waves in the inner ear fluid. **Low-Frequency Sounds:** Substrate-borne vibrations transmitted through front appendages and the pectoral girdle cause the **operculum** to vibrate, stimulating a second patch of sensory receptor cells.

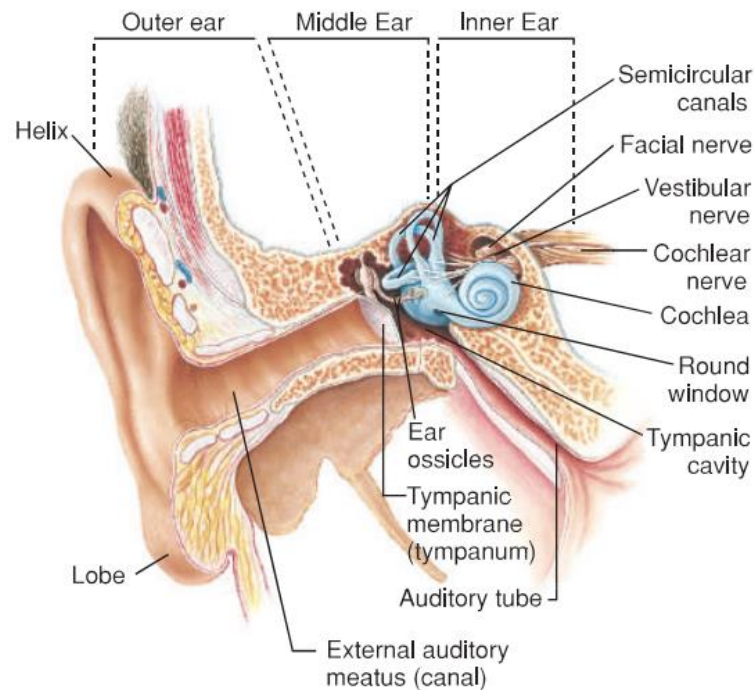


Adaptations in Other Vertebrates:

- **Salamanders:** Lack a tympanum and middle ear, sensing sounds transmitted through the substratum and skull.
- **Reptiles:** Ear structures vary, with some lacking a middle-ear cavity and tympanum. In snakes, a jaw bone articulates with the stapes to receive vibrations of the substratum.
- **Birds:** Well-developed hearing with external ear openings covered by delicate feathers.
- **Mammals:** Adaptations include an ear flap (auricle) and auditory tube, leading to the tympanum and middle ear with three ossicles (malleus, incus, and stapes). The cochlea in the inner ear provides greater sensitivity to pitch and volume.

Human Ear Structure:

- **Outer Ear:** Auricle and external auditory canal.
- **Middle Ear:** Begins at the tympanic membrane and ends at the oval and round windows.
- **Inner Ear:** Vestibule and semicircular canals for equilibrium, and cochlea for hearing.



Hearing Process:

1. **Sound waves** enter the outer ear, creating pressure waves that reach the tympanic membrane.
2. **Tympanic membrane** vibrates, moving the malleus.
3. **Malleus** articulates with the incus, vibrating it.
4. **Incus** moves the stapes against the oval window.
5. **Oval window** movements create pressure changes in the inner ear fluid, transmitted to the basilar membrane.
6. **Receptor hair cells** in the organ of Corti bend, causing a generator potential and an action potential that travels to the brain.
7. **Vibrations** dissipate through movements of the round window.

Equilibrium (Balance):

- **Static Equilibrium:** Sensing movement in one plane (vertical or horizontal).
- **Dynamic Equilibrium:** Sensing angular and/or rotational movement.
- **Otoliths:** Rest on hair cells in semicircular canals, displacing and bending hairs during movement, initiating a generator potential.

Motion Sickness: Continuous fluid movement in the semicircular canals can cause motion sickness or seasickness in humans.

Hearing and Equilibrium in Water

Inner Ear Functions in Bony Fishes: In bony fishes, the receptors for equilibrium, balance, and hearing are located in the inner ear, similar to other vertebrates.

- **Semicircular Canals:** Detect rotational movements.

- **Sensory Patches:** Assist with equilibrium and balance by detecting the direction of gravitational pull.

Vibration Detection: Fishes lack the outer and middle ear found in other vertebrates. Therefore, vibrations pass from the water through the bones of the skull to the inner ear.

- **Swim Bladder and Ossicles:** Some fishes have chains of bony ossicles (modified vertebrae) between the swim bladder and the back of the skull. Vibrations are amplified by the swim bladder and transmitted through the ossicles to the skull.

Skin Sensors of Damaging Stimuli

Pain Receptors (Nociceptors): Bare sensory nerve endings present throughout the body of mammals, except for the brain and intestines. These nerve endings respond to severe heat, cold, irritating chemicals, and strong mechanical stimuli (e.g., penetration), which the brain interprets as pain or itching.

Skin Sensors of Heat and Cold

Thermoreceptors: Bare sensory nerve endings in either the epidermis or dermis that respond to temperature changes.

- **Cold and Warm Spots:** Specific areas of the skin sensitive to either cold or warm stimuli.
- **Cold Receptors:** Respond to temperatures below skin temperature.
- **Heat Receptors:** Respond to temperatures above skin temperature.

Materials and Temperature Sensations: Materials coming into contact with the skin need not be warm or cold to produce temperature sensations.

- **Example:** Metal feels colder than wood because it absorbs more heat from the skin.

Heat-Sensitive Pit Organs in Snakes: Rattlesnakes and other pit vipers have heat-sensitive pit organs on each side of the face between the eye and nostril. These organs help snakes locate warm-blooded prey by detecting temperature differences from the surroundings.

Auditory Sonar System in Bats

Sophisticated Imaging: Bats use an auditory sonar system to create three-dimensional imaging for locating and catching prey. This system provides highly detailed information about the environment.

Skin Sensors of Mechanical Stimuli

Tactile Stimuli: Many animals rely on touch to get information about their environment. **Mechanical sensory receptors** in vertebrate skin detect stimuli that the brain interprets as light touch, touch-pressure, and vibration.

- **Light Touch:** Perceived when the skin is touched but not strongly deformed. Receptors for light touch include bare sensory nerve endings and **tactile (Meissner's) corpuscles**.
- **Bare Sensory Nerve Endings:** Widely distributed in the vertebrate body and involved with pain, thermal stimuli, and light touch.
- **Bulbs of Krause:** Found in the dermis in certain parts of the body, these receptors respond to some physical stimuli, such as position changes.
- **Touch-Pressure Receptors:** Include **Pacinian corpuscles** and **organs of Ruffini**.

- **Sensory Hairs (Vibrissae):** Many mammals have specialized sensory hairs on their wrists, snouts, and eyebrows. Nerves around the base of each vibrissa carry impulses from mechanoreceptors to the brain.

Sonar

Echolocation (Sonar): Bats, shrews, certain cave-dwelling birds, whales, and dolphins can determine distance and depth using echolocation.

- **High-Frequency Sounds:** These animals emit high-frequency sounds and determine how long it takes for the sounds to return after bouncing off objects.
- **Example:** Bats emit clicks that last 2-3 milliseconds, repeated several hundred times per second. The returning echo helps the bat locate and catch prey.
- **Sophisticated Imaging:** This auditory sonar system provides detailed three-dimensional imaging.

Smell (Olfaction)

Olfaction: The sense of smell is due to **olfactory neurons** in the roof of the vertebrate nasal cavity.

- **Olfactory Cells:** Specialized endings of fibers that make up the olfactory nerve, lying among supporting epithelial cells.
- **High Density:** For example, a dog has up to 40 million olfactory receptor cells per square centimeter.
- **Odor Perception:** Odor molecules interact with protein receptors on the cell membrane, altering membrane permeability and leading to a generator potential.

Olfaction in Fishes:

- **External Nares:** Openings in the snout lead to olfactory receptors.
- **Salmon and Lampreys:** Use their sense of smell to return to spawn in the same streams where they hatched, guided by characteristic odors.

Olfaction in Amphibians:

- Used in mate recognition, detecting noxious chemicals, and locating food.

Olfaction in Reptiles:

- **Jacobson's (Vomerinal) Organs:** Blind-ending pouches that open into the mouth, best developed in snakes and lizards. These organs help perceive odor molecules.
- **Forked Tongues:** Snakes and lizards use their tongues to sample airborne chemicals and bring them to Jacobson's organs.

Olfaction in Birds:

- Plays a minor role for most birds. **Vultures** are an exception, using smell to locate prey.

Olfaction in Mammals:

- Mammals can detect olfactory stimuli over long distances and use them to locate food, recognize members of the same species, and avoid predators.

Taste (Gustation)

Receptors for Taste: Chemoreceptors that may be on the body surface or in the mouth and throat.

- **Tongue Papillae:** Small protuberances covering the surface of the mammalian tongue.
- **Taste Buds:** Specialized receptors located in the crevices between the papillae, consisting of clusters of chemoreceptor cells (gustatory cells) and supporting cells.
- **Taste Sensations:** Generally recognized as sweet (sugars), sour (acids), bitter (alkaloids), and salty (electrolytes).
- **Taste Mechanism:** The exact mechanism that stimulates a chemoreceptor taste cell is not known. One theory suggests that different gustatory stimuli change the permeability of the receptor-cell membrane, leading to a generator potential.

Taste in Different Vertebrates:

- **Reptiles and Birds:** Taste buds are typically located in the pharynx rather than on the tongue.
- **Fishes and Amphibians:** Taste buds may also be found in the skin. For example, in sturgeons, taste buds are abundant on the head projection called the rostrum.
- **Distribution in Fishes:** Taste buds are distributed in various locations, including the pharynx, skin, and barbels (whiskers) of catfish. They help monitor the flow of water and detect potential food sources.

Vision (Photoreception)

Overview: Vision is the primary sense used by vertebrates in light-filled environments, facilitated by well-developed photoreceptive structures.

Eye Structure:

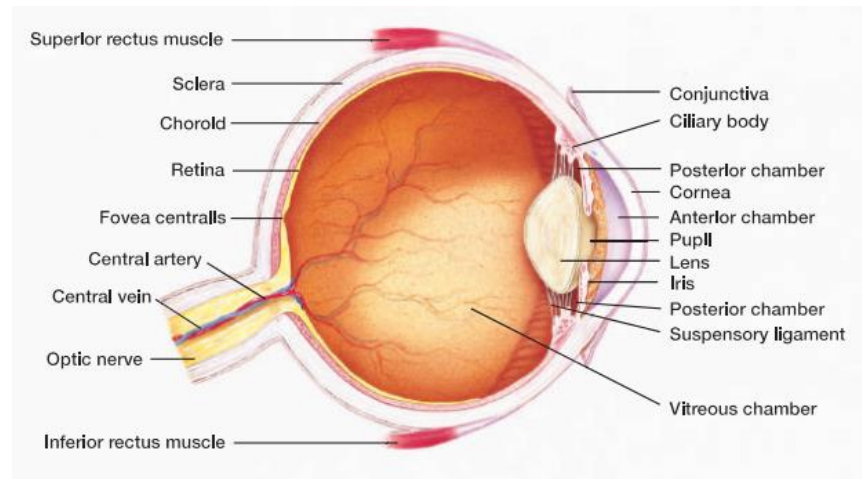
- **Lens:** Focuses light onto the retina.
- **Sclera:** Tough outer coat of the eyeball.
- **Choroid Layer:** Thin middle layer.
- **Retina:** Inner layer containing light-sensitive photoreceptor cells.
- **Cornea:** Transparent covering over the front of the eyeball.
- **Iris:** Contains pigments and muscles regulating light entering the pupil.
- **Aqueous Humor:** Fluid filling anterior and posterior chambers.
- **Vitreous Body:** Jellylike substance behind the lens.
- **Conjunctiva:** Moist mucous membrane covering the eyeball.

Accommodation:

- **Definition:** Adjustment of vision for light from close-up or distant objects.
- **Mechanism:** Coordinated stretching and relaxation of eye muscles and fibers for lens adjustment.

Adaptations Across Vertebrates:

- **Fishes:** Lidless eyes with a rounded lens close to the cornea, focusing by moving the lens forward or backward.
- **Amphibians:** Varied eye adaptations for terrestrial environments, including binocular vision for prey capture and lateral eyes for peripheral vision.
- **Reptiles:** Similar eye structure to amphibians, with upper and lower eyelids, a nictitating membrane, and a protective spectacle.
- **Birds:** Unique double-focusing mechanism for rapid adjustments during flight, with a nictitating membrane for eye protection.



Retinal Photoreceptors:

- **Rods:** Sensitive to dim light.
- **Cones:** Respond to high-intensity light and enable color perception.
- **Rhodopsin:** Pigment in rod cells absorbing light energy, triggering nerve impulses.
- **Color Perception:** Mediated by cones with red-, green-, or blue-absorbing pigments, translating light absorption into nerve impulses for brain interpretation.

Communication III: The Endocrine System and Chemical Messengers:

Chemical Messengers in Animal Development

The development of most animals starts with fertilization and the subsequent division of the zygote. Further development depends on continuous cell division, growth, and specialization. The integration of these events, as well as the communication and coordination of physiological processes like metabolism, breathing, waste removal, movement, and reproduction, depend on chemical messengers—molecules that specialized cells produce and release. Chemical messengers can be categorized as follows:

1. Local Chemical Messengers

Many cells release chemicals that change physiological conditions in the nearby area. Most of these chemicals act on nearby cells and do not accumulate in the blood. Vertebrate examples include some of the chemicals called lumones that the gut produces and that help regulate digestion. In a wound, mast cells release a substance called histamine that participates in the inflammatory response.

2. Neurotransmitters

As mentioned in chapter 24, neurons release chemicals called neurotransmitters (e.g., nitric oxide and acetylcholine) that act on immediately adjacent target cells. These chemical messengers reach high concentrations in the synaptic cleft, act quickly, and are actively broken down and recycled.

3. Neuropeptides

Some specialized neurons (called neurosecretory cells) release neuropeptides (neurohormones). The blood or other body fluids transport neuropeptides to distant target cells, where neuropeptides exert their effects. In mammals, for example, certain nerve cells in the hypothalamus release a neuropeptide that causes the pituitary gland to release the hormone oxytocin, which induces powerful uterine contractions during childbirth.

4. Hormones

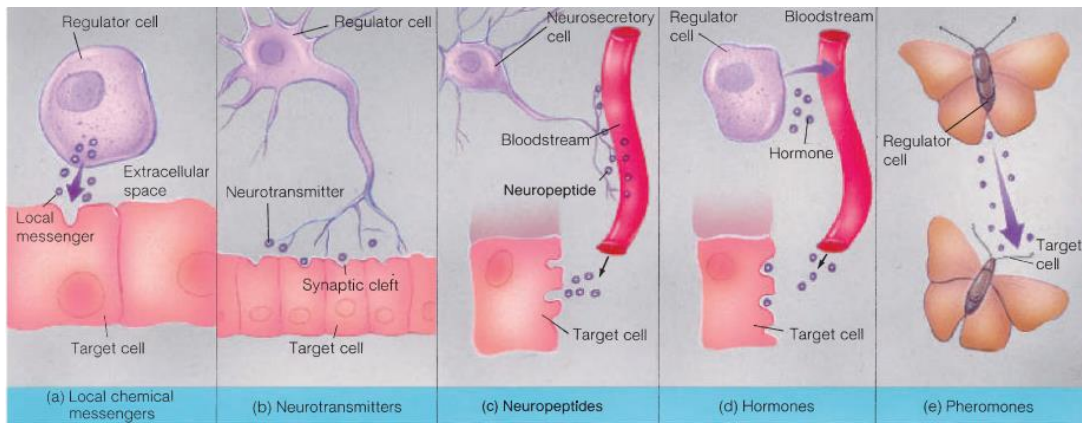
Endocrine glands or cells release hormones that the bloodstream transports to distant target cells. Many examples are given in the rest of this chapter.

5. Pheromones

Pheromones are chemical messengers released to the exterior of one animal that affect the behavior of another individual of the same species.

Neuroendocrine System

Scientists now recognize that the nervous and endocrine systems work together as a combined communicative and integrative network called the neuroendocrine system. In this system, feedback systems regulate chemical messengers in their short- and long-term coordination of animal body functions to maintain balance (homeostasis).



Hormones and Their Feedback Systems

A hormone (from the Greek word "hormaein," meaning to set in motion or to spur on) is a specialized chemical messenger produced and released by an endocrine gland or tissue. The study of endocrine glands and their hormones is called endocrinology. Hormones circulate through body fluids and affect the metabolic activity of a target cell or tissue in a specific way. By definition, a target cell has receptors that specifically bind to these chemical messengers or are affected by them.

Typically, one hormone influences, depends on, and balances another hormone within a controlled feedback network. It is rare for a hormone to operate independently.

Biochemistry of Hormones

Most hormones fall into one of the following categories:

- **Proteins (Polypeptides):** For example, hormones that the vertebrate pancreas secretes.
- **Derivatives of Amino Acids (Amines):** For instance, hormones secreted by the thyroid gland.
- **Steroids:** Secreted by the ovaries, testes, and the cortex of the adrenal glands.
- **Fatty Acid Derivatives:** Although less common, some hormones fall into this category.

In invertebrates, neurosecretory cells often produce polypeptides called neuropeptides.

Hormone Function and Effects

Hormones are effective in extremely small amounts. Just a few molecules of a hormone can produce a significant response in a target cell. In the target cell, hormones help control biochemical reactions in three primary ways:

1. **Regulating Substance Movement:** A hormone can increase the rate at which other substances enter or leave the cell.
2. **Stimulating Synthesis:** A hormone can stimulate a target cell to synthesize enzymes, proteins, or other substances.
3. **Activating or Suppressing Enzymes:** A hormone can prompt a target cell to activate or suppress existing cellular enzymes.

Like enzymes, hormones are not changed by the reactions they regulate.

Feedback Control System of Hormone Secretion

Although hormones are always present in some amount in endocrine cells or glands, they are not secreted continuously. Instead, glands secrete the amount of hormone needed to maintain balance in the body, known as homeostasis. A feedback control system monitors changes in the body or the external

environment and sends information to a central control unit (such as the central nervous system), which makes adjustments.

Types of Feedback Systems

- **Negative Feedback System:** This system produces a response that counteracts the initiating stimulus. It helps maintain stability and homeostasis. For example, if a process slows down, the feedback system will stimulate it to speed up and vice versa.
- **Positive Feedback System:** This system reinforces the initial stimulus, leading to an increase in the response. Positive feedback systems are rare in animals because they can lead to instability or pathological states.

Example of Negative Feedback

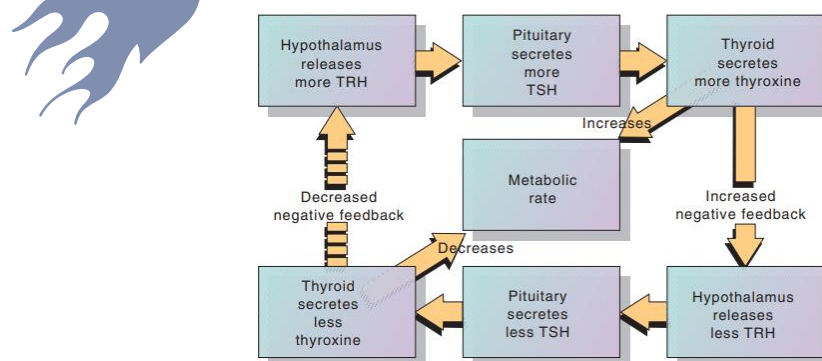
Suppose the metabolic rate (rate of chemical activity) in the body cells of a dog slows down:

1. **Detection:** The hypothalamus detects the slower metabolic rate.
2. **Response by Hypothalamus:** The hypothalamus releases more thyrotropin-releasing hormone (TRH).
3. **Response by Pituitary Gland:** TRH causes the pituitary gland to secrete more thyrotropin, also known as thyroid-stimulating hormone (TSH).
4. **Response by Thyroid Gland:** TSH stimulates the thyroid gland to secrete thyroxine.
5. **Effect of Thyroxine:** Thyroxine increases the metabolic rate, restoring homeostasis.

Conversely, if the metabolic rate speeds up:

1. **Detection:** The hypothalamus detects the faster metabolic rate.
2. **Response by Hypothalamus:** The hypothalamus releases less TRH.
3. **Response by Pituitary Gland:** The pituitary gland secretes less TSH.
4. **Response by Thyroid Gland:** The thyroid gland secretes less thyroxine.
5. **Effect of Reduced Thyroxine:** The metabolic rate decreases, restoring homeostasis.

In summary, negative feedback systems monitor hormone levels and adjust cellular activity as needed to maintain a stable internal environment.



Mechanisms of Hormone Action

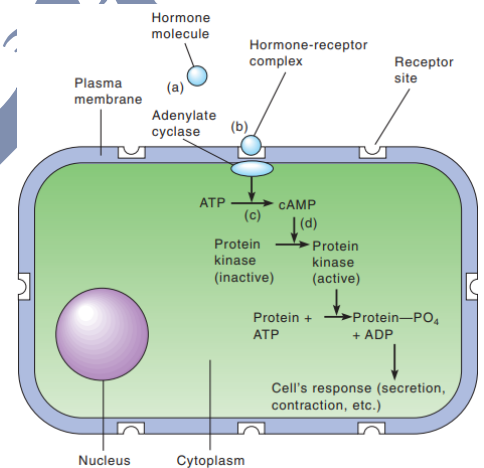
Hormones modify the biochemical activity of target cells or tissues through two basic mechanisms: the fixed-membrane-receptor mechanism and the mobile-receptor mechanism.

Fixed-Membrane-Receptor Mechanism

This mechanism applies to hormones that are proteins or amines. These hormones are water-soluble and cannot diffuse across the plasma membrane, so they initiate their response by means of specialized receptors on the plasma membrane of the target cell.

Steps of Fixed-Membrane-Receptor Mechanism

1. **Hormone Secretion:** An endocrine cell secretes a water-soluble hormone that circulates through the bloodstream.
2. **Binding to Receptor:** At the cells of the target organ, the hormone acts as a "first messenger" by binding to a specific receptor on the plasma membrane.
3. **Activation of Adenylate Cyclase:** The hormone-receptor complex activates the enzyme adenylate cyclase in the membrane.
4. **Formation of Cyclic AMP:** The activated enzyme converts ATP into cyclic AMP (cAMP), which acts as the "second messenger."
5. **Activation of Protein Kinase:** Cyclic AMP diffuses throughout the cytoplasm and activates an enzyme called protein kinase.
6. **Cellular Response:** Protein kinase causes the cell to respond with its specific physiological activity.
7. **Inactivation:** The enzyme phosphodiesterase inactivates cyclic AMP, and the receptor on the plasma membrane becomes available for a new reaction.



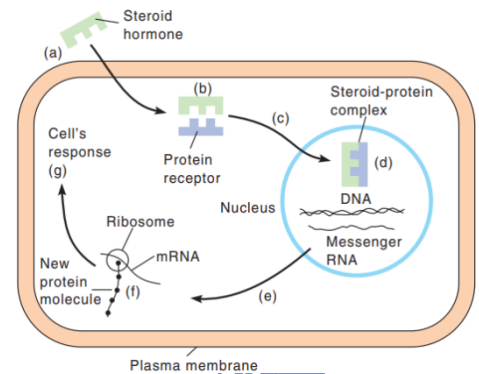
Mobile-Receptor Mechanism

This mechanism applies to steroid hormones, which are lipid-soluble and can diffuse easily into the cytoplasm of target cells, where they initiate their response by binding to cytoplasmic receptors.

Steps of Mobile-Receptor Mechanism

1. **Hormone Release:** The steroid hormone is released from a carrier protein in the bloodstream.
2. **Diffusion into Cell:** The steroid hormone enters the target cell by diffusion.
3. **Binding to Cytoplasmic Receptor:** The hormone binds to a specific protein receptor in the cytoplasm.

4. **Formation of Steroid-Protein Complex:** The steroid-protein complex gains an affinity for DNA and enters the nucleus.
5. **Binding to DNA:** The complex binds to DNA and regulates the transcription of specific genes to form messenger RNA (mRNA).
6. **mRNA Translation:** The newly transcribed mRNA leaves the nucleus and moves to the rough endoplasmic reticulum, where it initiates protein synthesis.
7. **Protein Synthesis:** The newly synthesized proteins, which may include enzymes, affect cellular metabolism and constitute the cellular response to the specific steroid hormone.



These mechanisms illustrate how hormones can significantly influence cellular activities, either by triggering enzyme actions through a fixed-membrane-receptor pathway or by regulating gene expression via the mobile-receptor mechanism.

Hormones of Invertebrates

The survival of animals depends on their ability to grow, mature, and reproduce at times when conditions such as climate and food supply are optimal. Invertebrates rely on various hormones to regulate these processes. Most invertebrate hormones are neurosecretions called neuropeptides, though some more complex invertebrates have additional types of hormones. Here's a brief overview of the hormonal systems in some invertebrate groups:

Porifera (Sponges)

- **Hormonal System:** Sponges do not have classical endocrine glands or neurons, and thus, they do not have neurosecretory cells.

Cnidarians (e.g., Hydra)

- **Growth-Promoting Hormone:** Cnidarians have nerve cells that contain a hormone stimulating budding, regeneration, and growth. This hormone, known as the "head activator," accelerates head regeneration and stimulates mitosis.

Platyhelminths (Flatworms)

- **Neurosecretory Cells:** These cells are found in the cerebral ganglion and along major nerve cords.
- **Functions:** The neuropeptides produced by these cells regulate regeneration, asexual reproduction, and gonad maturation. For instance, in tapeworms, neurosecretory cells in the scolex control the shedding of proglottids or the initiation of strobilization.

Nemerteans (Ribbon Worms)

- **Cephalization and Brain Structure:** Nemerteans have a higher degree of cephalization and a larger brain composed of dorsal and ventral ganglia connected by a nerve ring.
- **Neuropeptides:** The neuropeptides produced by these ganglia control gonadal development and water balance.

Nematodes (Roundworms)

- **Neurosecretory Cells:** These cells are associated with the central nervous system, even though nematodes lack classical endocrine glands.
- **Ecdysis Regulation:** The neuropeptide produced by these cells controls ecdysis (shedding of the cuticle). After a new cuticle is produced, the neuropeptide stimulates the excretory gland to secrete an enzyme (leucine aminopeptidase) into the space between the old and new cuticles. The fluid accumulation in this space causes the old cuticle to split and be shed.
- **Hormones of Invertebrates**
- Invertebrates rely on various hormones to regulate growth, maturation, and reproduction. Here's an overview of the hormonal systems in some invertebrate groups:

Molluscs

- **Neurosecretory Cells:** Molluscs have a central nervous system with neurosecretory cells.
- **Functions:** The neuropeptides regulate heart rate, kidney function, and energy metabolism.
- **Gastropods (e.g., Helix):** Specific hormones stimulate spermatogenesis, egg development, and the function of accessory sex organs. A growth hormone controls shell growth.
- **Cephalopods (e.g., octopus, squid):** The optic gland produces hormones that stimulate egg development, proliferation of spermatogonia, and development of secondary sexual characteristics.

Annelids

- **Endocrine System:** Annelids have a developed endocrine system due to their well-developed nervous system, circulatory system, and coelom.
- **Functions:** Involved in morphogenesis, development, growth, regeneration, and gonadal maturation.
- **Polychaetes:** Juvenile hormone inhibits gonads and stimulates growth and regeneration. Gonadotropin stimulates egg development.
- **Leeches:** A neuropeptide stimulates gamete development and triggers color changes.
- **Oligochaetes:** Osmoregulatory hormones are reported, and a hyperglycemic hormone maintains high blood glucose levels in Lumbricus.

Arthropods

- **Endocrine System:** Controls growth, maturation, and reproduction.
- **Crustaceans (e.g., crayfish):**

- **Ecdysis (Molting):** Controlled by X-organs and Y-organs. X-organs produce molt-inhibiting hormone (MIH), which keeps the Y-organ inactive. When MIH release stops, the Y-organ releases ecdysone, leading to molting.

- **Insects:**

- **Molting:** The central nervous system activates neurosecretory cells, which secrete ecdysiotropin. This leads to the production of thoracotropic hormone by the corpora cardiaca, stimulating the prothoracic glands to release ecdysone for molting.
- **Bursicon:** Influences epidermal development, such as tanning (hardening and darkening of the cuticle).
- **Juvenile Hormone (JH):** Produced by corpora allata. High JH levels inhibit differentiation. Reduced JH production leads to differentiation into a pupa and later into an adult.

Echinoderms

- **Hormonal System:** Different from chordates, despite being deuterostomes.
- **Neuropeptide:** Radial nerves contain a gonad-stimulating substance. This induces gamete shedding, spawning behavior, and meiosis in oocytes when injected into a mature sea star. It also releases a hormone called maturation-inducing substance, affecting the reproductive system.

AN OVERVIEW OF THE VERTEBRATE ENDOCRINE SYSTEM

The vertebrate endocrine system is the best-understood system of hormonal control due to extensive study. As vertebrates evolved, specialized hormone-producing cells and tissues emerged, controlled in various ways:

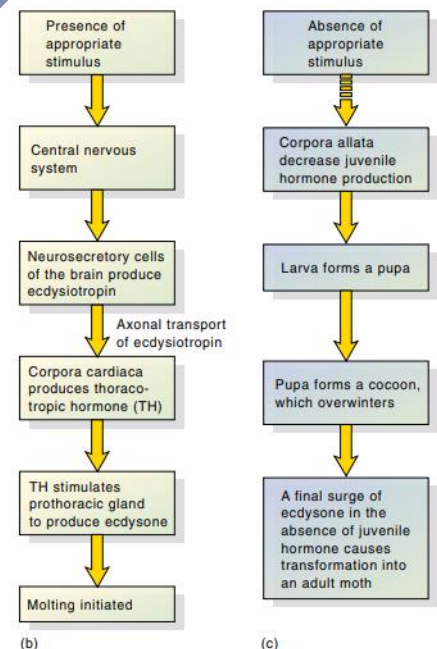
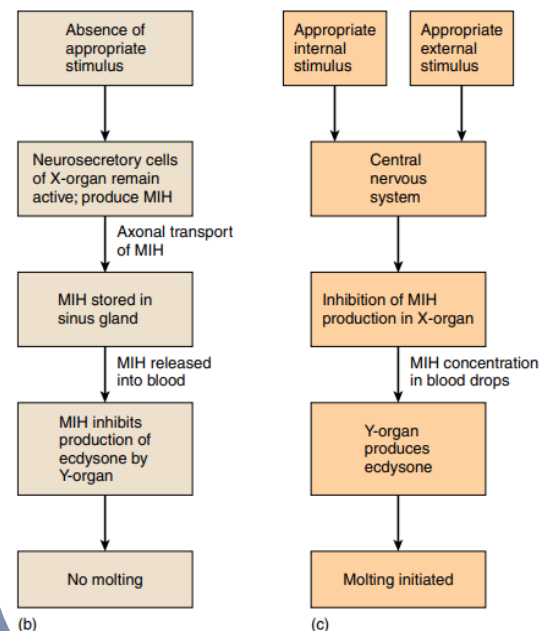
- **Nerve Cells and Brain Control:** Certain endocrine tissues, such as the medullary areas of the adrenal glands, are directed by sets of nerve cells in the brain.
- **Hypothalamus and Pituitary Gland Control:** The hypothalamus and the pituitary gland (hypophysis) exert control over many other endocrine functions.
- **Independent Endocrine Tissues:** Some endocrine tissues operate independently of both nerve signals and the pituitary gland.

- **Types of Glands in Vertebrates**

Vertebrates have two main types of glands:

Exocrine Glands:

- **Definition:** Secrete chemicals into ducts.



- **Function:** These ducts then empty the chemicals into body cavities or onto body surfaces.
- **Examples:** Mammary glands (milk), salivary glands (saliva), and sweat glands (sweat).

Endocrine Glands:

- **Definition:** Do not have ducts.
- **Function:** Secrete hormones directly into the surrounding tissue space. From there, hormones diffuse into the bloodstream, which transports them throughout the body to reach their target cells.

Hormonal Control Mechanisms

- The endocrine system uses hormones as chemical messengers, which are crucial for regulating various physiological processes. The hormones produced by endocrine glands reach their specific target cells through the bloodstream, influencing functions such as growth, metabolism, and reproduction.
- Understanding the vertebrate endocrine system highlights the complex interaction between different control mechanisms—neural, hypothalamic-pituitary, and independent endocrine functions—that collectively maintain homeostasis and coordinate bodily functions.

Endocrine Systems of Vertebrates Other Than Birds or Mammals

Since this text uses a phylogenetic approach to describe form and function in animals, endocrine regulation in vertebrates other than birds or mammals is now discussed. Birds and mammals are the subject of the last part of this chapter. Vertebrates other than birds or mammals have somewhat similar endocrine systems, but differences do exist.

Species Differences in Endocrine Systems

Recent research has revealed the following three aspects of endocrinology that relate to species differences among these vertebrates:

1. **Hormone Variations:** Hormones (or neuropeptides) with the same function in different species may not be chemically identical.
2. **Species-Specific Hormones:** Certain hormones are species-specific with respect to their function; conversely, some hormones produced in one species may be completely functional in another species.
3. **Different Responses to Hormones:** A hormone from one species may elicit a different response in the same target cell or tissue of a different species.

The examples that follow illustrate these three principles and also present a comparative survey of endocrine function in selected vertebrates. When more ancient groups of vertebrates are compared with more recent ones, one general tendency surfaces: older groups seem to have simpler endocrine systems. For example, many of the hormones present in mammals are absent in fishes.

TABLE 25.1
MAJOR VERTEBRATE ENDOCRINE TISSUES AND HORMONES

SOURCE	HORMONES	TARGET CELLS AND PRINCIPAL ACTIONS
Anterior lobe of pituitary (adenohypophysis)	Somatotropin (STH, or growth hormone [GH])	Stimulates growth of bone and muscle; promotes protein synthesis; affects lipid and carbohydrate metabolism; increases cell division
	Adrenocorticotrophic hormone (ACTH)	Stimulates secretion of adrenocortical steroids; is involved in stress response
	Thyrotropin (TSH) of thyroid-stimulating hormone	Stimulates thyroid gland to synthesize and release thyroid hormones concerned with growth, development, metabolic rate
	Endorphins	Decrease pain
	Gonadotropins: Luteinizing or interstitial cell-stimulating hormone (LH or ICSH)	In ovary: Forms corpora lutea; secretes progesterone; probably acts in conjunction with FSH In testis: Stimulates the interstitial cells, thus promoting the secretion of testosterone
	Follicle-stimulating hormone (FSH)	In ovary: Stimulates growth of follicles; functions with LH to cause estrogen secretion and ovulation In testis: Acts on seminiferous tubules to promote spermatogenesis
Intermediate or posterior lobe of pituitary	Prolactin (PRL)	Initiates milk production by mammary glands; acts on crop sacs of some birds; stimulates maternal behavior in birds
	Melanocyte-stimulating hormone (MSH)	Expands amphibian melanophores; contracts iridophores and xanthophores; promotes melanin synthesis; darkens the skin; responds to external stimuli
Posterior lobe of pituitary (neurohypophysis) releases these hormones produced by the hypothalamus	Antidiuretic hormone (ADH or vasopressin)	Elevates blood pressure by acting on arterioles; promotes reabsorption of water by kidney tubules
	Oxytocin	Affects postpartum mammary gland, causing ejection of milk; promotes contraction of uterus; has possible action in parturition and in sperm transport in female reproductive tract

Hormonal Control in Fish

In fishes, the brain and spinal cord are the most important producers of hormones, with other glands being rudimentary.

Major Hormone-Secreting Regions in Fish

1. Pineal Gland (Epithalamus):

- Produces neuropeptides affecting pigmentation and inhibiting reproductive development, both influenced by light.
- Produces melatonin, which affects metabolism by synchronizing activity patterns with light intensity and day length.

2. Preoptic Nuclei (Hypothalamus):

- Produces various neuropeptides controlling functions such as growth, sleep, and movement.

3. Urophysis:

- Located in the spinal cord of the tail.
- Produces neuropeptides that help control water and ion balance, blood pressure, and smooth muscle contractions.
- Its absence in more complex vertebrates is not well understood.

Pineal Gland and Skin Color

In many fishes, amphibians, and reptiles, hormones like melatonin from the pineal gland control variations in skin color. Injecting melatonin from one species into another can cause dramatic color changes, indicating chemical similarity among hormones despite distant evolutionary relationships.

Prolactin

Prolactin, produced by the pituitary gland, has multiple roles across species:

- Stimulates reproductive migrations in animals like salamanders.
- Causes brooding behavior in some fishes.
- Controls water and salt balances, essential for certain saltwater fishes to enter freshwater during spawning.

Evolution of the Thyroid Gland

Evidence suggests that the thyroid gland in the earliest vertebrates evolved from a pouchlike structure (endostyle) in the digestive tract. This explains its position in the neck, ventral to the pharynx, in all vertebrates.

Thyroid Hormones

1. Thyroxine (T4) and Triiodothyronine (T3):

- Regulate metabolism, growth, and tissue differentiation in vertebrates.
- In amphibians, they also control metamorphosis.

Hypothalamus	Thyroid-stimulating hormone (TSH)	Stimulates release of TSH by anterior pituitary
	Adrenocorticotropin-releasing hormone (CRH)	Stimulates release of ACTH by anterior pituitary
	Gonadotropin-releasing hormone (GnRH)	Stimulates gonadotropin release by anterior pituitary
	Prolactin-inhibiting factor (PIF)	Inhibits prolactin release by anterior pituitary
	Somatostatin	Inhibits release of STH by anterior pituitary
Thyroid gland	Thyroxine, triiodothyronine	Affect growth, amphibian metamorphosis, molting, metabolic rate in birds and mammals, development
	Calcitonin	Lowers blood calcium level by inhibiting calcium reabsorption from bone
Parathyroid glands	Parathormone	Regulates calcium concentration
Pancreas, islet cells	Insulin (from beta cells)	Promotes glycogen synthesis and glucose utilization and uptake from blood
	Glucagon (from alpha cells)	Raises blood glucose concentration
Adrenal cortex	Glucocorticoids (e.g., cortisol)	Promote synthesis of carbohydrates and breakdown of proteins; initiate antiinflammatory and antiallergic actions; mediate response to stress
	Mineralocorticoids (e.g., aldosterone)	Regulate sodium retention and potassium loss through kidneys, and water balance
Adrenal medulla	Epinephrine (adrenaline)	Mobilizes glucose; increases blood flow through skeletal muscle; increases oxygen consumption; increases heart rate
	Norepinephrine	Elevates blood pressure; constricts arterioles and venules
Testes	Androgens (e.g., testosterone)	Maintain male sexual characteristics; promote spermatogenesis
Ovaries	Estrogens (e.g., estradiol)	Maintain female sexual characteristics; promote oogenesis
Corpus luteum	Progesterone	Maintains pregnancy; stimulates development of mammary glands

Metamorphosis in Amphibians

In frogs, the concentrations of three hormones—prolactin, thyroxine, and triiodothyronine—control metamorphosis. Low thyroxine and triiodothyronine and high prolactin levels in young tadpoles prevent metamorphosis. As tadpoles grow, the hypothalamus releases hormones that trigger the pituitary gland to stop producing prolactin and start producing thyroid-stimulating hormone, leading to metamorphosis.

Ultimobranchial Glands and Calcitonin

In jawed fishes and primitive tetrapods, small ultimobranchial glands form near the esophagus and produce calcitonin, which helps regulate blood calcium levels.

Adrenal Glands and Stress Response

Specialized endocrine cells (chromaffin tissue) or glands (adrenal glands) near the kidneys prepare vertebrates for stress. They produce epinephrine (adrenaline) and norepinephrine (noradrenaline), which cause vasoconstriction, increased blood pressure, changes in heart rate, and increased blood glucose levels, contributing to "fight-or-flight" reactions.

Endocrine Systems of Birds and Mammals

With some minor exceptions, **birds and mammals** have a similar complement of endocrine glands (figure 25.12). Table 25.1 summarizes the major hormones these vertebrates produce.

Birds

The endocrine glands in birds include the **ovary, testes, adrenals, pituitary, thyroid, pancreas, parathyroids, pineal, hypothalamus, thymus, ultimobranchial gland, and bursa of Fabricius** (figure 25.12a). Since the hormones that most of these glands produce and their effects on target tissues are nearly the same as in mammals, they are discussed in the next section on mammals. A discussion of some unique hormones and their functions in birds follows.

1. **Prolactin:**

- Secreted by the **pituitary gland** in some birds (e.g., pigeons and doves).
- Stimulates the production of "**pigeon's milk**" by desquamation (sloughing off cells) in the pigeon's crop.
- Stimulates and regulates **broodiness** and other parental behaviors.
- Along with estrogen, stimulates full development of the **brood (incubation) patch**, which helps maintain the egg temperature between 33 and 37°C.

2. **Thyroxine:**

- Produced by the **thyroid gland**.
- Regulates the normal development of **feathers** and the **molt cycle**.
- Plays a role in the onset of **migratory behavior**.

3. **Testosterone:**

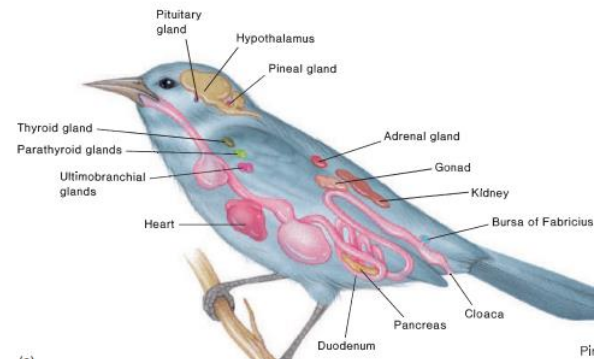
- Produced by the **testes** in male birds.
- Controls secondary sexual characteristics such as **bright plumage color, comb** (if present), and **spurs**.
- Strongly influences **sexual behavior**.

4. Calcitonin:

- Secreted by the **ultimobranchial glands**, small, paired structures in the neck just below the parathyroid glands.
- Involved in regulating **blood calcium concentrations**.

5. Bursa of Fabricius:

- A sac that lies just dorsal to the cloaca and empties into it.
- Well developed during the bird's embryological development, it begins to shrink soon after hatching.
- Produces secretions responsible for the maturation of **white blood cells (B lymphocytes)**, which play an important role in immunological reactions.



Mammals

Zoologists know more about the endocrine organs, hormones, and target tissues of mammals than of any other animal group. This is especially true for the human body. A brief overview of mammalian endocrinology follows.

Pituitary Gland (Hypophysis)

The **pituitary gland** (also known as the hypophysis) is directly below the hypothalamus. The pituitary has two distinct lobes: the **anterior lobe (adenohypophysis)** and the **posterior lobe (neurohypophysis)**. The two lobes differ in several ways:

1. The **adenohypophysis** is larger than the **neurohypophysis**.
2. Secretory cells called **pituicytes** are in the adenohypophysis but not in the neurohypophysis.
3. The **neurohypophysis** has a greater supply of nerve endings.

Pituicytes produce and secrete hormones directly from the adenohypophysis, whereas the neurohypophysis obtains its hormones from the neurosecretory cells in the hypothalamus, storing and releasing them when needed. These modified hypothalamic nerve cells project their axons down a stalk of nerve cells and blood vessels, called the **infundibulum**, into the pituitary gland, directly linking the nervous and endocrine systems.

The pituitary of many vertebrates (but not in humans, birds, and cetaceans) also has a functional **intermediate lobe (pars intermedia)** of mostly glandular tissue. Its secretions (e.g., melanophore-stimulating hormone) in response to external stimuli induce changes in the coloration of the body surface of many animals.

Hormones of the Neurohypophysis

The neurohypophysis does not manufacture any hormones. Instead, the neurosecretory cells of the hypothalamus synthesize and secrete two hormones, **antidiuretic hormone** and **oxytocin**, which move down nerve axons into the neurohypophysis, where they are stored in the axon terminals until released.

Antidiuretic Hormone (ADH)

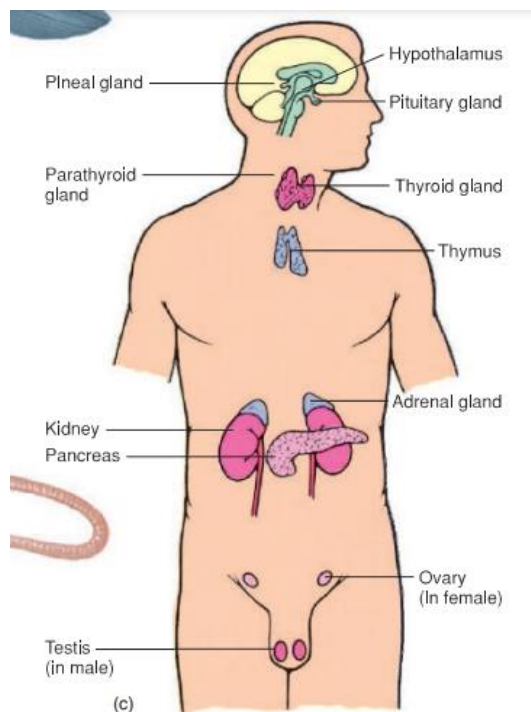
- **Function:** ADH (vasopressin) decreases urine secretion.
- **Mechanism:** Released when a mammal becomes dehydrated, ADH increases water absorption in the kidneys, reducing urine secretion and retaining water.
- **Homeostasis:** This negative feedback system restores water and solute balance.

Oxytocin

- **Function:** Plays a role in reproduction by affecting smooth muscle.
- **Mechanism:** Stimulates uterine contractions to aid in childbirth and promotes milk ejection from mammary glands for newborn nourishment.

Evolutionary Perspective

- **Common Ancestry:** ADH and oxytocin are thought to have evolved from a similar ancestral chemical messenger that controlled water loss and solute concentrations.
- **Adaptation:** The neurohypophysis is larger in animals living in arid regions, where water conservation is crucial.
- **Structural Similarity:** ADH and oxytocin have similar structures, differing in only two amino acids.



Hormones of the Adenohypophysis

The **adenohypophysis** is the true endocrine portion of the pituitary gland, synthesizing six different hormones. These hormones are all polypeptides, and all but two are true **tropic hormones** that target other endocrine glands. The two non-tropic hormones are **growth hormone** and **prolactin**.

Non-Tropic Hormones

1. Growth Hormone (GH) / Somatotropin (STH)

- **Function:** Affects all parts of the body concerned with growth.
- **Mechanism:** Directly induces cell division necessary for growth and protein synthesis by stimulating amino acid uptake, RNA synthesis, and ribosome activity.

2. Prolactin (PRL)

- **Function:** Plays a role in many aspects of reproduction.
- **Mechanism:** Stimulates reproductive migrations in mammals, enhances mammary gland development, and promotes milk production. (Oxytocin stimulates milk ejection from the mammary glands, but not its production.)

Tropic Hormones

1. Thyrotropin / Thyroid-Stimulating Hormone (TSH)

- **Function:** Stimulates the thyroid gland.
- **Mechanism:** Promotes synthesis and secretion of thyroxine, the main thyroid hormone.

2. Adrenocorticotrophic Hormone (ACTH)

- **Function:** Stimulates the adrenal gland.
- **Mechanism:** Promotes production and secretion of steroid hormones called glucocorticoids (cortisol). ACTH secretion is regulated by corticotropin-releasing factor from the hypothalamus, which is controlled by factors such as stress, insulin, ADH, and other hormones.

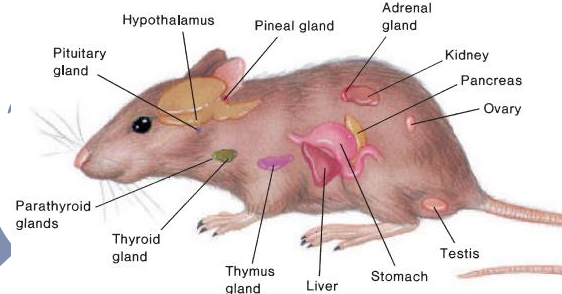
3. Gonadotropins

• Luteinizing Hormone (LH)

- **Function:** Stimulates the gonads.
- **Mechanism:** In females, LH stimulates ovulation and the secretion of estrogen and progesterone from the corpus luteum. In males, LH targets cells in the testes to secrete testosterone.

• Follicle-Stimulating Hormone (FSH)

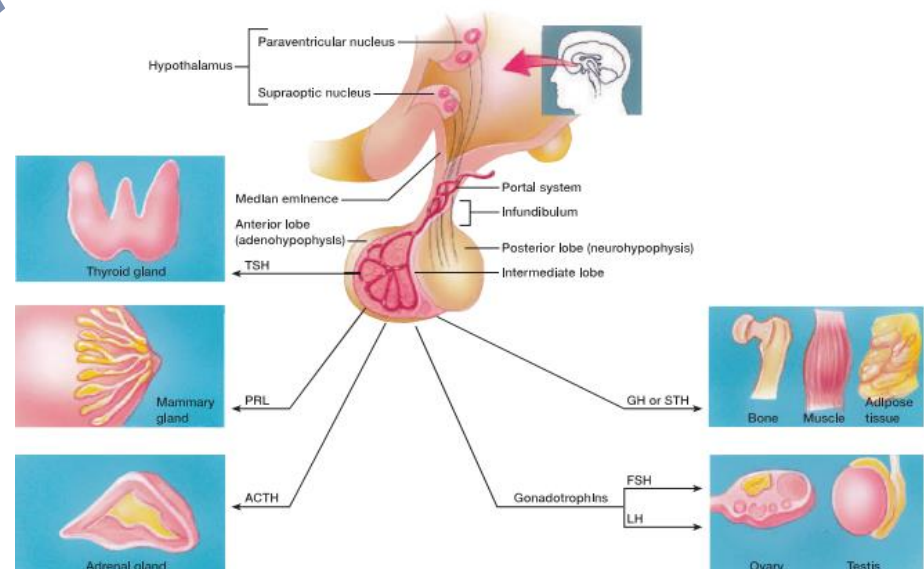
- **Function:** Stimulates the gonads.
- **Mechanism:** In females, FSH stimulates follicular cells in the ovaries to develop into mature eggs and produce estrogen. In males, FSH stimulates the cells of the testes to produce sperm.



Pineal Gland

The **pineal gland** (or pineal body) is named for its pine cone shape. Its distinctive cells evolved from the photoreceptors of lower vertebrates and synthesize **melatonin**. These cells are most active in the dark, as light inhibits the enzymes needed for melatonin synthesis.

- **Function:** Melatonin affects many physiological processes and adjusts them to diurnal and seasonal cycles.



- **Evolutionary Adaptation:** Melatonin ensures that periodic activities in mammals occur when environmental conditions are optimal.
- **Role in Humans:** Decreased melatonin secretion may help trigger the onset of puberty, the age at which reproductive structures start to mature.

Thyroid Gland and Parathyroid Glands

Thyroid Gland

The **thyroid gland** is located in the neck, in front of the trachea (see Figure 25.12). It produces two important hormones: **thyroxine** and **triiodothyronine**. These hormones affect overall **growth, development, and metabolism**.

Another hormone from the thyroid gland is **calcitonin**. Calcitonin helps control the level of **calcium ions (Ca^{2+})** in the blood. When calcium levels are high, calcitonin promotes the storage of calcium in the bones. When calcium levels return to normal, the thyroid gland reduces calcitonin production.

Parathyroid Glands

The **parathyroid glands** are small, pea-sized glands located in the thyroid lobes, usually two in each lobe (see Figure 25.12). They produce a hormone called **parathormone (PTH)**. PTH regulates the levels of **calcium (Ca^{2+})** and **phosphate (HPO_4^{2-})** ions in the blood.

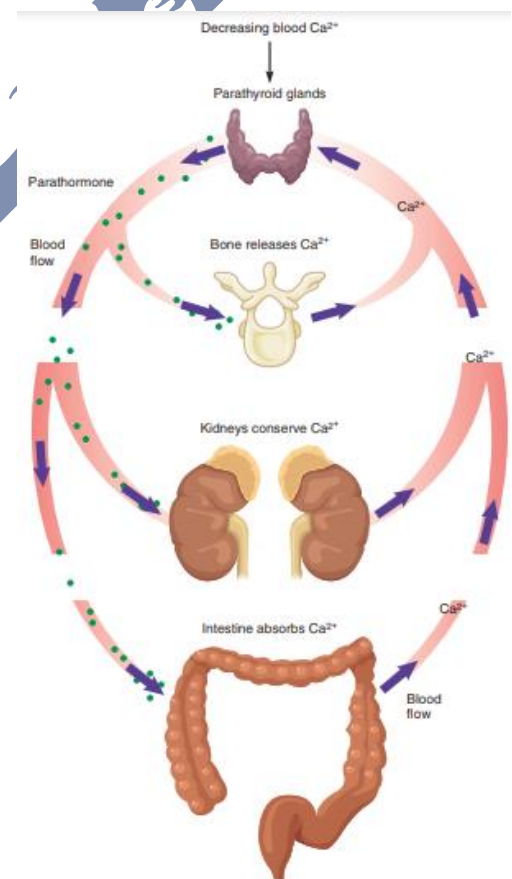
When blood calcium levels are low, the parathyroid glands increase the production of PTH. PTH has several effects:

- It stimulates bone cells to break down bone tissue and release **calcium ions** into the blood.
- It enhances calcium absorption from the **small intestine** into the blood.
- It promotes calcium reabsorption by the **kidneys**, reducing the amount of calcium lost in urine.

Figure 25.15 illustrates the **negative feedback system** for parathormone.

Adrenal Glands

In mammals, the **adrenal glands** are located on top of the kidneys. Each adrenal gland is made up of two different types of tissue: the **medulla** (inner portion) and the **cortex** (outer portion) (see Figure 25.16).



Adrenal Cortex

The **adrenal cortex** produces three types of steroid hormones: **glucocorticoids**, **mineralocorticoids**, and **sex hormones**.

Glucocorticoids

- **Cortisol** is the main glucocorticoid.
- **Functions:** Regulates overall **metabolism** and **blood sugar levels**. It also helps the body respond to **infections** and **injuries**.

Mineralocorticoids

- **Aldosterone** is the primary mineralocorticoid.
- **Functions:** Maintains the balance of **solutes** (like sodium) in the extracellular fluid. It also promotes **sodium reabsorption** in the kidneys, which leads to **water reabsorption**. This helps maintain the **homeostasis** of extracellular fluid.

Sex Hormones

- The adrenal cortex produces **sex hormones**, including **androgens** (weak male hormones) and **estrogens** (female hormones).
- **Functions:** These hormones usually have only a minor effect on the **gonads** (male and female reproductive organs).

Adrenal Medulla

The **adrenal medulla** is controlled by the nervous system. It contains **neurosecretory cells** that release two important hormones: **epinephrine (adrenaline)** and **norepinephrine (noradrenaline)**. These hormones help regulate **heart rate** and **carbohydrate metabolism**.

Neural Control

- **Brain Centers and Hypothalamus:** The brain centers and the hypothalamus control the adrenal medulla through **sympathetic nerves**.
- **Functions:** During times of **excitement**, **emergency**, or **stress**, the adrenal medulla is activated by the sympathetic nervous system.

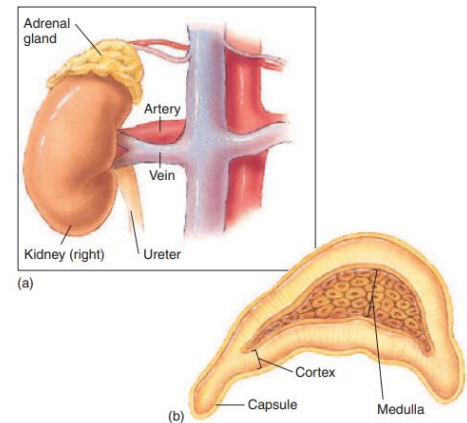
Fight-or-Flight Response

In response to **epinephrine** and **norepinephrine**:

- The **heart rate** increases.
- **Blood flow** increases to many vital organs.
- The **airways in the lungs** dilate.
- More **oxygen** is delivered to all cells of the body.

These changes prepare the body to react quickly and effectively to emergencies, a response known as the "**fight-or-flight**" response.

Pancreas



The **pancreas** is an elongated, fleshy organ located behind the stomach. It has both exocrine and endocrine functions.

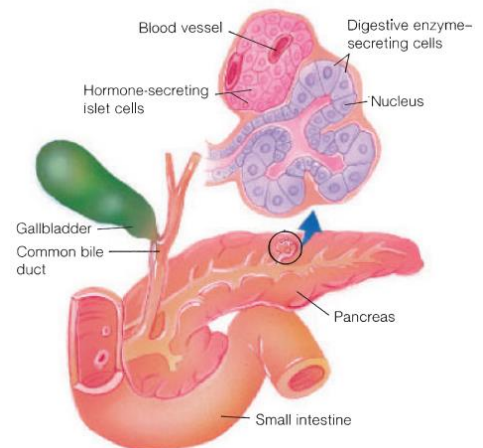
Exocrine Function

As an **exocrine gland**, the pancreas secretes digestive enzymes through ducts.

Endocrine Function

As an **endocrine gland** (ductless), the pancreas performs the following functions:

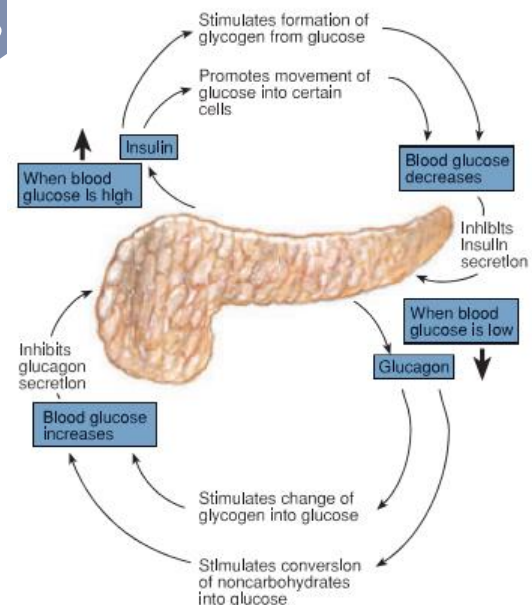
- The endocrine portion constitutes about 1% of the gland.
- It consists of clusters of cells called **pancreatic islets**.
- There are 200,000 to 2,000,000 pancreatic islets scattered throughout the gland.



Pancreatic Islets

Each pancreatic islet contains four types of specialized cells:

1. **Alpha (α) cells:**
 - Produce the hormone **glucagon**.
2. **Beta (β) cells:**
 - Produce **insulin**.
3. **Delta (δ) cells:**
 - Secrete **somatostatin**. Somatostatin inhibits the secretion of glucagon and insulin.
4. **F cells:**
 - Secrete **pancreatic polypeptide**. This polypeptide inhibits somatostatin secretion, gallbladder contraction, and the secretion of pancreatic digestive enzymes after a meal.



Regulation of Blood Glucose

Insulin

- When blood glucose levels are high (e.g., after a meal), **beta cells** secrete **insulin**.
- Insulin promotes the **uptake of glucose** by body cells, including liver cells, where excess glucose is converted to **glycogen** (a storage form of glucose).

Glucagon

- When blood glucose levels are low, **alpha cells** secrete **glucagon**.
- Glucagon stimulates the **breakdown of glycogen** into glucose, which is then released into the bloodstream to raise blood glucose levels to the homeostatic range.

Insulin and glucagon are crucial for regulating blood glucose concentrations through a **negative feedback system**. This system maintains appropriate blood glucose levels, ensuring the body's energy needs are met efficiently.

TABLE 25.2
SOME OTHER MAJOR SOURCES OF VERTEBRATE HORMONES

GLAND/ORGAN	HORMONE	FUNCTION	TARGET AREA
Placenta	Estrogens, progesterone, human chorionic gonadotropin (hCG), human chorionic somatomammotropin (hCS)	Maintain pregnancy	Ovaries, mammary glands, uterus
Digestive tract	Secretin	Stimulates release of pancreatic juice to neutralize stomach acid	Cells of pancreas
	Gastrin	Stimulates digestive enzymes and HCl in stomach	Stomach mucosa
	Cholecystokinin (CCK)	Stimulates release of pancreatic enzymes and bile from gallbladder	Pancreas, gallbladder
Heart	Atrialpeptin	Lowers blood pressure, maintains fluid balance	Blood vessels, kidneys
Kidneys	Erythropoietin	Stimulates red blood cell production	Bone marrow
	Urotensin	Stimulates constriction of arteries	Major arteries
	Calcitriol	Aids in the absorption of dietary calcium and phosphorus	Small intestine
Adipose tissue	Leptin	Suppresses appetite	Brain

Gonads

The **gonads** (ovaries and testes) secrete hormones that regulate reproductive functions.

Male Gonads (Testes)

- **Testosterone:**
 - Works with **luteinizing hormone (LH)** and **follicle-stimulating hormone (FSH)** from the adenohypophysis to stimulate **spermatogenesis**.
 - Necessary for the growth and maintenance of the **male sex organs**.
 - Promotes the development and maintenance of **sexual behavior**.
 - Stimulates the growth of **facial and pubic hair** and the **enlargement of the larynx**, which deepens the voice.
- **Inhibin:**
 - Inhibits the secretion of **FSH**.

Female Gonads (Ovaries)

- **Estrogens** (estrin, estrone, estradiol):
 - Regulate the **menstrual and estrus cycles**.
 - Aid in the development of the **mammary glands** and other female **secondary sexual characteristics**.
- **Progestins** (primarily progesterone):
 - Regulate the **menstrual and estrus cycles**.

- Aid in the development of the **mammary glands** and **placenta formation** during pregnancy.
- **Relaxin:**
 - Softens the **cervix** at the time of delivery.
- **Inhibin:**
 - Inhibits the secretion of **FSH**.

Thymus

The **thymus gland** is located near the heart. It is large and noticeable in young birds and mammals but shrinks in size throughout adulthood.

- Major hormonal products:
 - **Thymopoietin (TP)**
 - **Alpha1 thymosin**
 - **Beta4 thymosin**
- These hormones are essential for the normal development of the **immune system**.

Circulation and Gas Exchange:

Internal Transport and Circulatory Systems

Introduction

All animals need to maintain a stable internal environment, or homeostasis. This involves the circulation of nutrients, wastes, and gases throughout the body. Internal transport or circulatory systems reduce the distance that these substances must travel by diffusion. The complexity and nature of these systems vary depending on the size and lifestyle of the animal.

Transport Systems in Invertebrates

Protozoa

- **Diffusion:** Protozoa rely on simple diffusion due to their small size and high surface-area-to-volume ratio.
- **Medium:** Materials diffuse through the plasma membrane and cytoplasm.

Sponges

- **Water Circulation:** Sponges circulate external water through their bodies instead of using an internal fluid system.

Cnidarians

- **Gastrovascular Cavity:** Hydra and similar cnidarians use a fluid-filled gastrovascular cavity to supply nutrients and oxygen to cells and remove wastes.

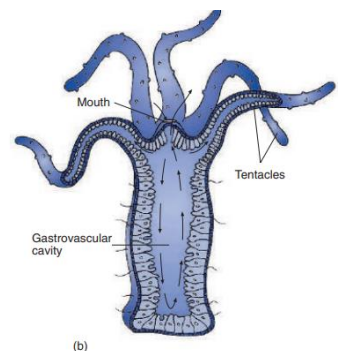
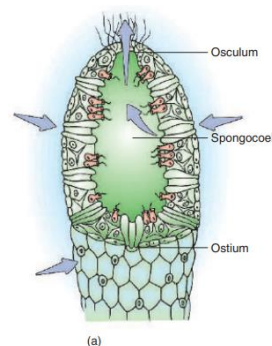
Flatworms

- **Complex Gastrovascular Cavity:** Planarians have a branched gastrovascular cavity that shortens diffusion distances for nutrients, gases, and wastes. Body movements help distribute materials.

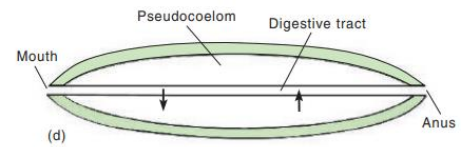
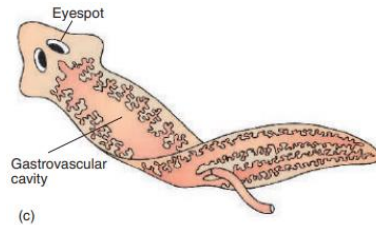
Pseudocoelomate Invertebrates

- **Coelomic Fluid:** Rotifers, gastrotrichs, and nematodes use the fluid in their body cavity for transport. Body movements ensure adequate distribution.

Molluscs and Beyond



- **Separate Circulatory Systems:** Beginning with molluscs, many invertebrates have developed separate circulatory systems.

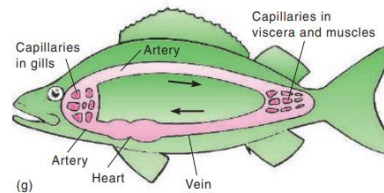
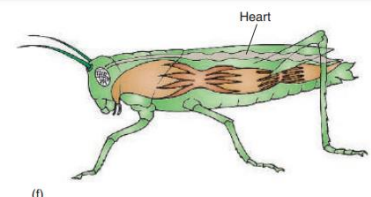
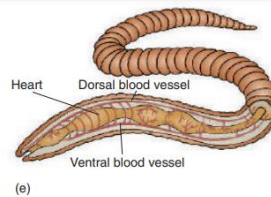


- **Types of Circulatory Systems:**

- **Open Circulatory System:** Hemolymph is pumped by the heart into the body cavity, where it bathes cells and tissues directly.
- **Closed Circulatory System:** Blood circulates within vessels, delivering nutrients and removing wastes.

Examples

- **Annelids (e.g., Earthworms):** Have a closed circulatory system with blood confined to vessels.
- **Molluscs and Arthropods:** Most have open circulatory systems where hemolymph bathes cells and tissues directly. For example, insects have hearts that pump hemolymph through vessels that open into a body cavity (hemocoel).



In summary, invertebrates use a variety of transport systems ranging from simple diffusion to complex circulatory systems, depending on their size and complexity. These systems ensure that nutrients, gases, and wastes are efficiently circulated to maintain homeostasis.

Characteristics of Invertebrate Coelomic Fluid, Hemolymph, and Blood Cells

Coelomic Fluid

- **Usage:** Some invertebrates like echinoderms, annelids, and sipunculans use coelomic fluid for transport within the body.
- **Composition:** Can be similar to interstitial fluids or may differ, especially in proteins and cells.
- **Functions:**
 - Transports gases, nutrients, and wastes.
 - Can act as a hydrostatic skeleton in certain invertebrates, such as annelids.

Hemolymph

- **Definition:** Hemolymph is the circulating fluid in animals with open circulatory systems, including most arthropods, ascidians, and many molluscs.
- **Circulation:**
 - Pumped by the heart at low pressures through vessels into tissue spaces (hemocoel) and sinuses.
 - Circulation is generally slow, and hemolymph volume is high.
- **Functions:**
 - Transports essential gases, nutrients, and wastes.
 - Noncirculatory roles in some invertebrates:
 - **Insects:** Hemolymph pressure assists in molting and wing inflation.
 - **Jumping Spiders:** Provides a hydraulic mechanism for limb extension.

Blood Cells (Hemocytes)

- **Presence:** Found in coelomic fluid, hemolymph, or blood of most invertebrates.
- **Types and Functions:**
 - **Erythrocytes (Red Blood Cells):** Contain respiratory pigments like hemoglobin to facilitate oxygen transport.
 - **Non-pigmented Cells:** Involved in various functions, such as blood clotting.

Examples by Invertebrate Group

1. Annelids:

- Blood contains phagocytic hemocytes.
- Coelomic fluid has coelomocytes (amoebocytes, eleocytes, lampocytes, linocytes) involved in:
 - Phagocytosis
 - Glycogen storage
 - Encapsulation
 - Defense responses
 - Excretion

2. Molluscs:

- Hemolymph contains two main types of hemocytes:
 - **Amoebocytes:** Involved in immune responses and nacre formation (pearl formation in some bivalves).
 - **Granulocytes:** Perform various functions including phagocytosis.

3. Insects:

- Hemolymph contains a variety of hemocyte types that function in:
 - Phagocytosis
 - Encapsulation
 - Clotting

In summary, the internal transport systems of invertebrates feature a variety of fluids and cells adapted to their specific needs, ranging from basic diffusion in simple organisms to more complex systems in larger or more active animals. These systems ensure efficient transport of essential substances and contribute to other physiological functions, aiding in the overall maintenance of homeostasis.

Transport Systems in Vertebrates

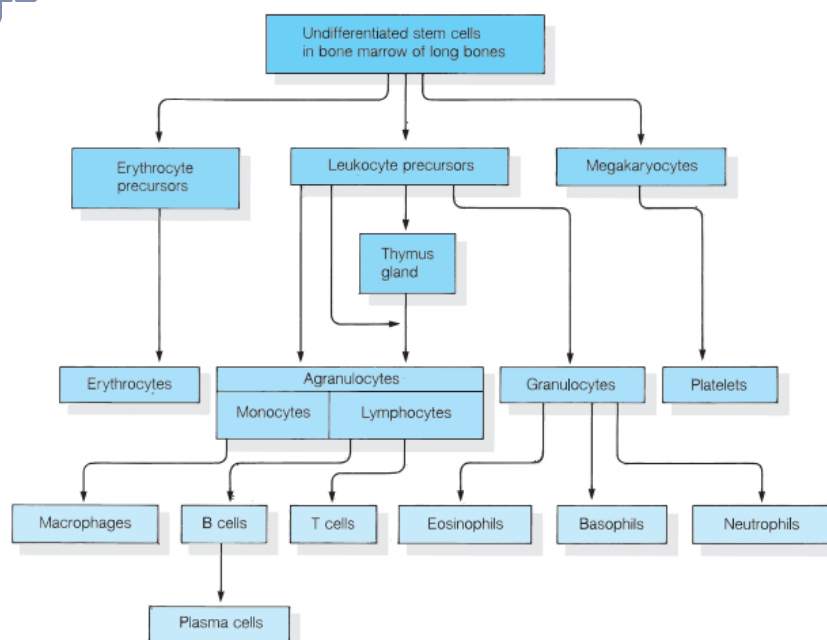
All vertebrates have a closed circulatory system in which the walls of the heart and blood vessels are always contracted, and blood never leaves the blood vessels (figure 26.1g). Blood moves from the heart, through arteries, smaller arteries (arterioles), tiny blood vessels (capillaries), small veins (venules), veins, and back to the heart. Exchange between the blood and the fluid outside the cells only occurs at the capillary level.

Characteristics of Vertebrate Blood and Blood Cells

Overall, vertebrate blood transports oxygen, carbon dioxide, and nutrients; defends against harmful microorganisms, cells, and viruses; prevents blood loss through clotting; and helps regulate body temperature and pH. Because it is a liquid, vertebrate blood is classified as a special type of connective tissue. Like other connective tissues, blood contains a fluid matrix called plasma and cellular parts called formed elements.

Plasma

Plasma (Gr., anything formed or molded) is the straw-colored, liquid part of blood. In mammals, plasma is about 90% water and provides the liquid for dissolving and transporting nutrients. A group of proteins (albumin, fibrinogen, and globulins) makes up another 7% of the plasma. The amount of these plasma proteins affects the movement of water between the blood and the fluid outside the cells. Because albumin makes up about 60% of the total plasma proteins, it plays important roles with respect to water movement. Fibrinogen is needed for blood clotting, and the globulins include the immunoglobulins and various metal-binding proteins. Serum is plasma from which the proteins involved in blood clotting have been removed. The gamma globulin part functions in the immune response because it consists mostly of antibodies. The remaining 3% of plasma is made up of electrolytes, amino acids, glucose and other nutrients, various enzymes, hormones, metabolic wastes, and traces of many inorganic and organic molecules.



Formed Elements

The formed-element part (cellular component) of vertebrate blood consists of red blood cells (erythrocytes), white blood cells (leukocytes), and platelets (thrombocytes) (figure 26.3). White blood cells are present in lower numbers than red blood cells, generally being 1 to 2% of the blood by volume. White blood cells are divided into agranulocytes (without granules in the cytoplasm) and granulocytes (have granules in the cytoplasm). The two types of agranulocytes are lymphocytes and monocytes. The three types of granulocytes are eosinophils, basophils, and neutrophils. Fragmented cells are called platelets (thrombocytes). Each of these cell types is now discussed in more detail.

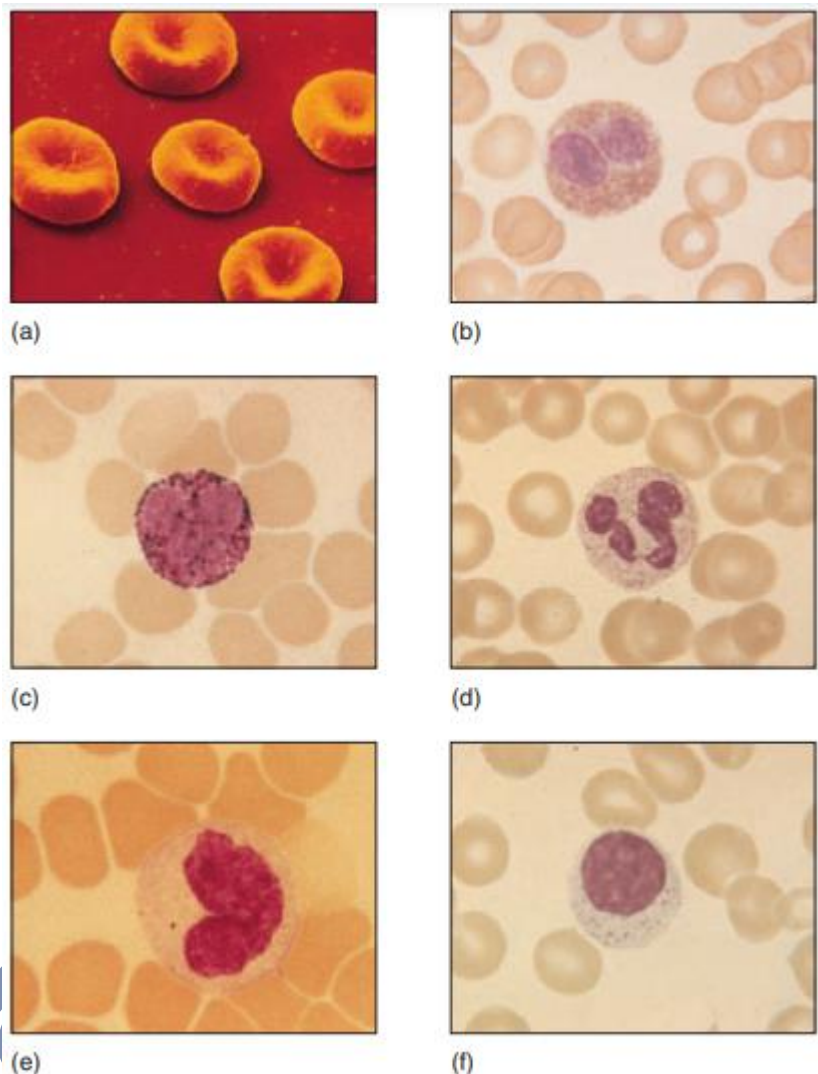
Red Blood Cells

Red blood cells (erythrocytes; Gr. erythros, red cells) vary greatly in size, shape, and number in different vertebrates (figures 26.4 and 26.5a). For example, the RBCs of most vertebrates are **nucleated** (have a nucleus), but mammalian RBCs are **enucleated** (without a nucleus). Some fishes and amphibians also have enucleated RBCs. Among all vertebrates, the **salamander *Amphiuma*** has the largest RBC (figure 26.4a). **Bird RBCs** (figure 26.4c) are oval-shaped, nucleated, and larger than mammalian RBCs. Among birds, the **ostrich** has the largest RBC. Most mammalian RBCs are **biconcave disks** (figure 26.5a); however, the **camel** (figure 26.4e) and **llama** have elliptical RBCs. The shape of a biconcave disk provides a larger surface area for gas diffusion than a flat disk or sphere. Generally, the lower vertebrates tend to have fewer but larger RBCs than the higher vertebrates.

Almost the entire mass of an RBC consists of **hemoglobin** (Gr. haima, blood L. globulus, little globe), an iron-containing protein. The major function of a red blood cell is to pick up oxygen from the environment, bind it to hemoglobin to form **oxyhemoglobin**, and transport it to body tissues. Blood rich in oxyhemoglobin is bright red. As oxygen diffuses into the tissues, blood becomes darker and appears blue when observed through the blood vessel walls. However, when this less oxygenated blood is exposed to oxygen (such as when a vein is cut and a mammal begins to bleed), it immediately turns bright red. Hemoglobin also carries waste carbon dioxide (in the form of **carbaminohemoglobin**) from the tissues to the lungs (or gills) for removal from the body.

White Blood Cells

White blood cells (leukocytes) (Gr. leukos, white cells) are scavengers that destroy microorganisms at infection sites, remove foreign chemicals, and clear debris resulting from dead or injured cells. All



WBCs are derived from immature cells (stem cells) in bone marrow through a process called **hematopoiesis** (Gr. hemato, blood poiein, to make; see figure 26.3).

Granulocytes

Among the granulocytes, **eosinophils** are phagocytic, ingesting foreign proteins and immune complexes rather than bacteria (figure 26.5b). In mammals, eosinophils also release chemicals that counteract the effects of certain inflammatory chemicals during allergic reactions. **Basophils** are the least numerous WBCs (figure 26.5c). When they encounter a foreign substance, their granules release histamine and heparin. Histamine causes blood vessels to widen and leak fluid at sites of inflammation, while heparin prevents blood clotting. **Neutrophils** are the most abundant of the white blood cells (figure 26.5d). They are chemically drawn to sites of inflammation and are active phagocytes.

Agranulocytes

The two types of agranulocytes are **monocytes** and **lymphocytes** (figure 26.5e,f). Two distinct types of lymphocytes are **B cells** and **T cells**, both crucial to the immune response. B cells originate in the bone marrow and mature in lymphoid tissue, while T cells are associated with the thymus gland before colonizing lymphoid tissue and contributing to the immune response. When B cells are activated, they divide and differentiate to produce plasma cells.

Platelets (Thrombocytes)

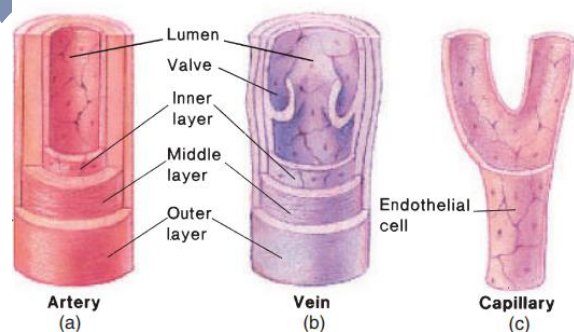
Platelets (named for their flat, plate-like shape), or **thrombocytes** (Gr. thrombus, clot cells), are disk-shaped cell fragments that initiate blood clotting. When a blood vessel is injured, platelets quickly move to the site and clump, adhering to the damaged area and initiating the process of blood coagulation.

Vertebrate Blood Vessels

Arteries are elastic blood vessels that carry blood away from the heart to the organs and tissues of the body. The central canal of an artery (and all blood vessels) is called a **lumen**. Surrounding the lumen of an artery is a thick wall composed of three layers, called **tunics** (L. tunica, covering) (figure 26.6a).

Most **veins** are relatively rigid, large vessels that carry blood from the body tissues back to the heart. The wall of a vein also contains the same three layers (tunics) as arterial walls, but the middle layer is much thinner, and one or more **valves** are present (figure 26.6b). Valves permit blood flow in only one direction, which is crucial for returning the blood to the heart.

Arteries lead to **terminal arterioles** (those closest to a capillary). The arterioles then branch to form **capillaries** (L. capillus, hair), which connect to **venules** and then to veins. **Capillaries** are generally made up of a single layer of endothelial cells and are the most numerous blood vessels in an animal's body (figure 26.6c). An abundance of capillaries provides an enormous surface area for the exchange of gases, fluids, nutrients, and wastes between the blood and nearby cells.



The Hearts and Circulatory Systems of Bony Fishes, Amphibians, and Reptiles

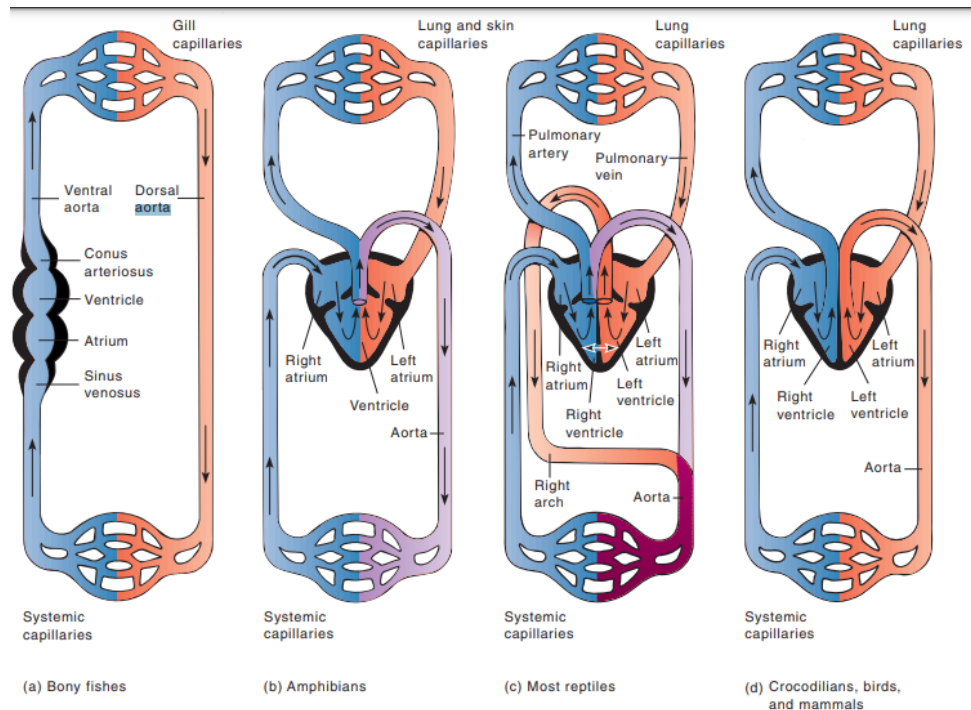
The heart and blood vessels underwent significant changes as vertebrates transitioned from water to land and as endothermy evolved. Let's delve into these transformations with examples from bony fishes, amphibians, and reptiles.

Bony Fishes:

The **bony fish heart** has two chambers—the atrium and ventricle (figure 26.7a). Blood exits the heart through the ventral aorta, which leads to the gills. In the gills, blood becomes oxygenated, releases carbon dioxide, and then enters the dorsal aorta. The dorsal aorta distributes blood to all body organs, and blood eventually returns to the heart through the venous system. This system, called a **single circulation circuit**, has the advantage of circulating oxygenated blood from the gills to the systemic capillaries in all organs almost simultaneously. However, the circulation of blood through the gill capillaries creates resistance to flow, leading to reduced blood pressure and flow rates to other organs. This arrangement may not support the high metabolic rates present in some birds and mammals.

Amphibians:

In **amphibians**, a **double circulatory circuit** evolved, where blood passes through the heart twice during its circuit through the body. Amphibians typically have hearts that are not fully divided into two chambers. A single ventricle pumps blood to both the lungs and the rest of the body (figure 26.7b). Since most amphibians absorb more oxygen through their skin than through their lungs or gills, oxygenated blood returning from the skin also contributes to the ventricle, resulting in highly oxygenated blood pumped out to the rest of the body.



Reptiles:

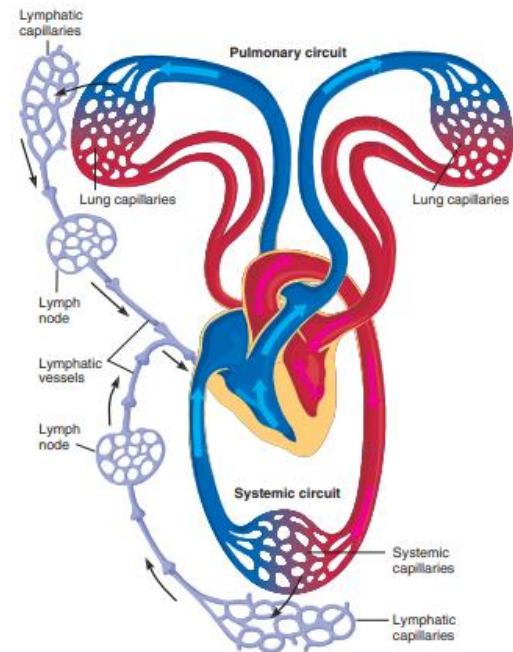
In most **reptiles**, such as turtles, the ventricle is partially divided into right and left sides (figure 26.7c). Oxygenated blood from the lungs returns to the left side of the heart via the pulmonary vein and does not mix much with deoxygenated blood in the right side of the heart. When the ventricles contract, blood is pumped out to two aortae for distribution throughout the body and to the lungs. This incomplete separation of the ventricles is an essential adaptation for reptiles, allowing blood to be diverted away from the pulmonary circulation during diving and when the turtle withdraws into its shell. This conserves energy and redirects blood to vital organs when the lungs cannot be ventilated.

The Hearts and Circulatory Systems of Birds and Mammals

While the physiological separation of blood in left and right ventricles is nearly complete in reptiles, the **complete anatomical separation** of ventricles only occurs in crocodilians, birds, and mammals (figure 26.7d). This enables the **double circulation** required to maintain high blood pressure. High blood pressure is crucial for the rapid delivery of oxygenated, nutrient-rich blood to tissues with high metabolic rates.

Blood circulates throughout the avian and mammalian body in two main circuits: the **pulmonary circuit** and the **systemic circuit** (figure 26.8).

- The **pulmonary circuit** supplies blood only to the lungs. It carries **oxygen-poor (deoxygenated) blood** from the heart to the lungs, where carbon dioxide is removed, and oxygen is added. Then, it returns the **oxygen-rich (oxygenated) blood** to the heart for distribution to the rest of the body.
- The **systemic circuit** supplies all cells, tissues, and organs of the body with **oxygen-rich blood** and returns **oxygen-poor blood** to the heart.



THE HUMAN HEART: A Powerful Pump

Pumping Capacity

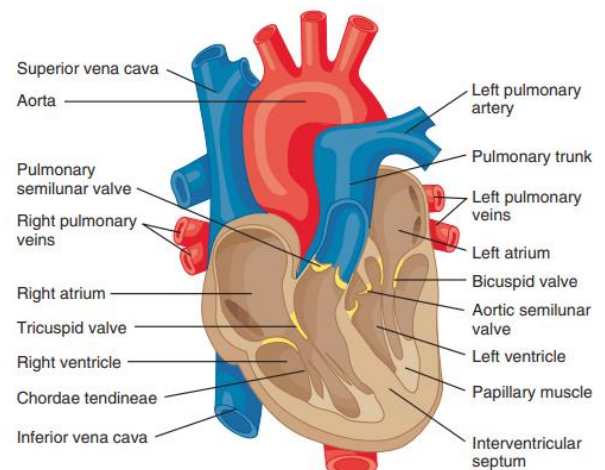
- **Volume and Rate:** Moves approximately 5 liters of blood per minute; handles around 8,000 liters daily through 96,000 km of blood vessels
- **Frequency:** Beats about 70 times per minute in an average adult, exceeding 100,000 times per day
- **Lifetime Activity:** Over 2.6 billion beats in a typical 70-year lifespan without tiring

Heart Structure

- **Myocardium:** Dominant tissue type, consisting of cardiac muscle
- **Epicardium:** Outer layer providing protective fibrous connective tissue
- **Endocardium:** Inner lining, formed by connective tissue and endothelium

Chamber Configuration and Valves

- **Atria and Ventricles:** Heart divided into left and right halves, each with an atrium and ventricle
- **Valves:** Ensure one-way blood flow; including tricuspid, bicuspid, pulmonary, and aortic valves



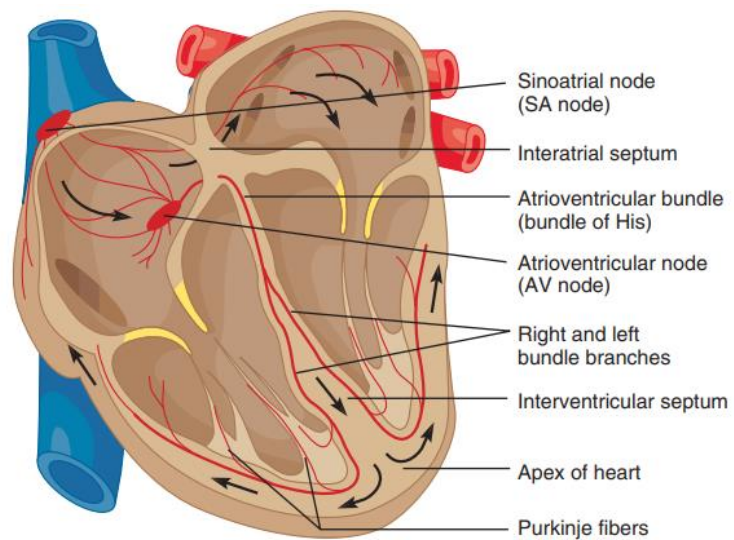
Cardiac Cycle

- **Initiation:** Sinoatrial node (SA node) initiates each heartbeat

- **Propagation:** Electrical signals travel through atrioventricular node (AV node) and bundle of His
- **Contraction:** Ventricles contract due to stimulation by Purkinje fibers

Electrical Activity and Monitoring

- **Pacemaker Function:** SA node generates electrical impulse, initiating cardiac cycle
- **Recording:** Electrocardiogram (ECG or EKG) monitors heart's electrical activity



Systole and Diastole

- **Contraction and Relaxation:** Atria and ventricles undergo rhythmic cycles of contraction (systole) and relaxation (diastole), facilitating blood movement

Blood Pressure Overview

Blood pressure is the fluid pressure generated by ventricular contraction that forces blood through the pulmonary and systemic circuits. It is the force that blood exerts against the inner walls of blood vessels and is a crucial measurement for assessing cardiovascular health.

Systemic Arterial Blood Pressure

Although blood pressure occurs throughout the vascular system, the term typically refers to systemic arterial blood pressure. This pressure rises and falls in a pattern that corresponds to the phases of the cardiac cycle:

1. **Ventricular (Contraction):** **Systole**

During this phase, the ventricles contract, forcing blood into the pulmonary arteries and the aorta. The pressure in these arteries rises sharply. The highest pressure achieved during this phase is known as the **systolic pressure**.

2. **Ventricular (Relaxation):** **Diastole**

TABLE 26.1

MAJOR STRUCTURAL AND FUNCTIONAL COMPONENTS OF THE LYMPHATIC SYSTEM IN VERTEBRATES

STRUCTURE	FUNCTION
Lymphatic capillaries	Collect excess extracellular fluid in tissues
Lymphatics	Carry lymph from lymphatic capillaries to veins in the neck, where lymph returns to the bloodstream
Lymph nodes	House the WBCs that destroy foreign substances; play a role in antibody formation
Spleen	Filters foreign substances from blood; manufactures phagocytic lymphocytes; stores red blood cells; releases blood to the body when blood is lost
Thymus gland (in mammals)	Site of antibodies in the newborn; is involved in the initial development of the immune system; site of T-cell differentiation
Bursa of Fabricius (in birds)	A lymphoid organ at the lower end of the alimentary canal in birds; the site of B-cell maturation

During this phase, the ventricles relax. The arterial pressure drops. The lowest pressure in the arteries before the next ventricular contraction is known as the **diastolic pressure**.

Normal Blood Pressure Values

Systolic Pressure: The normal systolic pressure for a young adult is about 120 mm Hg. This is the amount of pressure required to raise a column of mercury in a sphygmomanometer (sfig-mo-mah-nometer) to 120 mm.

Diastolic Pressure: The normal diastolic pressure is approximately 80 mm Hg.

These readings are conventionally expressed as **120/80 mm Hg**.

Importance of Blood Pressure Monitoring

Monitoring blood pressure is essential for detecting and managing conditions such as hypertension (high blood pressure) and hypotension (low blood pressure), both of which can lead to serious health complications if left unaddressed. Regular check-ups can help maintain optimal cardiovascular health and prevent long-term issues.

THE LYMPHATIC SYSTEM

The lymphatic system in vertebrates begins with lymphatic capillaries, which interact directly with the extracellular fluid surrounding tissues. This system performs four primary functions:

1. **Fluid Collection and Drainage:** It collects and drains most of the fluid that seeps out of the bloodstream and accumulates in the extracellular fluid.
2. **Protein Return:** It returns small amounts of proteins that have exited the cells.
3. **Lipid Transport:** It transports lipids absorbed from the small intestine.
4. **Transport of Foreign Particles and Debris:** It moves foreign particles and cellular debris to lymph nodes, which are disposal centers.

Lymphatic capillaries merge to form larger vessels known as lymphatics, which are thin-walled and equipped with valves to ensure the one-way flow of lymph. Lymph, the clear extracellular fluid in these vessels, travels through lymph nodes on its way back to the heart. Lymph nodes are concentrated in various body areas and play a crucial role in defending against disease.

In birds and mammals, the lymphatic system also includes lymphoid organs such as the spleen and, depending on the species, either the bursa of Fabricius in birds or the thymus gland, tonsils, and adenoids in mammals. This system is essential for an animal's defense against injury and infection.

Gas Exchange

To take advantage of the rich source of energy in earth's organic matter, animals must solve two practical problems. First, they must break down and digest the organic matter so that it can enter the cells that metabolize it. Second, they must provide cells with both an adequate supply of oxygen required for aerobic respiration and a way of eliminating the carbon dioxide that aerobic respiration produces. This process of gas exchange with the environment, also called **external respiration**, is the subject of the rest of this chapter.

Respiratory Surfaces

Protists and animals have five main types of respiratory systems (surfaces):

1. **Simple Diffusion across Plasma Membranes**
2. **Tracheae**
3. **Cutaneous (Skin) Exchange**
4. **Gills**
5. **Lungs**

Invertebrate Respiratory Systems

- **Single-celled Protists:** In single-celled protists, such as protozoa, diffusion across the plasma membrane moves gases into and out of the organism.
- **Multicellular Invertebrates:** Some multicellular invertebrates either have very flat bodies (e.g., flatworms) in which all body cells are close to the body surface or are thin-walled and hollow (e.g., Hydra). Gases diffuse into and out of these animals.
- **Integumentary Exchange:** Invertebrates such as earthworms that live in moist environments use integumentary exchange. Earthworms have capillary networks just under their skin and exchange gases with the air spaces among soil particles.
- **Gills:** Most aquatic invertebrates carry out gas exchange with gills. Sea stars have simple gills that are small skin projections. Marine and annelid worms have prominent lateral projections called parapodia that function as gills. Crustaceans and molluscs have compact gills protected by hard coverings, divided into highly branched structures to maximize the area for gas exchange.
- **Tracheal Systems:** Some terrestrial invertebrates (e.g., insects, centipedes, and some mites, ticks, and spiders) have tracheal systems consisting of highly branched tubes called tracheae. Tracheae open to the outside through spiracles, leading to branching tracheal trunks that give rise to smaller branches called tracheoles. Most insects have ventilating mechanisms that move air into and out of the trachea. Arachnids possess tracheae, book lungs, or both.
- **Book Lungs:** Book lungs are paired invaginations of the ventral body wall folded into a series of leaflike lamellae. Air enters the book lung through a slitlike opening called a spiracle and circulates between lamellae, with gases diffusing between the hemolymph and the air.
- **Pulmonate Lungs:** Land snails and slugs, part of the molluscan subclass Pulmonate, have a lung that opens to the outside via a pore called a pneumostome. This lung is derived from the mantle cavity common to molluscs. Primitive pulmonate snails are aquatic and close the pneumostome during submergence, while most higher pulmonates are terrestrial and rely on their lungs for gas exchange.

Vertebrate Respiratory Systems

Aquatic vertebrates (fish, amphibians, and some reptiles) rely on one or a combination of the following surfaces for gas exchange: the **cutaneous body surface**, **external filamentous gills**, and **internal lamellar gills**.

Bimodal Breathing is the ability of an organism to exchange respiratory gases simultaneously with both air and water. A **bimodal organism** (e.g., some salamanders, crabs, barnacles, bivalve molluscs, and fishes [lungfishes]) uses gills for water breathing and lungs for air breathing. However, some gas exchange is always cutaneous, and some bimodal breathers are actually trimodal (skin, gills, and lungs).

Bimodal breathing was an important respiratory adaptation that made possible the evolutionary transition between aquatic and terrestrial habitats. Fundamental changes in the structure and function of the respiratory organs accompanied the transition from water to air breathing.

In air-breathing terrestrial vertebrates (reptiles, birds, and mammals), **lungs** replaced gills. These vertebrate surfaces and transitions are now discussed.

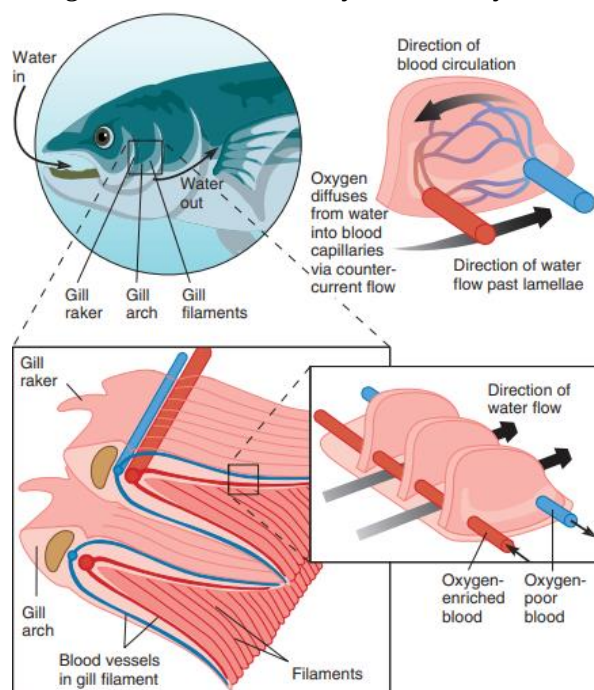
Cutaneous Exchange

Some vertebrates with lungs or gills, such as some aquatic turtles, salamanders with lungs, snakes, fishes, and mammals, use **cutaneous respiration** or **skin-based exchange** to help with gas exchange. However, cutaneous exchange is most developed in frogs, toads, lungless salamanders, and newts.

Amphibian skin has the simplest structure among the major vertebrate respiratory organs. In frogs, a uniform network of tiny blood vessels lies just below the skin. This arrangement helps gas exchange between the blood vessels and the environment through both diffusion and movement of air. A slimy mucus layer keeps amphibian skin moist and protects it from injury, aiding in gas exchange. Some amphibians get about 25% or more of their oxygen through this method, and lungless salamanders perform all of their gas exchange through the skin and the mouth-throat area.

Gills

Gills are respiratory organs with a thin, moist, vascularized layer of skin to allow gas exchange across thin gill membranes or a very thin skin layer over highly vascularized tissue. Larval forms of some fish and amphibians have **external gills** that project from their bodies. Adult fish have **internal gills**, where gas exchange is very efficient.



Gas Exchange in Gills occurs as blood and water move in opposite directions on either side of the gill membranes. For example, water passing over a gill first encounters blood vessels carrying low-oxygen blood into the body. Because the oxygen concentration is lower in the blood than in the water, oxygen diffuses into the blood. As water continues to pass over vessels carrying higher-oxygen blood from within the body, more oxygen diffuses inward since the blood still has less oxygen than the water. **Carbon dioxide** also diffuses into the water because its concentration is higher in the blood than in the water.

This **countercurrent exchange mechanism** provides efficient gas exchange by maintaining a concentration gradient between the blood and

water throughout the capillary bed.

Lungs

A **lung** is an internal, sac-shaped respiratory organ. In terrestrial vertebrates, the lung typically consists of one or more internal blind pouches into which air is either drawn or forced. The respiratory lining of the lungs is thin, well-supplied with blood vessels, and divided into many small units, significantly increasing the surface area for gas exchange between the lung air and the blood. However, this design

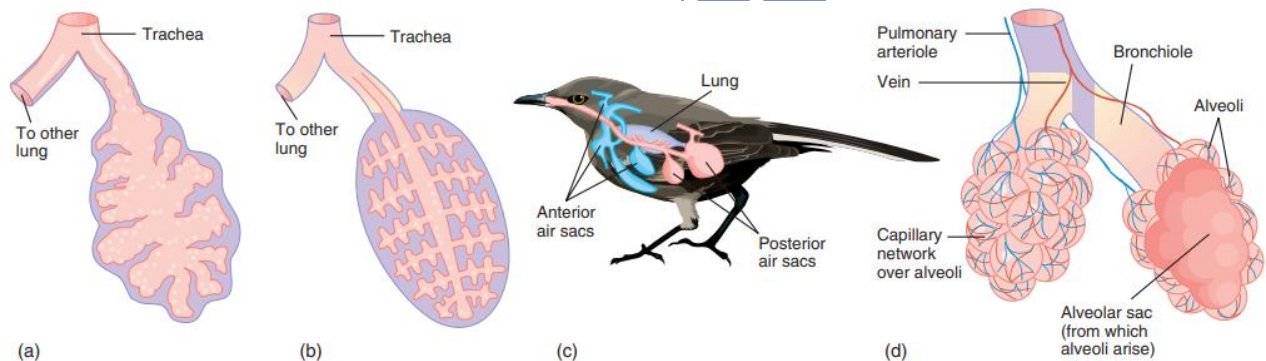
limits the efficiency of oxygen and carbon dioxide exchange with the atmosphere because only part of the lung air is replaced with each breath.

Bird Lungs are an exception, as they have a highly efficient one-way airflow system. For example, a mammal extracts about 25% of the oxygen from the air with each breath, while a bird extracts about 90%.

Evolution of Lungs and Swim Bladders: The evolution of vertebrate lungs is related to the evolution of the swim bladder. The **swim bladder** is an air sac located above the digestive tract in many modern fish. Evidence suggests that both lungs and swim bladders evolved from a lung-like structure present in primitive fish, ancestors of both modern fish and tetrapods (amphibians, reptiles, birds, and mammals). These ancestral fish likely had a ventral sac attached to their throat that served as a supplementary gas exchange organ when gills couldn't provide enough oxygen, such as in stagnant or low-oxygen water. By swimming to the surface and gulping air into this sac, the fish could exchange gases through its wall.

Evolutionary Paths: The further evolution of this sac followed two paths. In most modern bony fish, the swim bladder lies above the digestive tract. In contrast, the lungs are located below the digestive tract. A few present-day fish and all tetrapods have ventral lungs.

Adaptations in Lung Structure: The development of more complex lungs paralleled the evolution of larger body sizes and higher metabolic rates in warm-blooded vertebrates (birds and mammals). These adaptations required an increased lung surface area for gas exchange compared to the smaller body size and lower metabolic rates of cold-blooded vertebrates.



Lung Ventilation

Ventilation is based on several basic principles for all air-breathing animals with lungs:

1. **Air moves** by bulk flow into and out of the lungs.
2. **Oxygen and carbon dioxide** diffuse across the respiratory surface of the lung tissue from pulmonary capillaries.
3. At **systemic capillaries**, oxygen and carbon dioxide diffuse between the blood and interstitial fluid in response to concentration gradients.
4. **Oxygen and carbon dioxide** diffuse between the interstitial fluid and body cells.

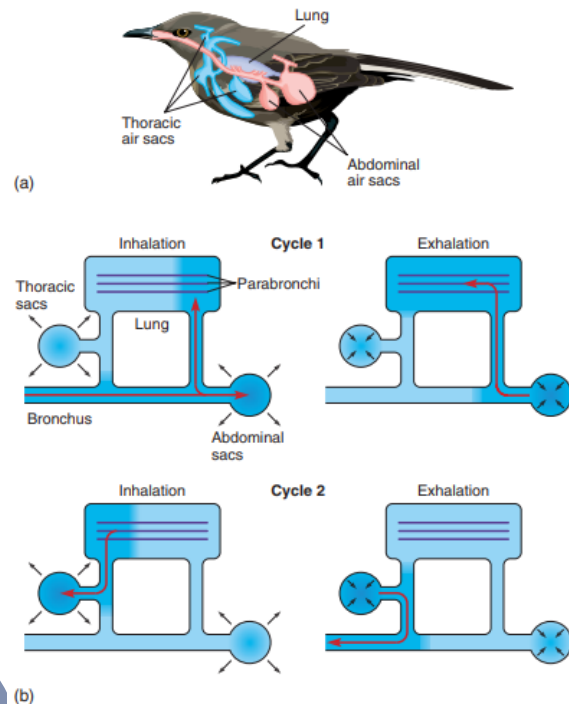
Vertebrates exhibit two different mechanisms for lung ventilation based on these principles:

- **Positive Pressure Pumping:** Amphibians and some reptiles push air into their lungs. The muscles of the mouth and throat create a positive pressure to force air into the lungs.

- **Negative Pressure System:** Most reptiles and all birds and mammals inhale by suction. They expand the body cavity to decrease pressure in the lungs, drawing air in. Elastic recoil of the lungs and movement of the ribs and body wall expel air.

In **reptiles** like snakes, lizards, and crocodiles, the body cavity expands with rib movement to ventilate the lungs. Turtles, with their ribs as part of the shell, exhale by contracting muscles that force the internal organs upward, compressing the lungs. They inhale by contracting muscles that increase the volume of the visceral cavity, creating negative pressure to draw air in.

Birds have the highest rate of oxygen consumption due to their high metabolic rate from flying. They use a negative pressure system for breathing, similar to mammals, but have a unique lung ventilation mechanism for one-way airflow over gas exchange surfaces. This makes bird lungs more efficient than mammal lungs. Birds have tunnel-like passages called **parabronchi** leading to air capillaries for gas exchange. A system of **air sacs** throughout the body cavity makes one-way airflow possible. Inhaled air bypasses the lungs and enters the **posterior air sacs**, passes through the lungs to the **anterior air sacs**, and is then exhaled. This process requires two complete breathing cycles.



Human Respiratory System

The structure and function of external respiration in humans are typical of mammals. Thus, the human respiratory system is used here to describe principles that apply to all air-breathing mammals.

Air-Conducting Portion

Nasal and Oral Cavities

- **Air Entry and Exit:** Air normally enters and leaves through either the nasal or oral cavities.

Pharynx

- **Common Area:** The pharynx is a common area for both the respiratory and digestive tracts.
- **Connections:** The pharynx connects to the larynx (voice box) and the esophagus (leading to the stomach).

Epiglottis

- **Function:** A flap of cartilage that directs air into the trachea during breathing and covers the trachea during swallowing to prevent food or water from entering.

Trachea

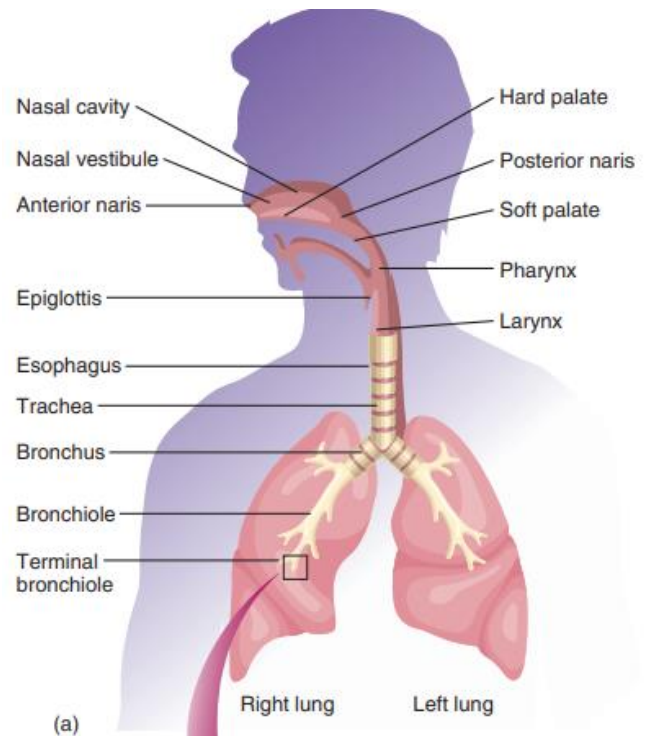
- **Windpipe:** Air moves from the larynx into the trachea.

Bronchi

- **Branches:** The trachea branches into a right and left bronchus (plural: bronchi), each entering a lung.

Bronchioles

- **Smaller Tubes:** Each bronchus branches into smaller tubes called bronchioles.
- **Terminal Bronchioles:** These branch further into even smaller tubes.
- **Respiratory Bronchioles:** Part of the gas-exchange portion of the respiratory system.



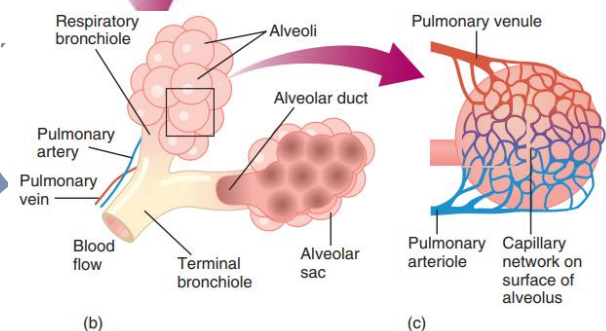
Gas-Exchange Portion

Alveolar Ducts

- **Connection:** Small tubes called alveolar ducts connect the respiratory bronchioles to grapelike outpouchings called **alveoli**.

Alveoli

- **Structure:** Alveoli cluster to form an **alveolar sac** surrounded by many capillaries.
- **Function:** Alveoli are the functional units of the lungs for gas exchange.
- **Gas Exchange:** Passive diffusion moves oxygen from the alveoli into the blood and carbon dioxide from the blood into the alveoli.
- **Surface Area:** The alveoli provide a large surface area for gas exchange. If spread out, the alveolar epithelium would cover the area of a tennis court.



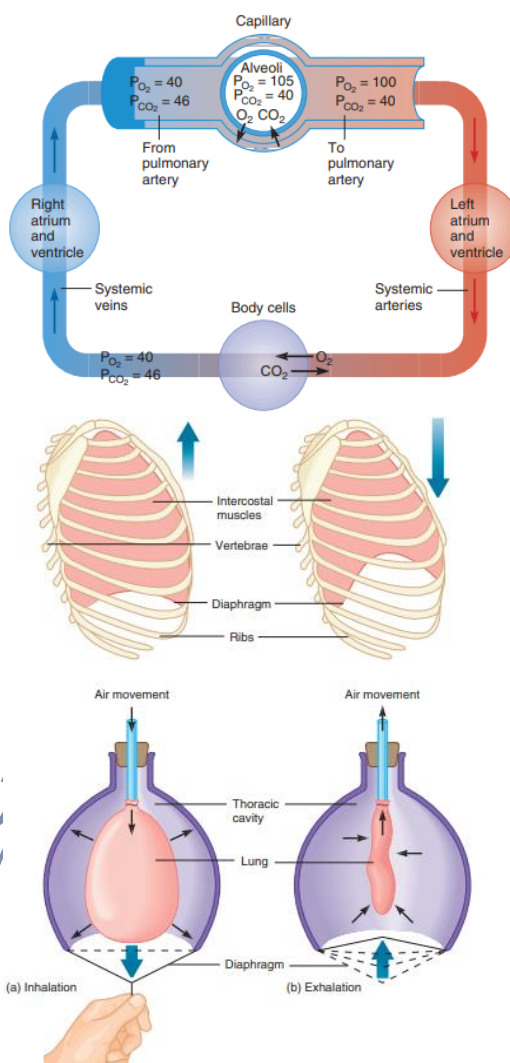
Ventilation

Breathing (Pulmonary Ventilation)

- **Phases:** Breathing has two phases: **inhalation** (intake of air) and **exhalation** (outflow of air).
- **Mechanism:** Air movements result from changes in thoracic cavity volume, which lead to pressure gradient reversals between the lungs and the atmosphere.

Inhalation

1. **Muscle Contraction:** Several muscle sets, mainly the **diaphragm** and **intercostal muscles**, contract. The intercostal muscles pull the ribs closer together, enlarging the thoracic cavity.
2. **Diaphragm Action:** The diaphragm contracts and flattens, further enlarging the thoracic cavity.
3. **Pressure Drop:** The increased size of the thoracic cavity causes pressure to drop below atmospheric pressure, allowing air to rush into the lungs and inflate them.



Exhalation

1. **Muscle Relaxation:** The intercostal muscles and the diaphragm relax, returning the thoracic cavity to its original size and increasing the pressure.
2. **Abdominal Muscles:** These muscles contract, pushing the abdominal organs against the diaphragm, increasing thoracic pressure.
3. **Lung Compression:** This action causes the elastic lungs to contract and compress the air in the alveoli. The increased alveolar pressure expels air from the lungs.

Gas Transport

As noted in the previous discussion, **oxygen** must be transported from the sites of environmental **gas exchange** to the **cells** of an animal's body. Various systems (e.g., **tracheae**, **cutaneous exchange**, **gills**, **lungs**) help accomplish this transport.

Limitations of Simple Diffusion

As animals became larger and acquired higher **metabolic rates**, simple **diffusion** became increasingly inadequate as a means of delivering **oxygen** to the **tissues**.

Specialized Circulatory Systems

Consequently, in most animals with high metabolic rates and tissues more than a few millimeters from **respiratory surfaces**, a specialized **circulatory system** circulates body fluids to aid in the internal distribution of **oxygen** (see figure 26.1).

Increased Oxygen Demand in Active Animals

In general, more active animals have an increased demand for **oxygen**.

Challenges of Oxygen Solubility

However, simply creating a **convection** of a water-based body fluid does not in itself guarantee internal transport of sufficient **oxygen** to meet this increased demand. The reason is the **low solubility** of **oxygen** in water-based body fluids.

Evolution of Respiratory Pigments

Thus, fluid-borne **respiratory pigments** specialized for reversibly binding large quantities of **oxygen** evolved in most phyla.

Functions of Respiratory Pigments

Respiratory pigments help the various transport systems satisfy this increased oxygen demand. In addition to **oxygen transport**, **respiratory pigments** may also function in short-term **oxygen storage**.

Respiratory Pigments

Respiratory pigments are organic compounds containing either metallic **copper** or **iron** that binds oxygen. These pigments may be dissolved in the blood or body fluids, or they may be contained within specific blood cells. Generally, pigments combine with oxygen in high oxygen concentrations and release oxygen in low oxygen concentrations.

Types of Respiratory Pigments

1. Hemoglobin

- **Metal:** Iron
- **Color:** Reddish
- **Distribution:** Common in a variety of invertebrates (e.g., protozoa, platyhelminths, nemertean, nematodes, annelids, crustaceans, some insects, and molluscs) and almost all vertebrates except a few fishes.
- **Location:** Carried within red blood cells (erythrocytes) or dissolved in blood or coelomic fluid.

2. Hemocyanin

- **Metal:** Copper
- **Color:** Bluish when oxygenated
- **Distribution:** Most common in molluscs and certain crustaceans.
- **Location:** Always dissolved in hemolymph.
- **Characteristic:** Releases oxygen easily, providing a ready source of oxygen to tissues as long as environmental oxygen concentrations are relatively high.

3. Hemerythrin

- **Metal:** Iron
- **Color:** Pink when oxygenated
- **Distribution:** Found in sipunculans, priapulids, a few brachiopods, and some polychaetes.
- **Location:** In nucleated cells, rather than free in body fluids or hemolymph.

4. Chlorocruorin

- **Metal:** Iron

- **Color:** Green in low oxygen concentrations, bright red in high oxygen concentrations
- **Distribution:** Found in several families of polychaete worms.

Function of Respiratory Pigments

Respiratory pigments significantly increase the **oxygen-carrying capacity** of body fluids beyond what would be achieved by simple transport in a dissolved state.

Carbon Dioxide Transport

Carbon dioxide (CO₂) concentrations in animal body fluids are higher than expected based on solubility due to transport mechanisms:

- **Bound to hemoglobin**
- **Dissolved state**
- **Carbonic acid (H₂CO₃) and bicarbonate ion (HCO₃⁻)** in reversible reactions:
 - $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3 \rightleftharpoons \text{H}^+ + \text{HCO}_3^-$

This "tying up" of carbon dioxide in other forms lowers its concentration in solution, thus increasing the overall carrying capacity of a body fluid like blood.

Evolutionary Perspective

The occurrence of respiratory pigments among various taxa lacks a clear phylogenetic explanation. Their sporadic distribution suggests that some pigments may have evolved multiple times through **parallel evolution**. Notably, respiratory pigments are rare among successful insects, reflecting their use of extensive **tracheal systems** to carry gases directly to the tissues. Insects without well-developed tracheae carry oxygen in solution in the hemolymph.

Nutrition and Digestion:

Animal Nutrition

Nutrition includes all processes by which an animal **takes in, digests, absorbs, stores, and uses food (nutrients)** to meet its **metabolic needs**.

Digestion

Digestion (L. *digestio*, from *dis*, apart + *gerere*, to carry) is the **chemical and/or mechanical breakdown of food** into particles that individual cells of an animal can absorb.

Overview of Animal Nutrition

This chapter discusses:

- **Animal nutrition**
- **Different strategies** animals use for consuming and using food
- **Various animal digestive systems**

Evolution of Nutrition

Nutrients in the food an animal eats provide the necessary chemicals for **growth, maintenance, and energy production**. Overall, the nutritional requirements of an animal are inversely related to its ability to make essential molecules for life. The fewer such **biosynthetic abilities** an animal has, the more kinds of nutrients it must get from its environment.

Autotrophs and Heterotrophs

Green plants and **photosynthetic protists** have the fewest nutritional requirements because they can make all their own complex molecules from simpler inorganic substances; they are **autotrophs** (Gr. *auto*, self + *trophe*, nourishing).

Animals, fungi, and bacteria that cannot make many of their own organic molecules and must get them by consuming other organisms or their products are **heterotrophs** (Gr. *heteros*, another or different + *trophe*, nourishing).

Types of Diets in Animals

Animals such as rabbits that live entirely on plant material are **herbivores** (L. *herba*, plant + *vorare*, to eat).

Carnivores (L. *caro*, flesh), such as hawks, are animals that eat only meat.

Omnivores (L. *omnius*, all), such as humans, bears, raccoons, and pigs, eat both plant and animal matter.

Insectivores, such as bats, eat mainly arthropods.

Evolutionary Loss of Biosynthetic Abilities

Losses of biosynthetic abilities have marked much of animal evolution. Once an animal regularly gets essential, complex organic molecules in its diet, it can afford to lose the ability to make those molecules.

Moreover, losing this ability gives the animal an advantage because it stops using energy and resources to make molecules that are already in its diet. Thus, as the diet of animals became more varied, they tended to lose their abilities to make such widely available molecules as some of the **amino acids**.

The Metabolic Fates of Nutrients in Heterotrophs

When a **heterotroph** consumes nutrients, they can be categorized into **macronutrients** and **micronutrients**.

Macronutrients are needed in large quantities and include **carbohydrates**, **lipids**, and **proteins**.

Micronutrients are required in small quantities and consist of **organic vitamins** and **inorganic minerals**. Together, these nutrients fulfill the animal's dietary requirements.

Calories and Energy

The **energy value** of food is measured in terms of **calories** or **Calories**.

A **calorie** (L. *calor*, heat) is the amount of energy needed to raise the temperature of 1 g of water by 1°C. A calorie, written with a small 'c', is also known as a **gram calorie**.

A **kilocalorie**, often referred to as a **Calorie** or **kilogram calorie** (kcal), is equal to 1,000 calories.

In everyday language, when we mention calories, we usually mean Calories because this larger unit is more practical for measuring food energy.

For instance, if an advertisement states that a "light beer" contains 95 calories per 12 oz, it actually means 95,000 calories, which translates to 95 Calories or 95 kcal.

Macronutrients

With a few exceptions, **heterotrophs** need **organic molecules** like **carbohydrates**, **lipids**, and **proteins** in their diets. Enzymes break down these molecules into components that can be used for **energy production** or as sources for life's **building blocks**.

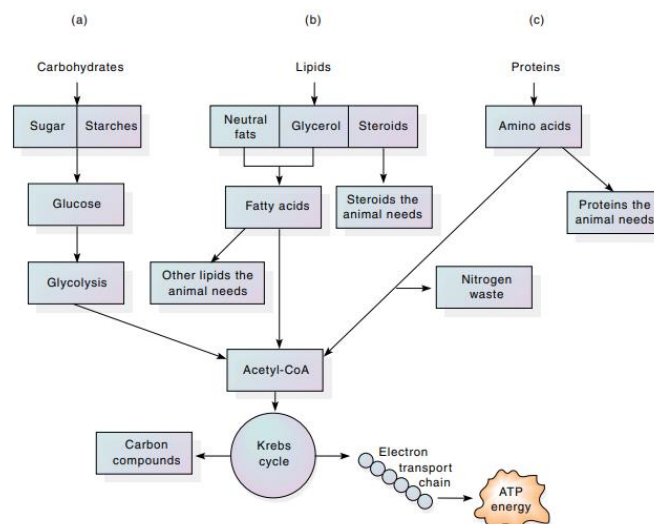
Carbohydrates: Carbon and Energy from Sugars and Starches

The primary source of energy for heterotrophs is **complex carbohydrates**. Most carbohydrates originate from **plants**. They can be in the form of various **polysaccharides**, **disaccharides**, or simple sugars (**monosaccharides**). Carbohydrates also serve as a major carbon source for important organic compounds.

Lipids: Highly Compact Energy Storage Nutrients

Neutral lipids (fats) or **triacylglycerols** are found in fats and oils, meat and dairy products, nuts, and certain fruits and vegetables rich in fats. Lipids are the most concentrated source of food energy, providing about 9 Calories (kcal) of usable energy per gram, more than twice the energy available from an equal mass of carbohydrate or protein.

Many heterotrophs have a definite dietary need for lipids, sometimes for specific types. For instance, many animals require **unsaturated fatty acids**, which serve as precursor molecules for the synthesis of **sterols**, such as cholesterol. Sterols are essential for the synthesis of **steroid hormones** and are incorporated into cell membranes. Other lipids function in insulating the bodies of some vertebrates and help maintain a constant temperature.



Proteins: Basic to the Structure and Function of Cells

Animal sources of protein include other animals and milk. Plant sources include beans, peas, and nuts. Proteins are needed for their amino acids, which heterotrophs use to build their own body proteins.

Micronutrients

Micronutrients typically consist of small ions, **organic vitamins**, **inorganic minerals**, and molecules crucial for enzymatic reactions or as parts of certain proteins (e.g., copper in hemocyanin and iron in hemoglobin). Despite being required in small amounts, animals cannot produce them rapidly, if at all; therefore, they must obtain them from their diet.

Minerals

Some minerals are necessary in relatively large quantities and are termed **essential minerals** or **macrominerals**. For instance, **sodium** and **potassium** are essential for the proper function of every nerve and muscle in an animal's body. Animals lose significant amounts of these minerals, particularly sodium, in their urine each day. Animals that sweat to regulate body temperature also lose sodium in their sweat. Adequate calcium intake is essential for muscular activity and, along with phosphorus, for bone formation. Table 27.2 outlines the functions of the major essential minerals.

TABLE 27.1
THE AVERAGE CALORIC VALUES OF
MACRONUTRIENTS

MACRONUTRIENT	CALORIES PER GRAM
Carbohydrates	4.1
Lipids	9.3
Proteins	4.4

Other minerals are referred to as **trace minerals**, **trace elements**, or **microminerals**. These are required in very small amounts for various enzymatic functions. Table 27.3 provides a list of the functions of some trace minerals.

Vitamins

Normal metabolic activity relies on small amounts of over a dozen organic substances known as **vitamins**. The term "vitamin" (L. *vita*, life) encompasses a variety of chemically unrelated organic compounds found in many foods in small quantities. They are necessary for normal metabolic functioning. Vitamins can be water-soluble or fat-soluble.

Water-Soluble Vitamins

Most **water-soluble vitamins**, such as the **B vitamins** and **vitamin C**, are **coenzymes** required in metabolism. These vitamins participate in various enzymatic reactions essential for energy production and other metabolic processes.

Fat-Soluble Vitamins

Fat-soluble vitamins have diverse functions. These vitamins include **vitamin A**, **vitamin D**, **vitamin E**, and **vitamin K**. They play roles in various bodily functions, including vision, bone health, antioxidant defense, and blood clotting.

Dietary Variations in Vitamin Requirements

The need for **vitamin C** and the fat-soluble vitamins (A, D, E, and K) tends to be limited to vertebrates. However, even within closely related groups, vitamin requirements can differ. For example, among vertebrates, humans and guinea pigs require vitamin C, while rabbits do not. Similarly, some birds require vitamin A, while others do not.

Digestion

In some of the simplest life forms like **protists** and **sponges**, certain cells directly take in whole food particles from the environment through processes like diffusion, active transport, and/or endocytosis. These cells then break down the food with enzymes to obtain nutrients. This approach is known as **intracellular digestion**.

Intracellular digestion eliminates the need for mechanical breakdown of food or for a gut or other cavity for chemical digestion. However, it limits an animal's size and complexity because only very small pieces of food can be utilized. Intracellular digestion provides nutrients in protozoa, sponges, cnidarians, platyhelminths, rotifers, bivalve molluscs, and primitive chordates.

Extracellular Digestion

TABLE 27.2

PHYSIOLOGICAL ROLES OF THE ESSENTIAL MINERALS (MACROMINERALS) ANIMALS REQUIRE IN LARGE AMOUNTS

MINERAL	MAJOR PHYSIOLOGICAL ROLES
Calcium (Ca)	Component of bone and teeth; essential for normal blood clotting; needed for normal muscle, neuron, and cellular function
Chlorine (Cl)	Principal negative ion in extracellular fluid; important in acid-base and fluid balance; needed to produce stomach HCl
Magnesium (Mg)	Component of many coenzymes; needed for normal neuron and muscle function, as well as carbohydrate and protein metabolism
Potassium (K)	Major constituent of bones, blood plasma; needed for energy metabolism
Phosphorus (P)	Major positive ion in cells; influences muscle contraction and neuron excitability; part of DNA, RNA, ATP, energy metabolism
Sodium (Na)	Principal positive ion in extracellular fluid; important in fluid balance; essential for conduction of action potentials, active transport
Sulfur (S)	Protein structure; detoxification reactions and other metabolic activity

TABLE 27.3

SOME PHYSIOLOGICAL ROLES OF TRACE MINERALS (MICROMINERALS) IN ANIMALS

MINERAL	PHYSIOLOGICAL ROLES
Cobalt (Co)	Component of vitamin B ₁₂ ; essential for red blood cell production
Copper (Cu)	Component of many enzymes; essential for melanin and hemoglobin synthesis; part of cytochromes
Fluorine (F)	Component of bone and teeth; prevents tooth decay
Iodine (I)	Component of thyroid hormones
Iron (Fe)	Component of hemoglobin, myoglobin, enzymes, and cytochromes
Manganese (Mn)	Activates many enzymes; an enzyme essential for urea formation and parts of the Krebs cycle
Molybdenum (Mo)	Constituent of some enzymes
Selenium (Se)	Needed in fat metabolism
Zinc (Zn)	Component of at least 70 enzymes; needed for wound healing and fertilization

Larger animals have evolved structures and mechanisms for **extracellular digestion**: the enzymatic breakdown of larger food pieces into constituent molecules, typically in a specialized organ or cavity. Nutrients from the food then pass into body cells lining the organ or cavity and can be used for energy metabolism or biosynthesis.

Animal Strategies for Getting and Using Food

In most animals, nutrients cannot be absorbed directly from the external environment via intracellular digestion. Therefore, animals must actively acquire their nutrients. The diversity of specializations for food procurement (feeding) and extracellular digestion across animal species is vast. Here's a brief overview of the major feeding strategies animals employ:

Continuous vs. Discontinuous Feeders

One aspect related to the structure of digestive systems is whether an animal is a **continuous** or **discontinuous feeder**.

- **Continuous Feeders:** These animals are often slow-moving or sessile (remaining permanently in one place). For instance, aquatic suspension feeders like tube worms and barnacles stay in one place and continuously filter small food particles from the water.

- **Discontinuous Feeders:**

Typically, these animals are active and sometimes highly mobile. They often possess more digestive specializations than continuous feeders because they consume large meals that require grinding up or storage, or both.

TABLE 27.5
FAT-SOLUBLE VITAMINS

VITAMIN	CHARACTERISTICS	FUNCTIONS	SOURCES
Vitamin A	Occurs in several forms; synthesized from carotenes; stored in liver; stable in heat, acids, and alkalis; unstable in light	Necessary for synthesis of visual pigments, mucoproteins, and mucopolysaccharides; for normal development of bones and teeth; and for maintenance of epithelial cells	Liver, fish, whole milk, butter, eggs, leafy green vegetables, and yellow and orange vegetables and fruits
Vitamin D	A group of sterols; resistant to heat, oxidation, acids, and alkalis; stored in liver, skin, brain, spleen, and bones	Promotes absorption of calcium and phosphorus; promotes development of teeth and bones	Produced in skin exposed to ultraviolet light; in milk, egg yolk, fish-liver oils, fortified foods
Vitamin E	A group of compounds; resistant to heat and visible light; unstable in presence of oxygen and ultraviolet light; stored in muscles and adipose tissue	An antioxidant; prevents oxidation of vitamin A and polyunsaturated fatty acids; may help maintain stability of cell membranes	Oils from cereal seeds, salad oils, margarine, shortenings, fruits, nuts, and vegetables
Vitamin K	Occurs in several forms; resistant to heat, but destroyed by acids, alkalis, and light; stored in liver	Needed for synthesis of prothrombin; needed for blood clotting	Leafy green vegetables, egg yolk, pork liver, soy oil, tomatoes, cauliflower

Carnivores

Many carnivores pursue and capture relatively large prey. When successful, they consume large meals, enabling them to avoid continuous hunting. Carnivores have digestive systems that allow for the storage and gradual digestion of large, infrequent meals.

Herbivores

Herbivores spend more time eating than carnivores but are also discontinuous feeders. They must move from one area to another when food is depleted and, in natural environments, must limit their grazing time to avoid excessive exposure to predators. Herbivores have digestive systems that facilitate relatively rapid food gathering and gradual digestion.

Suspension Feeders

Suspension feeding involves the removal of suspended food particles from the surrounding water using a capture, trapping, or filtration structure. This feeding strategy includes three steps:

1. Transport of water past the feeding structure

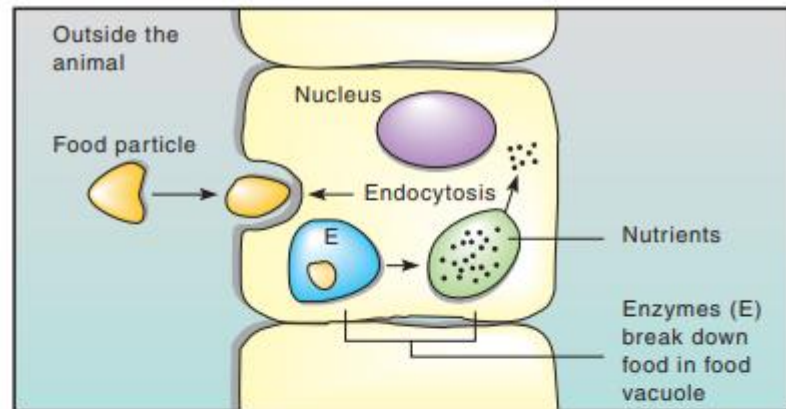
TABLE 27.4
WATER-SOLUBLE VITAMINS

VITAMIN	CHARACTERISTICS	FUNCTIONS	SOURCES
Thiamin (vitamin B ₁)	Destroyed by heat and oxygen, especially in alkaline environment	Part of coenzyme needed for oxidation of carbohydrates, and coenzyme needed in synthesis of ribose	Lean meats, liver, eggs, whole-grain cereals, leafy green vegetables, legumes
Riboflavin (vitamin B ₂)	Stable to heat, acids, and oxidation; destroyed by alkalis and light	Part of enzymes and coenzymes needed for oxidation of glucose and fatty acids and for cellular growth	Meats, dairy products, leafy green vegetables, whole-grain cereals
Niacin (nicotinic acid)	Stable to heat, acids, and alkalis; converted to niacinamide by cells; synthesized from tryptophan	Part of coenzymes needed for oxidation of glucose and synthesis of proteins, fats, and nucleic acids	Liver, lean meats, poultry, peanuts, legumes
Vitamin B ₆	Group of three compounds; stable to heat and acids; destroyed by oxidation, alkalis, and ultraviolet light	Coenzyme needed for synthesis of proteins and various amino acids, for conversion of tryptophan to niacin, for production of antibodies, and for synthesis of nucleic acids	Liver, meats, fish, poultry, bananas, avocados, beans, peanuts, whole-grain cereals, egg yolk
Pantothenic acid	Destroyed by heat, acids, and alkalis	Part of coenzyme needed for oxidation of carbohydrates and fats	Meats, fish, whole-grain cereals, legumes, milk, fruits, vegetables
Cyanocobalamin (vitamin B ₁₂)	Complex, cobalt-containing compound; stable to heat; inactivated by light, strong acids, and strong alkalis; absorption regulated by intrinsic factor from gastric glands; stored in liver	Part of coenzyme needed for synthesis of nucleic acids and for metabolism of carbohydrates; plays role in synthesis of myelin	Liver, meats, poultry, fish, milk, cheese, eggs
Folate (folic acid)	Occurs in several forms; destroyed by oxidation in acid environment or by heat in alkaline environment; stored in liver, where it is converted into folinic acid	Coenzyme needed for metabolism of certain amino acids and for synthesis of DNA; promotes production of normal red blood cells	Liver, leafy green vegetables, whole-grain cereals, legumes
Biotin	Stable to heat, acids, and light; destroyed by oxidation and alkalis	Coenzyme needed for metabolism of amino acids and fatty acids and for synthesis of nucleic acids	Liver, egg yolk, nuts, legumes, mushrooms
Ascorbic acid	Closely related to monosaccharides; stable in acids, but destroyed by oxidation, heat, light, and alkalis	Needed for production of collagen, conversion of folate to folinic acid, and metabolism of certain amino acids; promotes absorption of iron and synthesis of hormones from cholesterol	Citrus fruits, citrus juices, tomatoes, cabbage, potatoes, leafy green vegetables, fresh fruits

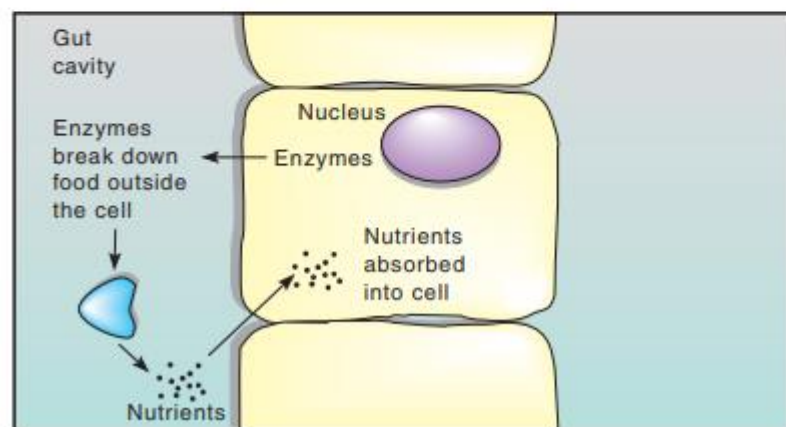
2. Removal of nutrients from the water
3. Transport of the nutrients to the mouth of the digestive system

Suspension feeders include:

- **Sponges**
- **Ascidians**
- **Branchiopods**
- **Ectoprocts**
- **Entoprocts**
- **Phoronids**
- Most **bivalves**
- Many **crustaceans**
- **Polychaetes**
- **Gastropods**
- Some nonvertebrate **chordates**



(a)



(b)

Deposit Feeders

Deposit feeding primarily involves omnivorous animals obtaining nutrients from sediments of soft-bottom habitats or terrestrial soils. There are two main types of deposit feeding:

- **Direct deposit feeding:** Animals swallow large quantities of sediment, digest usable nutrients, and excrete the remains.
- **Indirect deposit feeding:** Animals use tentacle-like structures to consume sediment.

Examples of deposit feeders include:

- **Polychaete annelids**
- Some **snails**
- Some **sea urchins**
- Most **earthworms**
- **Sea cucumbers**
- Most **sipunculans**
- Certain **clams**
- Several types of **polychaetes**

Herbivory

Herbivory involves the consumption of macroscopic plants. This feeding strategy requires the ability to bite and chew large pieces of plant matter (macroherbivory). Invertebrates that evolved macroherbivory include:

- **Molluscs**
- **Polychaete worms**
- **Arthropods**
- **Sea urchins**

Examples of herbivores include insects and crustaceans, which have large, powerful mandibles capable of biting off plant material and grinding and chewing it before ingestion.

Predation

Predation (L. *praedator*, a plunderer, pillager) is one of the most sophisticated feeding strategies as it involves the capture of live prey. Predators can be classified based on how they capture their prey:

- **Motile Stalkers:** These predators actively pursue their prey. Examples include ciliate protozoa, nemerteans, polychaete worms, gastropods, octopuses, squids, crabs, sea stars, and many vertebrates.
- **Lurking Predators:** These predators sit and wait for their prey to come within seizing distance. Examples include certain species of praying mantises, shrimp, crabs, spiders, polychaetes, and many vertebrates.
- **Sessile Opportunists:** These predators are usually not very mobile and can only capture prey when the prey organism comes into contact with them. Examples include certain protozoa, barnacles, and cnidarians.
- **Grazing Carnivores:** These predators move about the substrate picking up small organisms, primarily sessile and slow-moving animals like sponges, ectoprocts, tunicates, snails, worms, and small crustaceans.

Surface Nutrient Absorption

Some highly specialized animals have evolved to absorb nutrients directly from the external medium across their body surfaces, dispensing entirely with mechanisms for prey capture, ingestion, and digestive processes. Examples include:

- **Free-living Protozoa:** Some protozoa, such as *Chilomonas*, absorb all their nutrients across their body surface.
- **Endoparasitic Organisms:** Endoparasitic protozoa, cestode worms, endoparasitic gastropods, and crustaceans lack mouths and digestive systems, absorbing all nutrients across their body surface.
- **Nonparasitic Multicellular Animals:** Certain animals like gutless bivalves and pogonophoran worms also lack a mouth and digestive system, absorbing nutrients across their body surface. Pogonophoran worms, for instance, also supplement their nutrition with organic carbon fixed by symbiotic bacteria within their tissues.

Fluid Feeders

Fluid feeding involves consuming the biological fluids of animals and plants, which are rich sources of nutrients. This feeding strategy is particularly characteristic of some parasites and external parasites.

- **Parasites:** Intestinal nematodes and other parasites feed by biting, rasping off host tissue, or sucking blood.
- **External Parasites (Ectoparasites):** Leeches, ticks, mites, lampreys, and certain crustaceans are examples of external parasites that use various mouthparts to feed on body fluids.

Examples of Fluid Feeders:

- **Sea Lamprey:** This parasite has a funnel-like structure surrounding its mouth, lined with over 200 rasping teeth and a rasplike tongue. It grips its fish host with the funnel, rasps a hole in the fish's body wall with its tongue, and sucks blood and body fluids from the wound.
- **Insects:** Insects have highly developed sucking structures for fluid feeding. Butterflies, moths, aphids, and blood-sucking mosquitoes are examples. Mosquitoes have complex mouthparts with piercing stylets, while butterflies and moths have tubelike mouthparts to suck up plant fluids.
- **Birds:** Pollen- and nectar-feeding birds, like some butterflies, have long bills and tongues specialized for particular types of flowers. Some birds have brushlike or hollow-tipped tongues to collect nectar, while others make holes in flowers to access nectar with their tongues.
- **Mammals:** The only mammals that feed exclusively on blood are vampire bats, such as *Desmodus*, found in tropical South and Central America. They use sharp front teeth to pierce surface blood vessels and lap at the wound. Nectar-feeding bats and mammals like the honey possum also have specialized adaptations for fluid feeding.

Diversity in Digestive Structures: Invertebrates

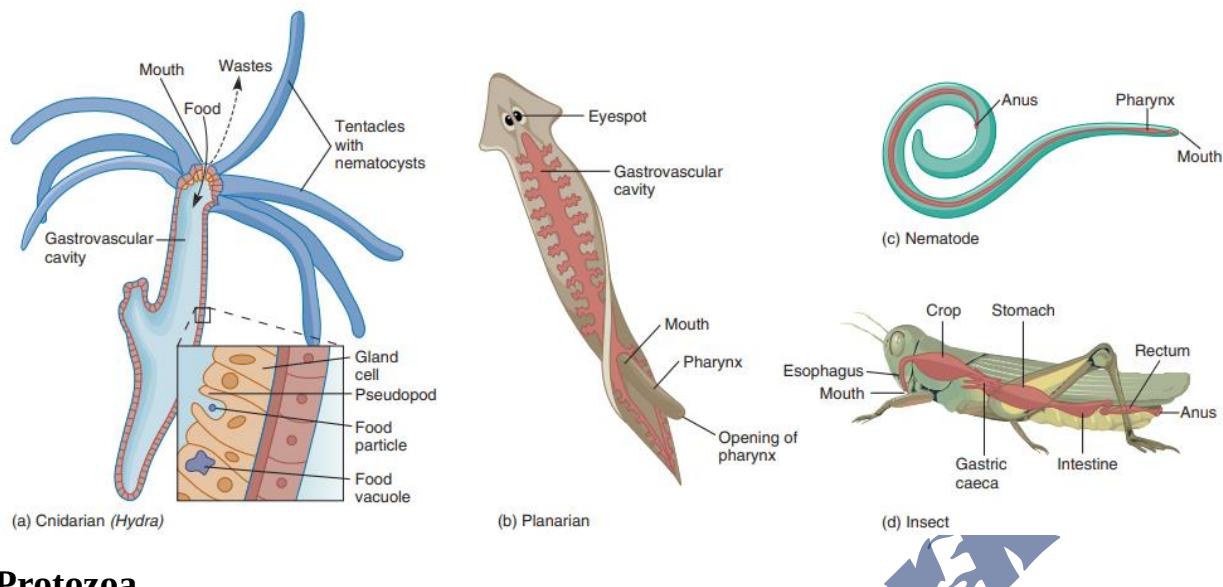
In primitive multicellular animals like cnidarians, the gut is a blind (closed) sac known as a **gastrovascular cavity**, possessing only one opening serving as both entrance and exit. This structure constitutes an incomplete digestive tract. Some specialized cells within the cavity secrete digestive enzymes to initiate extracellular digestion, while other phagocytic cells lining the cavity engulf food material and continue intracellular digestion within food vacuoles. A similar digestive pattern is observed in some flatworms.

An evolutionary breakthrough in digestive structures occurred with the development of the anus and complete digestive tract in the aschelminths. A complete digestive tract facilitates the one-way flow of ingested food without mixing it with previously ingested food or waste. Moreover, complete digestive tracts allow for progressive digestive processing in specialized regions along the system, enabling efficient digestion in a series of distinctly different steps. Variations in the basic plan of a complete digestive tract correlate with different food-gathering mechanisms and diets.

Examples of Digestive Systems in Protozoa and Invertebrates:

1. **Incomplete Digestive System in Ciliated Protozoan:** This represents an example of an intracellular digestive system.
2. **Bivalve Mollusc:** Bivalve molluscs showcase both intracellular and extracellular digestion.
3. **Insect:** Insects exhibit extracellular digestion and possess a complete digestive tract.

These examples highlight the diversity of digestive structures in protozoa and invertebrates, each adapted to specific feeding mechanisms and dietary needs.

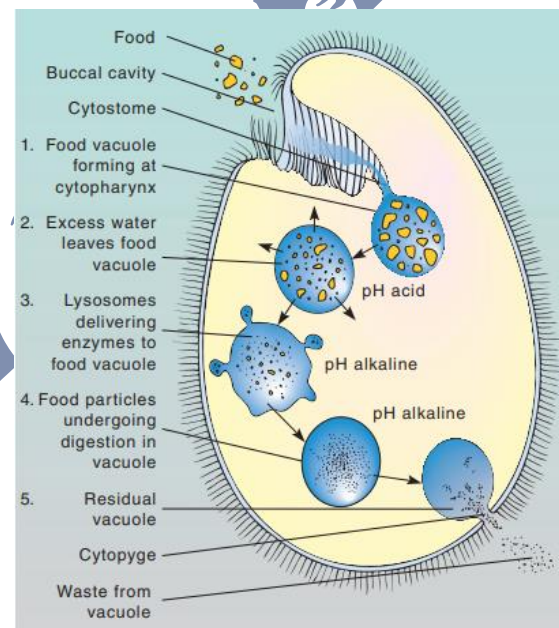


Protozoa

protozoa exhibit various nutritional modes, including autotrophic, saprozoic, or **heterotrophic** (ingestion of food particles). Among them, ciliated protozoa serve as prime examples of organisms employing **heterotrophic nutrition**.

Heterotrophic Nutrition in Ciliated Protozoa:

Ciliary action plays a crucial role in directing food from the environment into the **buccal cavity** and **cytostome**. The cytostome, in turn, leads to the **cytopharynx**, which expands as food enters and then pinches off a food-containing **vacuole**. As the detached food vacuole moves through the cytoplasm, excess water is eliminated, and the contents are subjected to acidification followed by alkalization. Additionally, **digestive enzymes** are added by **lysosomes**. Within the vacuole, food particles undergo digestion, and the resulting nutrients are absorbed into the cytoplasm. Waste products are excreted via the **cytopyge** from the residual vacuole.



This process illustrates the intricacies of heterotrophic nutrition in ciliated protozoa, highlighting the coordinated action of various cellular components for food ingestion, digestion, and waste excretion.

Bivalve Molluscs

Many **bivalve molluscs** engage in **suspension feeding**, ingesting small food particles from their environment. Their digestive tract consists of several parts, including a short esophagus, stomach, midgut, hindgut, and rectum.

Digestive Tract Structure:

- **Esophagus:** Connects the mouth to the stomach.
- **Stomach:** Contains a **crystalline style**, **gastric shield**, and **diverticulated region**.
- **Diverticulae:** Blind-ending sacs that increase surface area for absorption and intracellular digestion.

- **Midgut, Hindgut, and Rectum:** These parts function in **extracellular digestion** and absorption.

Digestive Processes:

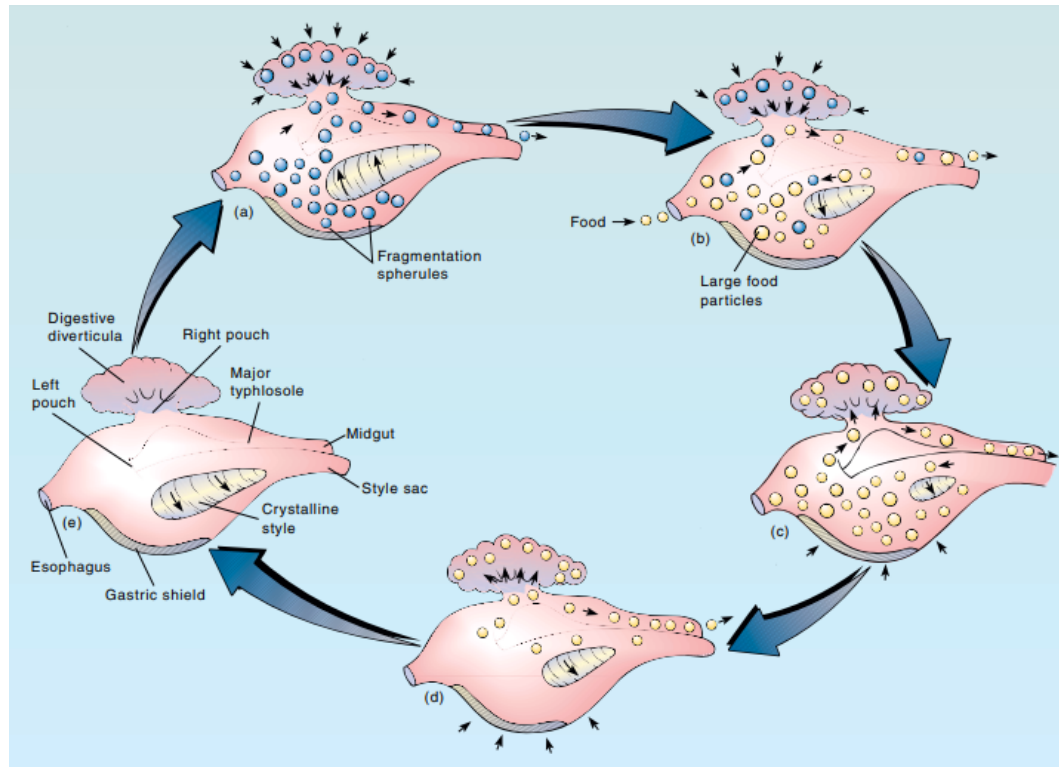
- **Feeding Cycle:** Involves the mechanical and enzymatic breakdown of food.

- **Extracellular Digestion Cycle:** Prepares for digestion and absorption.

- **Intracellular Digestion Cycle:** Releases nutrients into the blood and produces fragmentation spherules for waste excretion.

Integration with Tidal Patterns:

- These cycles are synchronized with the tidal immersion and emersion of the mollusc, ensuring efficient feeding and digestion based on environmental conditions.



This coordination of feeding and digestion processes reflects the adaptation of bivalve molluscs to their suspension-feeding lifestyle, optimizing nutrient acquisition in their aquatic habitat.

Insects: Grasshopper Digestive System

Overview: The **grasshopper** serves as a typical example of an insect with a **complete digestive tract** and **extracellular digestion**.

Feeding Process:

1. **Mastication:** Food is broken down by the **mandibles** and **maxillae**.
2. **Salivary Secretion:** Salivary glands secrete **saliva** containing the enzyme **amylase** to begin carbohydrate digestion.
3. **Transportation to Crop:** Food is passed into the **crop** via the esophagus for temporary storage.
4. **Enzymatic Digestion:** Additional enzymes (carbohydrases, lipases, proteases) from the **midgut** aid in digestion.
5. **Stomach Processing:** Food particles are mechanically reduced and sorted in the stomach.
6. **Extracellular Digestion:** Small particles move to the **gastric caecae** for completion of extracellular digestion.
7. **Nutrient Absorption:** Most nutrient absorption occurs in the **intestine**.

8. **Water and Ion Absorption:** In the **rectum**, water and ions are absorbed from undigested food.
9. **Fecal Pellet Formation:** Solid fecal pellets are formed and expelled through the **anus**.

Regulation and Control:

- Throughout the feeding process, the **nervous system** and **endocrine system** play crucial roles in regulating enzyme production at different stages of digestion.
- **Presence of food** also influences enzyme secretion and digestive activity in various parts of the digestive tract.

The coordinated action of the nervous and endocrine systems, along with the digestive processes, ensures efficient digestion and nutrient absorption in grasshoppers and other insects.

Diversity in Digestive Structures: Vertebrates

The vertebrate digestive tract, also known as the gut tube, is specialized in both structure and function to digest a wide variety of foods. The basic structures of the gut tube include the buccal cavity, pharynx, esophagus, stomach, small intestine, large intestine, rectum, and anus or cloaca. Additionally, three important glandular systems are associated with the digestive tract: the salivary glands, the liver (including the gallbladder and bile duct), and the pancreas (including the pancreatic duct).

Feeding as a Universal Activity

Most vertebrates spend the majority of their time acquiring food, making feeding a universal pastime. The structure of the oral cavity, teeth, intestines, and other major digestive structures usually reflects how an animal gathers food, the type of food it eats, and the way it digests that food. These major digestive structures illustrate the diversity of form and function among different vertebrates.

Tongues

A tongue or tongue-like structure develops in the floor of the oral cavity in many vertebrates:

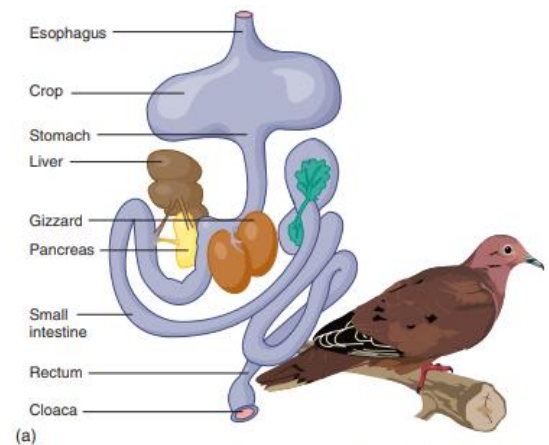
- **Lampreys:** Have a protrusible tongue with horny teeth that rasp the flesh of their prey.
- **Fishes:** May have a primary tongue that bears teeth to help hold prey, though it is not muscular.
- **Tetrapods:** Have evolved mobile tongues for gathering food. For example, frogs, salamanders, and some lizards can rapidly project part of their tongue to capture insects. Woodpeckers have long, spiny tongues for gathering insects and grubs. Ant- and termite-eating mammals use long, sticky tongues. Carnivores like cats have spiny papillae on their tongues to help rasp flesh from bones.

Teeth

Most vertebrates have teeth, except for birds, turtles, and baleen whales:

- **Birds:** Lack teeth, likely to reduce body weight for flight.

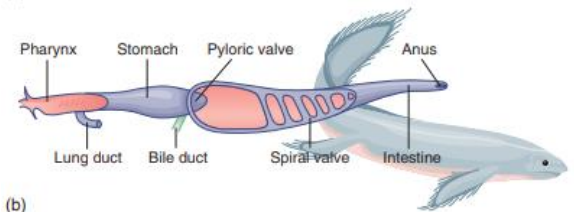
- **Snakes:** Have backward-sloping teeth to aid in retaining prey while swallowing.
- **Wolves:** Have specialized canine teeth for ripping food.
- **Herbivores:** Such as deer, have predominantly grinding teeth. Beavers use their front teeth for chiseling trees and branches. Elephants have two upper front teeth specialized as tusks for weapons and moving objects.
- **Omnivores:** Like humans, pigs, bears, and raccoons have versatile teeth capable of tearing, ripping, chiseling, and grinding.



Salivary Glands

The presence and function of salivary glands vary among vertebrates:

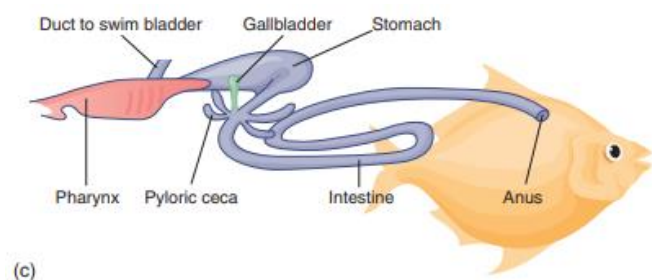
- **Fishes:** Generally lack salivary glands, with the exception of lampreys, which have glands that secrete an anticoagulant to keep prey's blood flowing.
- **Snakes:** Some have modified salivary glands that produce venom to immobilize prey.
- **Amphibians and Reptiles:** Typically lack significant salivary glands since oral digestive enzymes are not crucial.
- **Birds:** Mostly lack salivary glands.
- **Mammals:** Universally possess salivary glands for moistening food and initiating digestion.



Esophagi

The length of the esophagus varies:

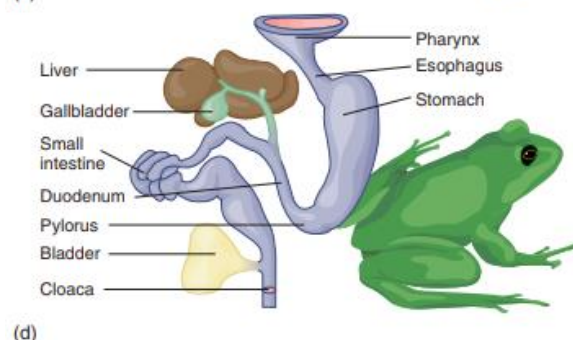
- **Fishes and Amphibians:** Have short esophagi.
- **Amniotes:** Possess longer esophagi due to their longer necks. Grain- and seed-eating birds have a crop that develops from the caudal portion of the esophagus, ensuring a continuous supply of food to the stomach and intestine, thus reducing the frequency of feeding while maintaining a high metabolic rate.



Stomachs

The stomach is an ancestral vertebrate structure:

- **Function:** Evolved to store food and possibly to kill bacteria with hydrochloric acid (HCl). The enzyme pepsinogen, essential for digestion, may have evolved later.



Gizzards

Some vertebrates possess a gizzard:

- **Fishes, Reptiles (like crocodilians), and Birds:** Have a gizzard for grinding food. In birds, the gizzard develops from the posterior part of the stomach (ventriculus). Swallowed pebbles (grit) retained in the gizzard facilitate the grinding process, especially in grain-eating birds.

These adaptations illustrate how vertebrates have evolved diverse digestive structures to efficiently process the wide variety of foods available in their environments.

Ruminant Mammals

Ruminant mammals—animals that “**chew their cud**”, such as cows, sheep, and deer—exhibit some of the most unusual modifications of the stomach. This digestion method allows animals to **eat large amounts of food quickly** and chew it later at a more comfortable or safer location. The ruminant stomach also supports large populations of **microorganisms** that digest the cellulose in grass and other vegetation.

Cellulose Digestion

Cellulose contains a significant amount of energy, but animals generally cannot produce **cellulase**, the enzyme needed to digest cellulose. **Gut microorganisms** can produce cellulase, making the herbivorous lifestyle more effective.

Stomach Structure

In ruminants, the stomach has four compartments:

1. **Rumen:** The upper portion that expands to form a large pouch where food encounters microorganisms.
2. **Reticulum:** A smaller compartment where partially digested food is regurgitated as “**cud**” for thorough chewing.
3. **Omasum:** A small antechamber where the more liquid digested food moves after being swallowed again.
4. **Abomasum:** The “**true**” stomach, where digestive enzymes are first encountered, and digestion continues.

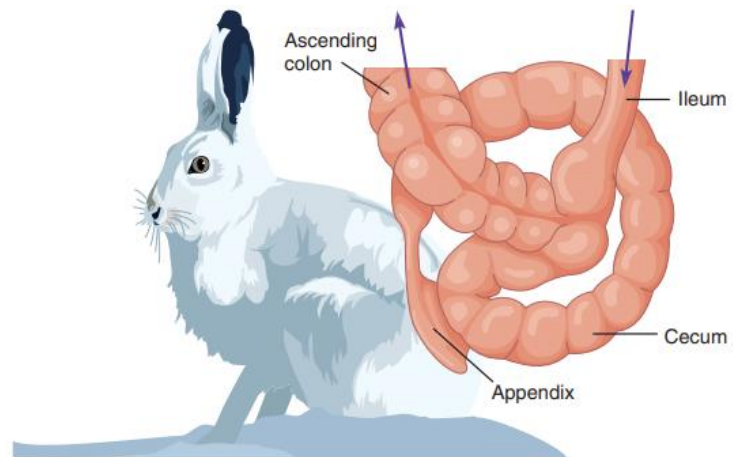
Digestive Process

1. **Food enters the rumen:** Encounters microorganisms that begin partial digestion, aided by fluid secretions, body heat, and churning.
2. **Moves to the reticulum:** Mouthfuls are regurgitated as cud and thoroughly chewed.
3. **Reswallowed food reenters the rumen:** Becomes more liquid in consistency.
4. **Flows to the omasum:** Digested material moves to the omasum as it becomes very liquid.
5. **Enters the abomasum:** Encounters digestive enzymes, continuing digestion.

CECA

In **nonruminant herbivores**, microbial action on cellulose occurs **after digestion** in the **cecum**:

- **Rabbits, horses, and rats:** Maintain a population of microorganisms in their **large cecum**, the blind pouch extending from the colon.
- **Coprophagy:** Some, like mice and rabbits, eat their own feces to reprocess remaining materials such as vitamins.



LIVERS AND GALLBLADDERS

In vertebrates with a **gallbladder**, it is closely associated with the **liver**:

- **Liver:** Manufactures **bile**, containing **bile salts** and **bile pigments**.
- **Bile salts:** Emulsify dietary fat, aiding digestion by **lipase**.
- **Bile pigments:** Result from phagocytosis of red blood cells, producing **bilirubin** ("red bile") and **biliverdin** ("green bile").
- **Gallbladder size:** Large in **carnivores** and vertebrates with fat-rich diets; reduced or absent in **bloodsuckers** (e.g., lamprey) and primarily **plant-eating animals** (e.g., some teleosts, many birds, and rats).

PANCREATA

All vertebrates have a **pancreas**:

- In **lampreys and lungfishes:** Embedded in the wall of the intestine, not visible.
- **Pancreatic fluid:** Contains many enzymes, empties into the small intestine via the **pancreatic duct**.

INTESTINES

Configuration and divisions of the small and large intestines vary greatly among vertebrates, influenced by:

- **Type of food**
- **Body size**
- **Levels of activity**

Examples of Intestinal Variations

- **Cyclostomes, chondrichthian fishes, primitive bony fishes:** Have short, nearly straight intestines extending from the stomach to the anus.
- **More advanced bony fishes:** Intestines increase in length and begin to coil.
- **Amphibians and reptiles:** Possess moderately long intestines.
- **Birds and mammals:** Have longer intestines with more surface area.

Specific Features

- **Birds:** Typically have **two ceca**.
- **Mammals:** Have a **single cecum** at the beginning of the large intestine.
- **Large intestine:** Much longer in mammals than in birds, emptying into the **cloaca** in most vertebrates.

The Mammalian Digestive System

Omnivores such as humans, pigs, bears, and raccoons possess a digestive system capable of processing a variety of foods. The following sections discuss gastrointestinal motility control, major parts of the alimentary canal, and accessory organs of digestion.

Digestive Process Steps

1. **Ingestion:** Eating
2. **Peristalsis:** Involuntary, sequential muscular contractions moving nutrients along the digestive tract
3. **Segmentation:** Mixing the contents in the digestive tract
4. **Secretion:** Release of hormones, enzymes, and specific ions and chemicals involved in digestion
5. **Digestion:** Conversion of large nutrient particles or molecules into smaller particles or molecules

6. **Absorption:** Passage of usable nutrient molecules from the small intestine into the bloodstream and lymphatic system to body cells
7. **Defecation:** Elimination of undigested and unabsorbed material as waste

Key Components

- **Control of gastrointestinal motility:** Regulation of movements within the digestive system
- **Major parts of the alimentary canal:** Includes the mouth, pharynx, esophagus, stomach, small intestine, and large intestine
- **Accessory organs:** Salivary glands, liver, gallbladder, and pancreas

This comprehensive process ensures the efficient breakdown, absorption, and excretion of nutrients and waste in mammals.

Gastrointestinal Motility and Its Control

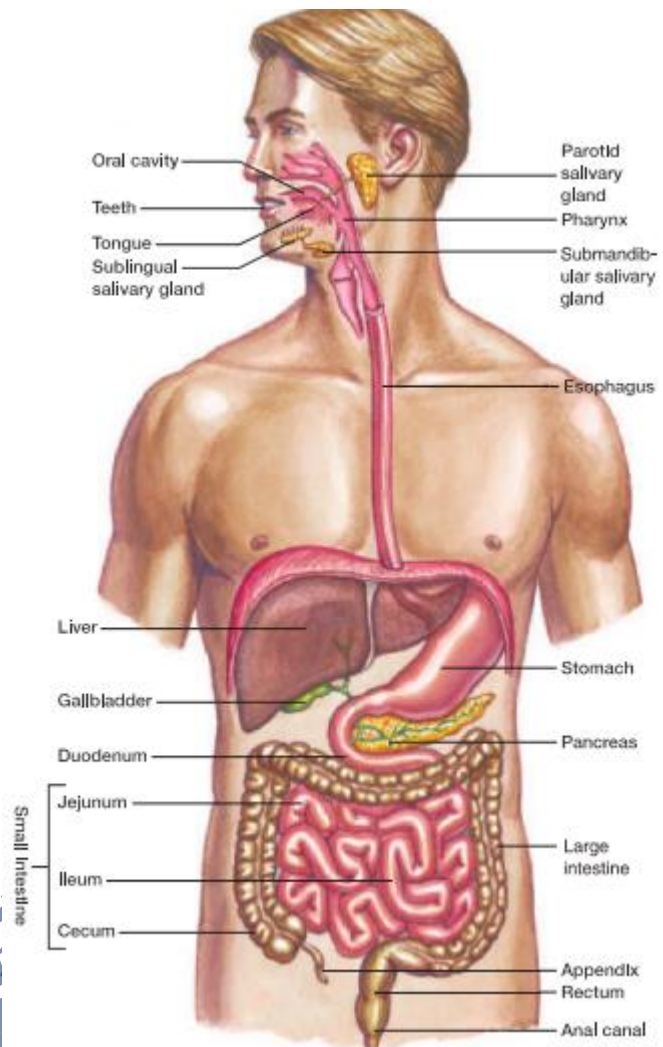
The function of the **gastrointestinal (GI) tract** is determined by its tissue types, which maintain a consistent anatomical structure throughout its length.

Anatomical Structure

1. **Serosa:** Outer layer of connective tissue, forming the moist epithelial sheet called the **peritoneum**.
 - The peritoneum lines the **abdominal cavity** and covers internal organs, encompassing the **coelom**.
2. **Muscle Layers:**
 - **Longitudinal smooth muscle layer**
 - **Circular smooth muscle layer**
3. **Submucosa:** Connective tissue containing blood and lymphatic vessels.
4. **Mucosa:** Faces the central opening called the **lumen**.

Types of Movements

1. **Peristalsis:**
 - **Function:** Advances food through the GI tract.
 - **Mechanism:** Rings of circular smooth muscle contract behind and relax in front of the food, similar to squeezing icing from a pastry tube.

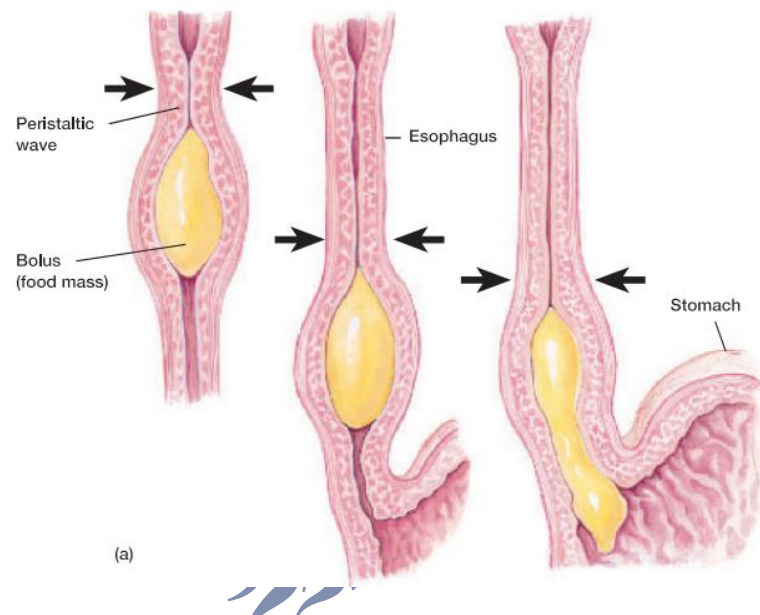


2. Segmentation:

- **Function:** Mixes food with digestive secretions, enhancing absorption efficiency.
- **Mechanism:** Rings of smooth muscles contract and relax, creating an oscillating back-and-forth movement in the same place.

Sphincters

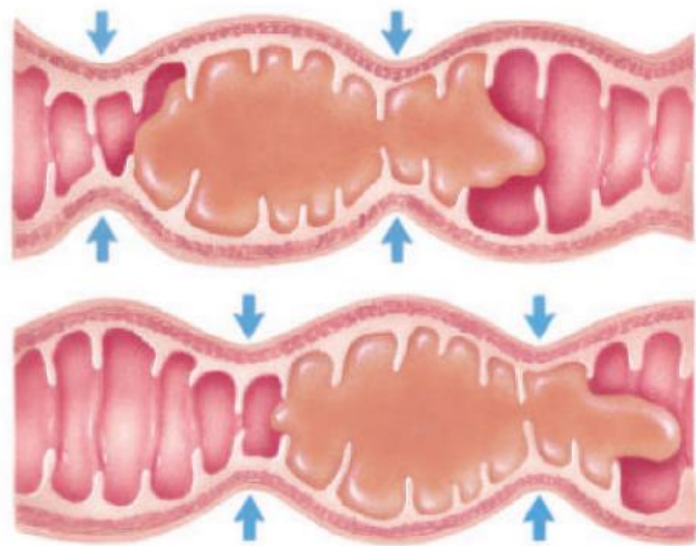
- **Role:** Control the flow of material and prevent backflow.
- **Examples:**
 - **Cardiac sphincter:** Between esophagus and stomach.
 - **Pyloric sphincter:** Between stomach and small intestine.



Control Mechanisms

Local Control:

- **Mechanical receptors:** Stimulated by gut distension due to ingested food.
- **Chemical receptors:** Stimulated by digestion of carbohydrates, lipids, and proteins.
- **Nerve Plexuses:** Signals travel through these in the gut wall to control muscular contractions (peristalsis and segmentation) and secretion of substances (e.g., mucus, enzymes).



Central Nervous System (CNS) Control:

- **Long-distance nerve pathways:** Connect receptors and effectors with the CNS, functioning to maintain homeostasis.

Endocrine Control:

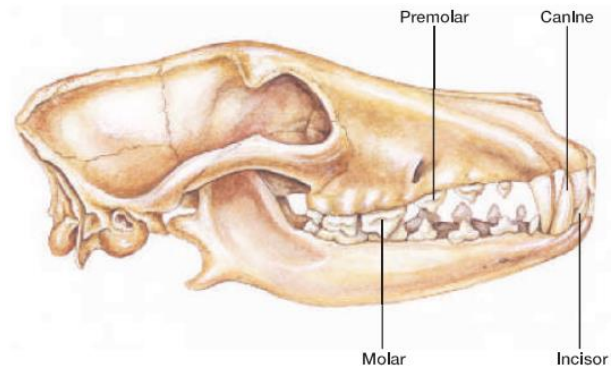
- **Hormones:** Produced by endocrine cells in the GI tract to regulate secretion, digestion, and absorption.

Oral Cavity

The **oral cavity** (mouth) is protected by a pair of lips, highly vascularized skeletal muscle tissue rich in sensory nerve endings. Lips aid in food retention during chewing and contribute to sound modification during phonation.

Components

1. **Lips:** Highly vascularized, skeletal muscle tissue with sensory nerve endings.
2. **Tongue and Teeth:** Mammals can mechanically process a wide range of foods due to enamel-covered teeth, the hardest material in the body, and strong jaw and tooth forces.
3. **Saliva:** Continuously bathes the oral cavity, secreted by at least three pairs of salivary glands.
 - **Functions:** Moistens food, binds it with mucins (glycoproteins), forming a moist mass (bolus). Contains bicarbonate ions (HCO_3) for buffering mouth chemicals, thiocyanate ions (SCN) and lysozyme for microbial control, and amylase for carbohydrate digestion initiation.



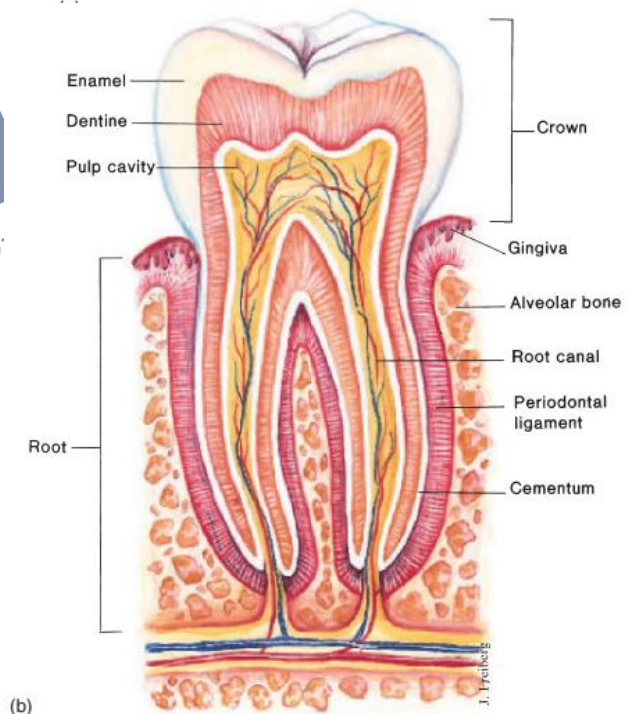
Pharynx and Esophagus

Swallowing Process

- **Initiation:** Voluntary or involuntary swallowing reflex.
- **Mechanism:** Sequential, involuntary contractions of smooth muscles in the esophageal walls propel the bolus or liquid to the stomach.
- **Epiglottis:** Temporarily seals off the tracheal opening (glottis) to prevent food entry into the trachea during swallowing.

Digestive Contribution

- **Neither the pharynx nor the esophagus** contribute to digestion. They serve as the common passageway for both the digestive and respiratory tracts.



Stomach

The mammalian **stomach** is a muscular, distensible sac with three main functions:

1. **Storage and Mixing:** Receives food bolus from the esophagus and mixes it.
2. **Secretion:** Releases substances (enzymes, mucus, hydrochloric acid [HCl]) initiating protein digestion.

3. **Control of Food Movement:** Regulates the rate of food passage into the small intestine via the pyloric sphincter.

Structure and Function

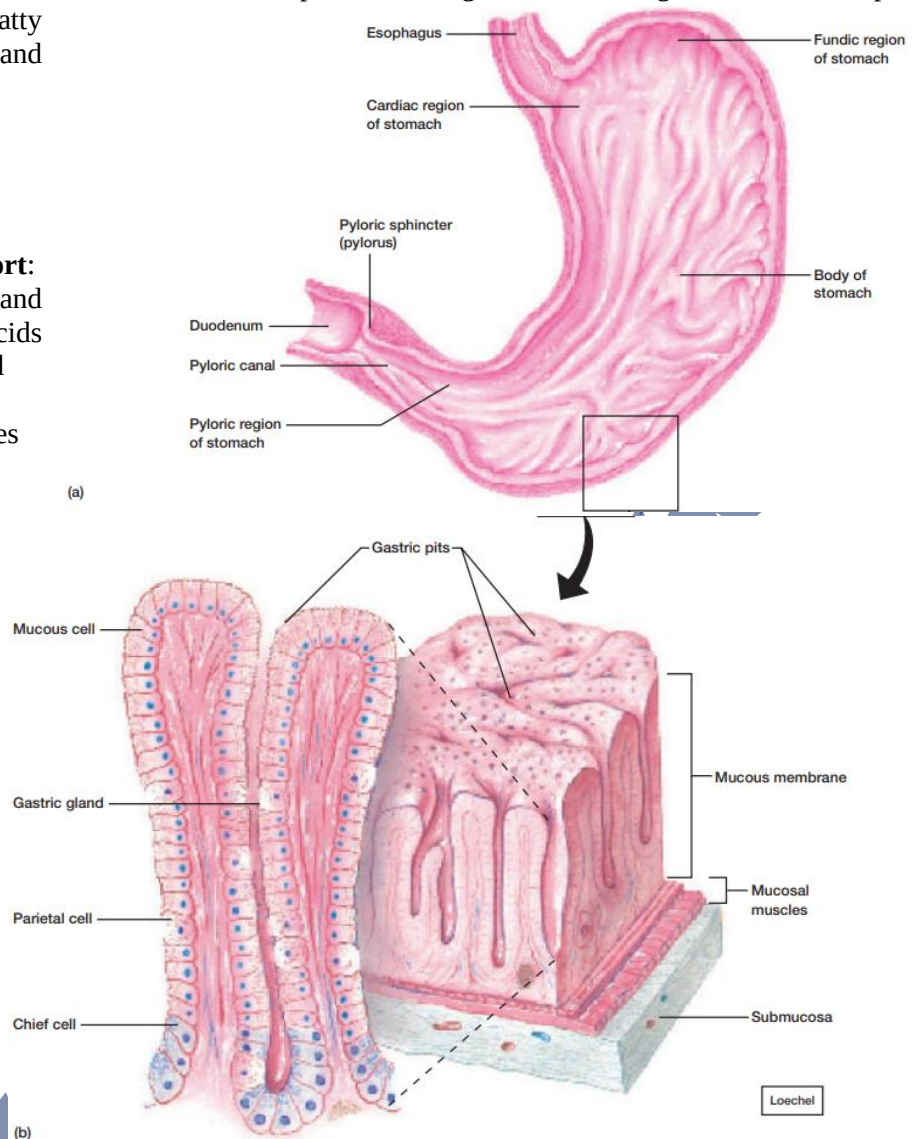
- **Mucous Membrane:** Contains thousands of gastric glands.
 - **Cell Types:**
 - **Parietal Cells:** Secrete a solution containing HCl.
 - **Chief Cells:** Secrete pepsinogen, the precursor of pepsin.
 - **Mucous Cells:** Secrete mucus to protect the stomach from HCl and digestive enzymes.
- **Hormonal Regulation:** Endocrine cells release gastrin hormone, stimulating gastric gland activity.
- **Digestive Process:**
 - **Bolus Entry:** Food bolus distends stomach walls, stimulating gastric gland secretion of HCl and pepsinogen.
 - **Activation:** HCl converts pepsinogen into active enzyme pepsin.
 - **Mixing and Churning:** Smooth mucosal muscles contract, churning and mixing the food bolus, initiating protein breakdown.
- **Chyme Formation:** After about three to four hours, stomach contents become semiliquid mass called chyme (Gr. chymos, juice).
- **Pyloric Sphincter:** Regulates release of chyme into the small intestine.
- **Peristaltic Waves:** Cease when stomach is empty, but may occur after about 10 hours of fasting, causing "hunger pangs" due to sensory nerve impulses to the brain.

Small Intestine: Main Site of Digestion

Most of the ingested food in mammals is digested and absorbed in the **small intestine**. Here's a breakdown of its structure and function:

- **Dimensions:** Approximately 4 cm in diameter and 7 to 8 m in length in humans, providing a substantial surface area for nutrient absorption.
- **Surface Area:** Enhanced by circular folds and villi, with each square centimeter of mucosa bearing thousands of fingerlike villi.
- **Epithelial Cells:** Covered with simple columnar epithelial cells, each with numerous microvilli, further increasing surface area.
- **Functional Segments:**
 - **Duodenum:** Primarily functions in digestion, containing many digestive enzymes secreted by intestinal glands and the pancreas.

- **Jejunum and Ileum:** Absorb end products of digestion, including amino acids, simple sugars, fatty acids, and glycerol.
- **Absorption Mechanisms:**
 - **Active Transport:** Sugars and amino acids absorbed into capillaries of villi.



- **Chylomicron Formation:** Free fatty acids enter epithelial cells of villi, recombine with glycerol to form triglycerides, coated with proteins to form chylomicrons, which enter lacteals of villi and eventually the bloodstream.
- **Water and Mineral Absorption:** Small intestine absorbs about 9 liters of water per day, with the large intestine absorbing the rest.

Large Intestine

- **Structure:** Lacks circular folds, villi, or microvilli, resulting in a smaller surface area compared to the small intestine.
- **Cecum:** Blind-ended sac where the small intestine joins the large intestine; serves as a storage site and possibly an evolutionary remnant.
- **Appendix:** Contains abundant lymphoid tissue, possibly functioning as part of the immune system.
- **Functions:**
 - **Water and Mineral Reabsorption:** Essential for maintaining hydration and electrolyte balance.

- **Feces Formation:** Stores and compacts waste material.
- **Peristalsis:** Moves food residue along, facilitating water and mineral absorption.
- **Microbiota:** Contains symbiotic bacteria and fungi that further break down organic molecules and secrete amino acids and vitamin K.
- **Feces:** Mixture of bacteria, fungi, undigested plant fiber, sloughed-off intestinal cells, and other waste products.

Role of the Pancreas in Digestion

The pancreas is a vital organ located just ventral to the stomach, serving both endocrine and exocrine functions. Here's an overview of its role in digestion:

- **Exocrine Function:**
 - **Secretion of Digestive Enzymes:** Exocrine cells in the pancreas release digestive enzymes into the pancreatic duct.
 - **Composition of Enzymes:**
 - *Protein Digestion:* Trypsin, carboxypeptidase, and chymotrypsin break down proteins into small peptides and individual amino acids.
 - *Lipid Digestion:* Pancreatic lipases split triglycerides into absorbable glycerol and free fatty acids.
 - *Carbohydrate Digestion:* Pancreatic amylase converts polysaccharides into disaccharides and monosaccharides.
- **Bicarbonate Secretion:**
 - The pancreas also secretes bicarbonate (HCO_3^-) ions into the pancreatic duct.
 - **Neutralization of Acidity:** Bicarbonate helps neutralize the acidic food residue from the stomach, raising the pH from 2 to 7 for optimal digestion.
 - **Facilitates Enzyme Function:** Without neutralization, pancreatic enzymes would not function effectively.

TABLE 27.6
MAJOR DIGESTIVE GLANDS, SECRETIONS, AND ENZYMES IN MAMMALS

PLACE OF DIGESTION	SOURCE	SECRETION	ENZYME	DIGESTIVE FUNCTION
Mouth	Salivary glands	Saliva	Salivary amylase	Begins the digestion of carbohydrates; inactivated by stomach HCl
	Mucous glands	Mucus	—	Lubricates food bolus
Esophagus	Mucous glands	Mucus	—	Lubricates food bolus
Stomach	Gastric glands	Gastric juice	Lipase	Digests lipids into fatty acids and glycerol
			Pepsin	Digests proteins into polypeptides
	Gastric mucosa	HCl	—	Converts pepsinogen into active pepsin; kills microorganisms
	Mucous glands	Mucus	—	Lubricates
Small intestine	Liver	Bile	—	Emulsifies lipids; activates lipase
	Pancreas	Pancreatic juice	Amylase	Digests starch into maltose
			Chymotrypsin	Digests proteins into peptides and amino acids
			Lipase	Digests lipids into fatty acids and glycerol (requires bile salts)
			Nuclease	Digests nucleic acids into mononucleotides
			Trypsin	Digests proteins into peptides and amino acids
	Intestinal glands	Intestinal juice	Enterokinase	Digests inactive trypsinogen into active trypsin
			Lactase	Digests lactose into glucose and galactose
			Maltase	Digests maltose into glucose
			Peptidase	Digests polypeptides into amino acids
			Sucrase	Digests sucrose into glucose and fructose
	Mucous glands	Mucus	—	Lubricates
Large intestine	Mucous glands	Mucus	—	Lubricates

Role of the Liver and Gallbladder in Digestion

Liver:

- **Location:** The liver, the largest organ in the mammalian body, is situated just under the diaphragm.
- **Function of Hepatocytes:**
 - **Nutrient Processing:** Millions of specialized cells called hepatocytes in the liver absorb nutrients from the intestines and release them into the bloodstream.
 - **Blood Protein Synthesis:** Hepatocytes manufacture blood proteins like prothrombin and albumin.
- **Major Metabolic Functions:**
 - Removal of amino acids from organic compounds.
 - Formation of urea from proteins and conversion of excess amino acids into urea to reduce ammonia levels in the body.
 - Synthesis of plasma proteins, formation of fetal erythrocytes, and destruction of worn-out erythrocytes.
 - Synthesis of nonessential amino acids.
 - Conversion of galactose and fructose to glucose.
 - Oxidation of fatty acids.

- Formation of lipoproteins, cholesterol, and phospholipids.
- Conversion of carbohydrates and proteins into fat.
- Modification and detoxification of waste products, drugs, and toxins.
- Synthesis of vitamin A from carotene and participation in vitamin D activation.
- Maintenance of stable body temperature by raising the blood temperature.
- Manufacture of bile salts used for fat emulsification in the small intestine.

Gallbladder:

- **Location:** The gallbladder is a small organ near the liver.
- **Function:**
 - **Bile Storage:** The gallbladder stores bile produced continuously by liver cells.
 - **Composition of Bile:** Bile is alkaline and contains pigments, cholesterol, lecithin, mucin, bilirubin, and bile salts.
 - **Emulsification of Fats:** Bile salts act as detergents to emulsify fats, aiding in fat digestion and absorption.
 - **Formation of Micelles:** Bile salts combine with end products of fat digestion to form micelles, facilitating fat absorption by crossing the intestinal epithelium.

The liver and gallbladder work together to produce and store bile, which is essential for the digestion and absorption of fats in the small intestine. Additionally, the liver performs numerous metabolic functions crucial for overall physiological homeostasis.

Cell Biology:

Introduction to Cell Biology

What is Cell Biology?

Cell biology, also known as cytology, is a branch of biology that focuses on the structure, function, and behavior of cells. Cells are the fundamental units of life, the building blocks from which all living organisms are constructed. Understanding cells, their components, and how they function is essential for all biological sciences.

Historical Background

The study of cells began with the invention of the microscope in the 17th century. Robert Hooke, in 1665, was the first to coin the term "cell" when he observed the cell walls in cork tissue. Later, Antonie van Leeuwenhoek's improvements in microscopy allowed for the observation of living cells, including bacteria and protozoa.

The Cell Theory

The development of cell theory in the 19th century was a pivotal moment in biology. The cell theory states:

1. **All living organisms are composed of one or more cells.**
2. **The cell is the basic unit of life.**
3. **All cells arise from pre-existing cells.**

Types of Cells

Cells come in various forms and functions, but they are primarily categorized into two types:

1. **Prokaryotic Cells:** These cells lack a true nucleus and membrane-bound organelles. Bacteria and archaea are examples of prokaryotic organisms.
2. **Eukaryotic Cells:** These cells have a true nucleus enclosed by a nuclear membrane and contain various membrane

Eukaryotic Cell Definition

"Eukaryotic cells are the cells that contain a membrane bound nucleus and organelles."

What is a Eukaryotic Cell?

Eukaryotic cells have a nucleus enclosed within the nuclear membrane and form large and complex organisms. Protozoa, fungi, plants, and animals all have eukaryotic cells. They are classified under the kingdom Eukaryota.

They can maintain different environments in a single cell that allows them to carry out various metabolic reactions. This helps them grow many times larger than the prokaryotic cells.

Characteristics of Eukaryotic Cells

The features of eukaryotic cells are as follows:

1. Eukaryotic cells have the nucleus enclosed within the nuclear membrane.
2. The cell has mitochondria.
3. Flagella and cilia are the locomotory organs in a eukaryotic cell.
4. A cell wall is the outermost layer of the eukaryotic cells.
5. The cells divide by a process called mitosis.
6. The eukaryotic cells contain a cytoskeletal structure.
7. The nucleus contains a single, linear DNA, which carries all the genetic information.

Structure Of Eukaryotic Cell

The eukaryotic cell structure comprises the following:

Plasma Membrane

- The plasma membrane separates the cell from the outside environment.
- It comprises specific embedded proteins, which help in the exchange of substances in and out of the cell.

Cell Wall

- A cell wall is a rigid structure present outside the plant cell. It is, however, absent in animal cells.
- It provides shape to the cell and helps in cell-to-cell interaction.
- It is a protective layer that protects the cell from any injury or pathogen attacks.
- It is composed of cellulose, hemicellulose, pectins, proteins, etc.

Cytoskeleton

The cytoskeleton is present inside the cytoplasm, which consists of microfilaments, microtubules, and fibres to provide perfect shape to the cell, anchor the organelles, and stimulate the cell movement.

Endoplasmic Reticulum

It is a network of small, tubular structures that divides the cell surface into two parts: luminal and extraluminal.

- Rough Endoplasmic Reticulum contains ribosomes.
- Smooth Endoplasmic Reticulum that lacks ribosomes and is therefore smooth.

Nucleus

- The nucleoplasm enclosed within the nucleus contains DNA and proteins.
- The nuclear envelop consists of two layers- the outer membrane and the inner membrane. Both the membranes are permeable to ions, molecules, and RNA material.
- Ribosome production also takes place inside the nucleus.

Golgi Apparatus

- It is made up of flat disc-shaped structures called cisternae.
- It is absent in red blood cells of humans and sieve cells of plants.
- They are arranged parallel and concentrically near the nucleus.
- It is an important site for the formation of glycoproteins and glycolipids.

Ribosomes

These are the main site for protein synthesis and are composed of proteins and ribonucleic acids.

Mitochondria

- These are also known as “powerhouse of cells” because they produce energy.
- It consists of an outer membrane and an inner membrane. The inner membrane is divided into folds called cristae.
- They help in the regulation of cell metabolism.

Lysosomes

They are known as “suicidal bags” because they possess hydrolytic enzymes to digest protein, lipids, carbohydrates, and nucleic acids.

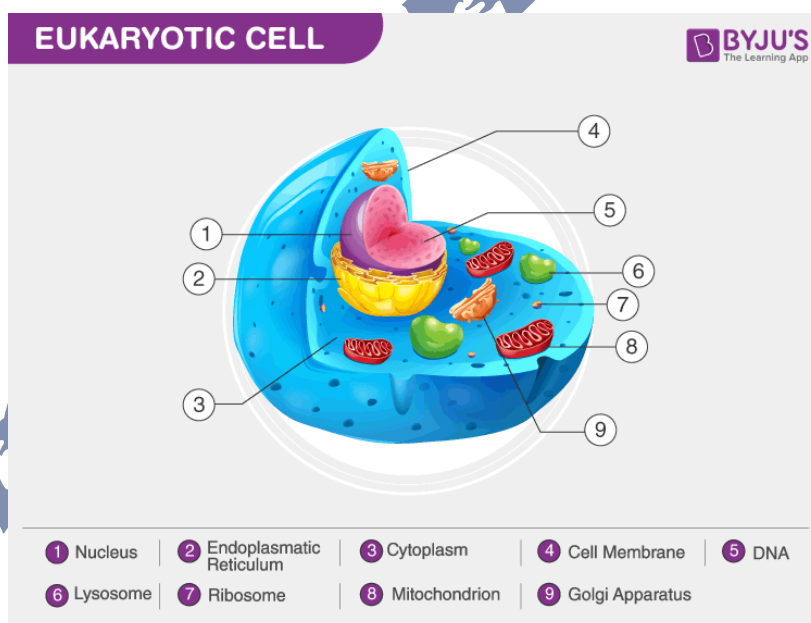
Plastids

These are double-membraned structures and are found only in [plant cells](#). These are of three types:

- **Chloroplast** that contains chlorophyll and is involved in photosynthesis.
- **Chromoplast** that contains a pigment called carotene that provides the plants yellow, red, or orange colours.
- **Leucoplasts** that are colourless and store oil, fats, carbohydrates, or proteins.

Eukaryotic Cell Diagram

Eukaryotic cell diagram mentioned below depicts different cell organelles present in eukaryotic cells. The nucleus, endoplasmic reticulum, cytoplasm, mitochondria, ribosomes, lysosomes are clearly mentioned in the diagram.



Prokaryotic Cell Definition

“Prokaryotic cells are the cells that do not have a true nucleus and membrane-bound organelles.”

What is a Prokaryotic Cell?

Prokaryotic cells are single-celled microorganisms known to be the earliest on earth. Prokaryotes include Bacteria and Archaea. The photosynthetic prokaryotes include cyanobacteria that perform photosynthesis.

A prokaryotic cell consists of a single membrane and therefore, all the reactions occur within the cytoplasm. They can be free-living or parasites.

Characteristics of Prokaryotic Cell

Prokaryotic cells have different characteristic features. The characteristics of the prokaryotic cells are mentioned below.

1. They lack a nuclear membrane.
2. Mitochondria, Golgi bodies, chloroplast, and lysosomes are absent.
3. The genetic material is present on a single chromosome.
4. The histone proteins, the important constituents of eukaryotic chromosomes, are lacking in them.
5. The cell wall is made up of carbohydrates and amino acids.
6. The plasma membrane acts as the mitochondrial membrane carrying respiratory enzymes.
7. They divide asexually by binary fission. The sexual mode of reproduction involves conjugation.

Prokaryotic Cell Structure

A prokaryotic cell does not have a nuclear membrane. However, the [genetic material](#) is present in a region in the cytoplasm known as the nucleoid. They may be spherical, rod-shaped, or spiral. A prokaryotic cell structure is as follows:

1. **Capsule**– It is an outer protective covering found in the bacterial cells, in addition to the cell wall. It helps in moisture retention, protects the cell when engulfed, and helps in the attachment of cells to nutrients and surfaces.
2. **Cell Wall**– It is the outermost layer of the cell which gives shape to the cell.
3. **Cytoplasm**– The cytoplasm is mainly composed of enzymes, salts, cell organelles and is a gel-like component.
4. **Cell Membrane**– This layer surrounds the cytoplasm and regulates the entry and exit of substances in the cells.
5. **Pili**– These are hair-like outgrowths that attach to the surface of other bacterial cells.
6. **Flagella**– These are long structures in the form of a whip, that help in the locomotion of a cell.
7. **Ribosomes**– These are involved in protein synthesis.
8. **Plasmids**– Plasmids are non-chromosomal DNA structures. These are not involved in reproduction.
9. **Nucleoid Region**– It is the region in the cytoplasm where the genetic material is present.

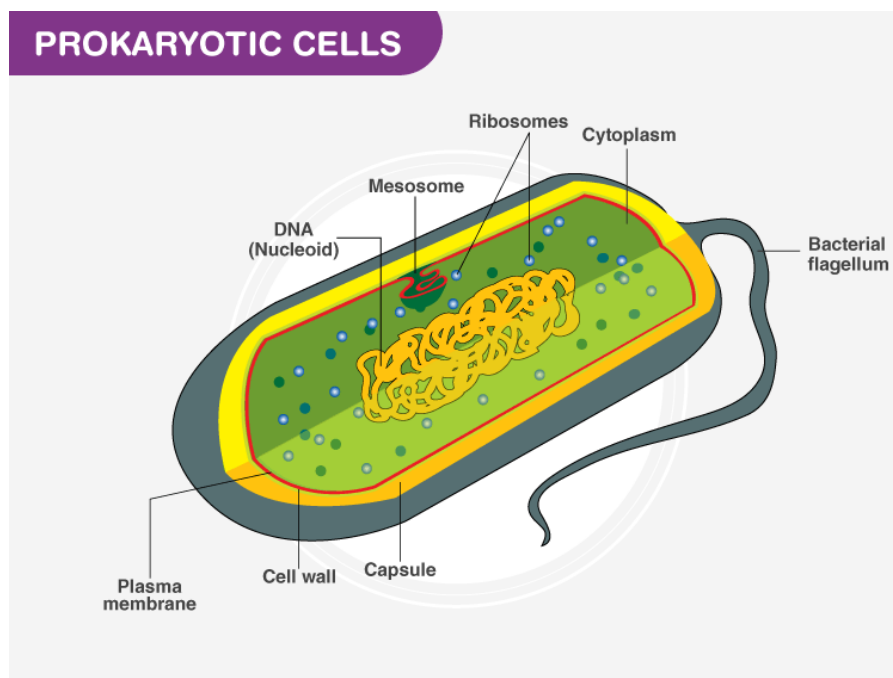
A prokaryotic cell lacks certain organelles like mitochondria, endoplasmic reticulum, and Golgi bodies.

Prokaryotic Cell Diagram

The prokaryotic cell diagram given below represents a bacterial cell. It depicts the absence of a true nucleus and the presence of a flagellum that differentiates it from a eukaryotic cell.

Difference between Prokaryotic and Eukaryotic Cells

Though these two classes of cells are quite different, they do possess some common characteristics. For instance, both possess cell membranes and ribosomes, but the similarities end there. The complete list of differences between prokaryotic and eukaryotic cells is summarized as follows:



	Prokaryotes	Eukaryotes
Type of Cell	Always unicellular	Unicellular and multi-cellular
Cell size	Ranges in size from 0.2 μm – 2.0 μm in diameter	Size ranges from 10 μm – 100 μm in diameter
Cell wall	Usually present; chemically complex in nature	When present, chemically simple in nature
Nucleus	Absent. Instead, they have a nucleoid region in the cell	Present
Ribosomes	Present. Smaller in size and spherical in shape	Present. Comparatively larger in size and linear in shape
DNA arrangement	Circular	Linear
Mitochondria	Absent	Present
Cytoplasm	Present, but cell organelles absent	Present, cell organelles present

Endoplasmic reticulum	Absent	Present
Plasmids	Present	Very rarely found in eukaryotes
Ribosome	Small ribosomes	Large ribosomes
Lysosome	Lysosomes and centrosomes are absent	Lysosomes and centrosomes are present
Cell division	Through binary fission	Through mitosis
Flagella	The flagella are smaller in size	The flagella are larger in size
Reproduction	Asexual	Both asexual and sexual
Example	Bacteria and Archaea	Plant and Animal cell

What are Cell Organelles?

The cellular components are called cell organelles. These cell organelles include both membrane and non-membrane bound organelles, present within the cells and are distinct in their structures and functions. They coordinate and function efficiently for the normal functioning of the cell. A few of them function by providing shape and support, whereas some are involved in the locomotion and reproduction of a cell. There are various organelles present within the cell and are classified into three categories based on the presence or absence of membrane.

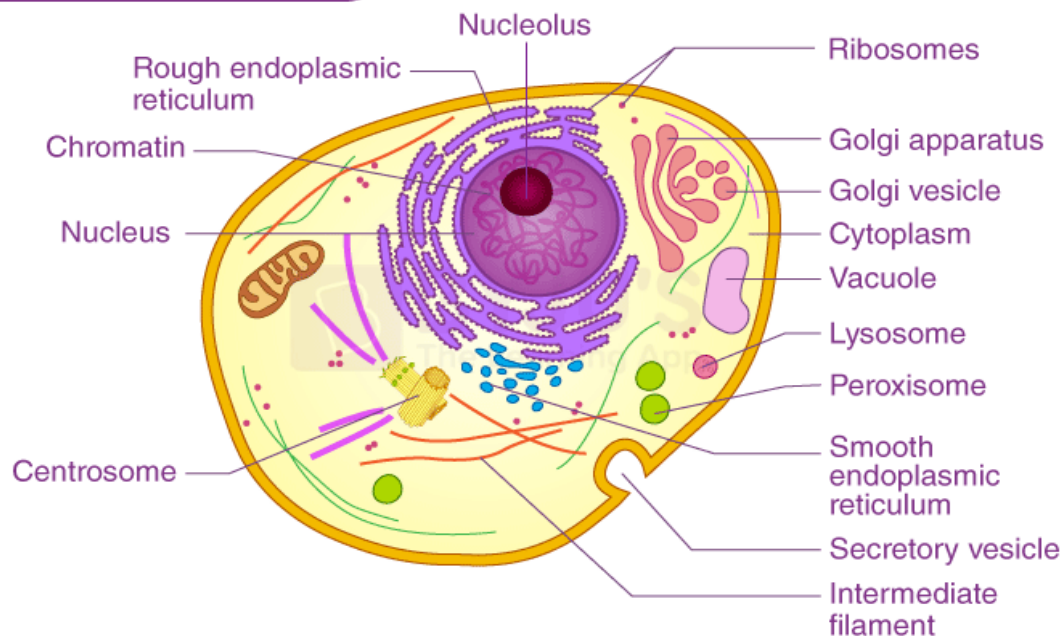
Organelles without membrane: The Cell wall, Ribosomes, and Cytoskeleton are non-membrane-bound cell organelles. They are present both in the [prokaryotic cell](#) and the eukaryotic cell.

Single membrane-bound organelles: Vacuole, Lysosome, Golgi Apparatus, Endoplasmic Reticulum are single membrane-bound organelles present only in a eukaryotic cell.

Double membrane-bound organelles: Nucleus, mitochondria and chloroplast are double membrane-bound organelles present only in a eukaryotic cell.

Let us learn more in detail about the different cell organelles in brief.

CELL ORGANELLES



Ultrastructure of the Cell

The ultrastructure of a cell refers to its fine structural details, typically observed using electron microscopy. This level of detail reveals the intricate organization and function of various cellular components. Here's an overview of the ultrastructure of key cell organelles:

1. Plasma Membrane

- **Structure:**
 - **Phospholipid Bilayer:** Composed of two layers of phospholipids with hydrophilic (water-attracting) heads facing outward and hydrophobic (water-repelling) tails facing inward.
 - **Proteins:** Integral (span the bilayer) and peripheral (attached to the surface) proteins involved in transport, signaling, and structural support.
 - **Cholesterol:** Interspersed within the bilayer, providing fluidity and stability.
 - **Carbohydrates:** Attached to proteins and lipids on the extracellular surface, forming glycoproteins and glycolipids that play roles in cell recognition and adhesion.

2. Nucleus

- **Structure:**
 - **Nuclear Envelope:** Double membrane with nuclear pores that control the movement of molecules in and out of the nucleus.
 - **Nucleoplasm:** Semi-fluid matrix inside the nucleus containing chromatin and the nucleolus.
 - **Chromatin:** Complex of DNA and histone proteins, which condenses to form chromosomes during cell division.

- **Nucleolus:** Dense region within the nucleus where ribosomal RNA (rRNA) synthesis and ribosome assembly occur.

3. Endoplasmic Reticulum (ER)

- **Rough Endoplasmic Reticulum (RER):**
 - **Structure:** Network of membranous sacs and tubules studded with ribosomes.
 - **Function:** Synthesis of proteins destined for secretion, incorporation into the plasma membrane, or use in lysosomes.
- **Smooth Endoplasmic Reticulum (SER):**
 - **Structure:** Network of membranous tubules without ribosomes.
 - **Function:** Synthesis of lipids, metabolism of carbohydrates, detoxification of drugs and poisons, and storage of calcium ions.

4. Golgi Apparatus

- **Structure:**
 - **Cisternae:** Stacked, flattened membranous sacs.
 - **Vesicles:** Small, membrane-bound compartments that transport materials to and from the Golgi.
- **Function:** Modifies, sorts, and packages proteins and lipids received from the ER for secretion or delivery to other organelles.

5. Mitochondria

- **Structure:**
 - **Double Membrane:** Outer membrane is smooth, while the inner membrane is highly folded into cristae to increase surface area.
 - **Matrix:** Inner space containing mitochondrial DNA, ribosomes, and enzymes involved in metabolic processes.
- **Function:** Site of cellular respiration and ATP production through oxidative phosphorylation.

6. Lysosomes

- **Structure:**
 - **Membrane-bound Vesicles:** Contain hydrolytic enzymes.
- **Function:** Degrade macromolecules, damaged organelles, and foreign particles through enzymatic digestion.

7. Peroxisomes

- **Structure:**
 - **Membrane-bound Organelles:** Contain enzymes such as catalase and oxidases.
- **Function:** Break down fatty acids and detoxify harmful substances, such as hydrogen peroxide.

8. Cytoskeleton

- **Structure:**
 - **Microfilaments:** Thin filaments composed of actin, involved in cell movement and shape.
 - **Intermediate Filaments:** Rope-like fibers providing mechanical support.
 - **Microtubules:** Hollow tubes composed of tubulin, involved in cell division, intracellular transport, and maintaining cell shape.
- **Function:** Provides structural support, facilitates cell movement, and aids in intracellular transport and cell division.

9. Ribosomes

- **Structure:**
 - **Composed of rRNA and Proteins:** Consist of two subunits (large and small).
 - **Free or Bound:** Found free in the cytoplasm or bound to the RER.
- **Function:** Site of protein synthesis, translating mRNA into polypeptide chains.

10. Chloroplasts (in Plant Cells)

- **Structure:**
 - **Double Membrane:** Outer and inner membranes enclose the stroma.
 - **Thylakoids:** Membranous sacs stacked into grana where photosynthesis occurs.
 - **Stroma:** Fluid containing enzymes, ribosomes, and chloroplast DNA.
- **Function:** Conduct photosynthesis to convert light energy into chemical energy stored in glucose.

11. Vacuoles

- **Structure:**
 - **Membrane-bound Sacs:** Larger in plant cells (central vacuole) than in animal cells.
- **Function:** Store nutrients, waste products, and maintain turgor pressure in plant cells.

Endoplasmic Reticulum

“Endoplasmic Reticulum is a complex network of tubular membranes exclusively present in the cytoplasm of the eukaryotic cell.”

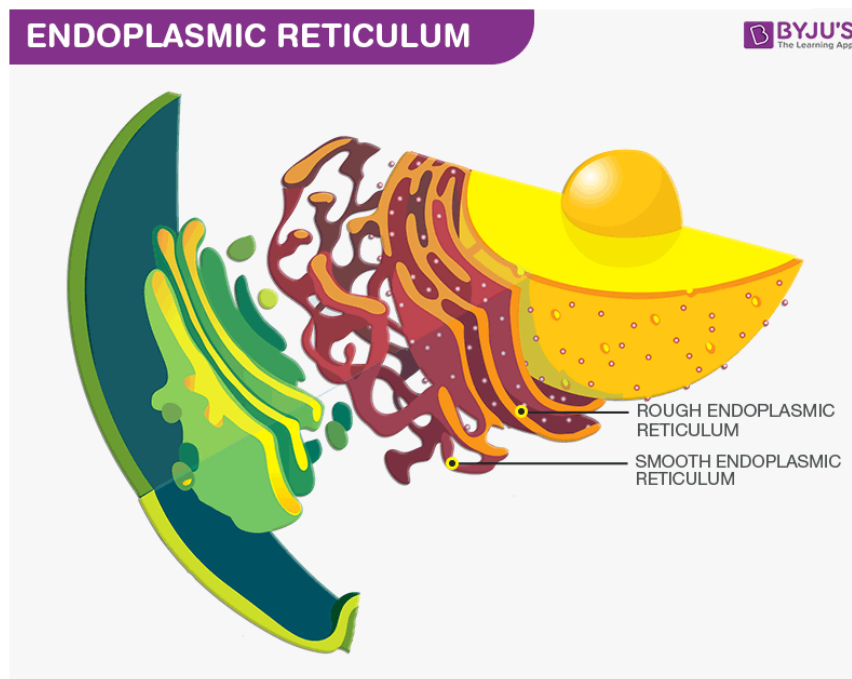
The endoplasmic reticulum transpires in two forms: a type with a ribosome-studded surface and another with a smooth surface. The latter is called the **smooth endoplasmic reticulum**, and the former is called the **rough endoplasmic reticulum**. These membranes form continuous folds, eventually joining the outer layer of the [nuclear membrane](#). Except for sperm cells and red blood cells, the endoplasmic reticulum is observed in every other type of eukaryotic cell.

Endoplasmic Reticulum Diagram

The below diagram shows the variants of the endoplasmic reticulum:

- Rough ER
- Smooth ER

Rough endoplasmic reticulum has ribosomes embedded within its structure, giving a “rough” appearance. The smooth endoplasmic reticulum does not have these ribosomes, hence appearing “smooth.”



Structure of Endoplasmic Reticulum

The structure of the endoplasmic reticulum is shaped like a sac. Since ER is of two types, each has its own distinguishing features:

Rough Endoplasmic Reticulum Structure

- The rough endoplasmic reticulum is named so because of its appearance.
- It is a series of connected flattened sacs having several ribosomes on its outer surface, hence the name.
- It synthesizes and secretes proteins in the liver, hormones and other substances in the glands.
- Rough ER is prominent in cells where protein synthesis happens (such as hepatocytes)

Role in Protein Synthesis

- **Ribosome Attachment:** Ribosomes on the RER synthesize proteins destined for secretion, incorporation into the plasma membrane, or use in lysosomes.
- **Co-translational Translocation:** As a protein is synthesized by a ribosome, it is threaded into the lumen of the RER through a translocon channel.
- **Protein Folding:** Inside the RER lumen, proteins fold into their correct three-dimensional shapes with the assistance of chaperone proteins.
- **Post-translational Modifications:** Proteins undergo initial modifications, such as glycosylation (addition of carbohydrate groups), forming glycoproteins.
- **Quality Control:** The RER ensures that only properly folded and assembled proteins proceed to the Golgi apparatus. Misfolded proteins are retained and eventually degraded.

Smooth Endoplasmic Reticulum Structure

- The smooth endoplasmic reticulum, on the other hand, does not have [ribosomes](#).
- The smooth endoplasmic reticulum has a tubular form.
- It participates in the production of phospholipids, the chief lipids in cell membranes and are essential in the process of metabolism.
- Smooth ER transports the products of the rough ER to other cellular organelles, especially the Golgi apparatus.

Role in Lipid Metabolism and Drug Detoxification

- **Lipid Synthesis:** The SER is involved in the synthesis of lipids, including phospholipids and steroids, which are essential components of cellular membranes and hormones.
- **Carbohydrate Metabolism:** The SER helps in the metabolism of carbohydrates, including the conversion of glycogen to glucose in liver cells.
- **Calcium Storage:** The SER stores calcium ions, which are crucial for muscle contraction and other cellular signaling processes.
- **Detoxification:** The SER contains enzymes, such as cytochrome P450, that are involved in the detoxification of drugs, alcohol, and other harmful substances. These enzymes catalyze reactions that make these substances more water-soluble and easier to excrete from the body.

Functions of Endoplasmic Reticulum

As stated above, the endoplasmic reticulum is categorised into two types, and both these types of ER perform specific functions:

Smooth Endoplasmic Reticulum Function:

- Smooth ER is responsible for the synthesis of essential lipids such as phospholipids and cholesterol.
- Smooth ER is also responsible for the production and secretion of steroid hormones.
- It is also responsible for the metabolism of [carbohydrates](#).
- The smooth ER store and releases calcium ions. These are quite important for the nervous system and muscular systems.

Rough Endoplasmic Reticulum Function:

- The majority of the functions of rough ER is associated with protein synthesis.
- The rough endoplasmic reticulum also plays a vital role in protein folding.
- Also ensures quality control (regarding correct protein folding).
- The second most important function after protein synthesis and protein folding is protein sorting.

Golgi Apparatus

The Golgi apparatus, also known as the Golgi complex or Golgi body, is a vital organelle in eukaryotic cells responsible for modifying, sorting, and packaging proteins and lipids for secretion or delivery to

other organelles. It plays a crucial role in the post-translational processing of proteins and the formation of lysosomes.

Structure

- **Cisternae:** The Golgi apparatus consists of a series of flattened, membrane-bound sacs called cisternae. These stacks are typically organized into three main regions:
 - **Cis-Golgi Network (CGN):** The entry face, located nearest the endoplasmic reticulum (ER). Vesicles from the ER fuse with the CGN.
 - **Medial Golgi:** The central layers where most of the processing occurs.
 - **Trans-Golgi Network (TGN):** The exit face, where modified molecules are sorted and packaged into vesicles for transport to their final destinations.
- **Vesicles:** Small, membrane-bound sacs that transport materials between the ER, Golgi apparatus, and other cellular compartments. Vesicles bud off from and fuse with the cisternae.

Role in synthesis of glycoprotein.

The Golgi apparatus is integral to the synthesis and maturation of glycoproteins. Glycoproteins are proteins that have carbohydrate groups covalently attached to them, which are crucial for proper protein folding, stability, signaling, and cell-cell recognition.

Steps in Glycoprotein Synthesis

1. Initial Glycosylation in the Endoplasmic Reticulum (ER):

- **N-linked Glycosylation:** The process begins in the ER, where an oligosaccharide is attached to the nitrogen atom (N) of an asparagine residue in the nascent protein. This oligosaccharide is a pre-assembled 14-sugar structure (often referred to as a "core glycan") that is transferred from a lipid carrier (dolichol phosphate) to the protein.
- **Quality Control:** The ER ensures that only properly folded proteins proceed to the Golgi apparatus. Misfolded proteins are retained and eventually degraded.

2. Transport to the Golgi Apparatus:

- **Vesicular Transport:** Once initial glycosylation and folding are complete, the glycoproteins are packaged into COPII-coated vesicles and transported from the ER to the cis-Golgi network (CGN).

3. Processing in the Golgi Apparatus:

- **Cis-Golgi Network (CGN):** The CGN receives the vesicles containing glycoproteins from the ER. Initial modifications may begin here.
- **Medial Golgi:** As the glycoproteins move through the medial Golgi, extensive remodeling of the oligosaccharide chains occurs. This can include the removal of mannose residues and the addition of various sugars such as N-acetylglucosamine, galactose, and sialic acid.
- **Trans-Golgi Network (TGN):** In the TGN, further modifications and sorting occur. Terminal sugars are often added here, and the glycoproteins are sorted into different vesicles based on their final destinations.

Specific Modifications in the Golgi Apparatus

- **Trimming and Addition of Sugars:**

- Enzymes in the Golgi apparatus trim the initial oligosaccharide added in the ER and sequentially add new sugars in a highly regulated manner.

- **Enzymes Involved:**

- **Glycosidases:** Remove specific sugar residues.
- **Glycosyltransferases:** Add sugars to specific positions on the growing oligosaccharide chain.

- **Complex and Hybrid N-glycans:**

- In the Golgi, N-linked glycans can be processed into complex or hybrid forms, which have different functional properties compared to the high-mannose forms typically found in the ER.

- **O-linked Glycosylation:**

- Unlike N-linked glycosylation, O-linked glycosylation occurs entirely in the Golgi apparatus. In this process, sugars are attached to the hydroxyl groups (O) of serine or threonine residues in the protein. This type of glycosylation is common in mucins and certain hormones.

Functional Importance of Glycoprotein Synthesis

- **Protein Stability and Folding:**

- Glycosylation helps in proper folding of the protein and increases its stability by protecting it from proteolytic degradation.

- **Cell-Cell and Cell-Matrix Interactions:**

- Glycoproteins play critical roles in cell-cell recognition, signaling, and adhesion processes. For example, they are important in immune responses where glycoproteins on the surface of cells can act as antigens.

- **Targeting and Trafficking:**

- Proper glycosylation is crucial for the targeting of glycoproteins to their correct destinations, such as lysosomes, plasma membrane, or for secretion outside the cell.

Mitochondria: Cell Respiration and Significance as a Semi-Autonomous Organelle

Structure of Mitochondria

Double Membrane:

Outer Membrane:

- Smooth and permeable to small molecules and ions.

Inner Membrane:

- Highly folded into structures called cristae, increasing the surface area for cellular respiration.

Intermembrane Space:

The region between the outer and inner membranes.

Matrix:

- The innermost compartment containing mitochondrial DNA, ribosomes, and enzymes necessary for the Krebs cycle and other metabolic pathways.

Cell Respiration

- Mitochondria are the powerhouses of the cell, responsible for producing ATP through the process of cellular respiration. This process consists of three main stages:

Glycolysis (Cytoplasm):

- Glucose is broken down into pyruvate, yielding a small amount of ATP and NADH.
- Pyruvate is transported into the mitochondria for further processing.

Krebs Cycle (Citric Acid Cycle) (Matrix):

- Acetyl-CoA, derived from pyruvate, enters the Krebs cycle.
- This cycle produces ATP, NADH, FADH₂, and releases CO₂ as a byproduct.
- NADH and FADH₂ are electron carriers that transfer electrons to the electron transport chain (ETC).

Electron Transport Chain (ETC) and Oxidative Phosphorylation (Inner Membrane):

- NADH and FADH₂ donate electrons to the ETC, which is composed of a series of protein complexes and electron carriers embedded in the inner mitochondrial membrane.
- As electrons move through the ETC, protons are pumped from the matrix into the intermembrane space, creating a proton gradient (proton motive force).
- Protons flow back into the matrix through ATP synthase, driving the synthesis of ATP from ADP and inorganic phosphate (Pi).
- Oxygen serves as the final electron acceptor, combining with electrons and protons to form water.

Significance as a Semi-Autonomous Organelle

- Mitochondria exhibit several unique features that classify them as semi-autonomous organelles:

Own DNA:

- Mitochondria contain their own circular DNA (mtDNA), which encodes some of the proteins and RNAs required for mitochondrial function.
- mtDNA is inherited maternally and is similar to bacterial DNA, supporting the endosymbiotic theory.

Ribosomes:

- Mitochondria possess their own ribosomes (similar to bacterial ribosomes), allowing them to synthesize some of their own proteins.

Replication and Division:

- Mitochondria can replicate independently of the cell cycle through a process similar to binary fission in bacteria.
- The number of mitochondria within a cell can change in response to the cell's energy needs.

Protein Synthesis:

- While mitochondria can produce some proteins, the majority of mitochondrial proteins are encoded by nuclear DNA, synthesized in the cytoplasm, and imported into the mitochondria.
- Mitochondrial biogenesis requires the coordinated expression of both nuclear and mitochondrial genomes.

Dual Genetic Control:

- Mitochondrial function is governed by both mitochondrial and nuclear genomes, necessitating intricate coordination between these genetic systems.

Functional Significance**Energy Production:**

- Mitochondria generate most of the cell's ATP through oxidative phosphorylation, providing the energy necessary for various cellular processes.

Metabolic Integration:

- Mitochondria play key roles in metabolic pathways, including the Krebs cycle, fatty acid oxidation, and amino acid metabolism.

Apoptosis Regulation:

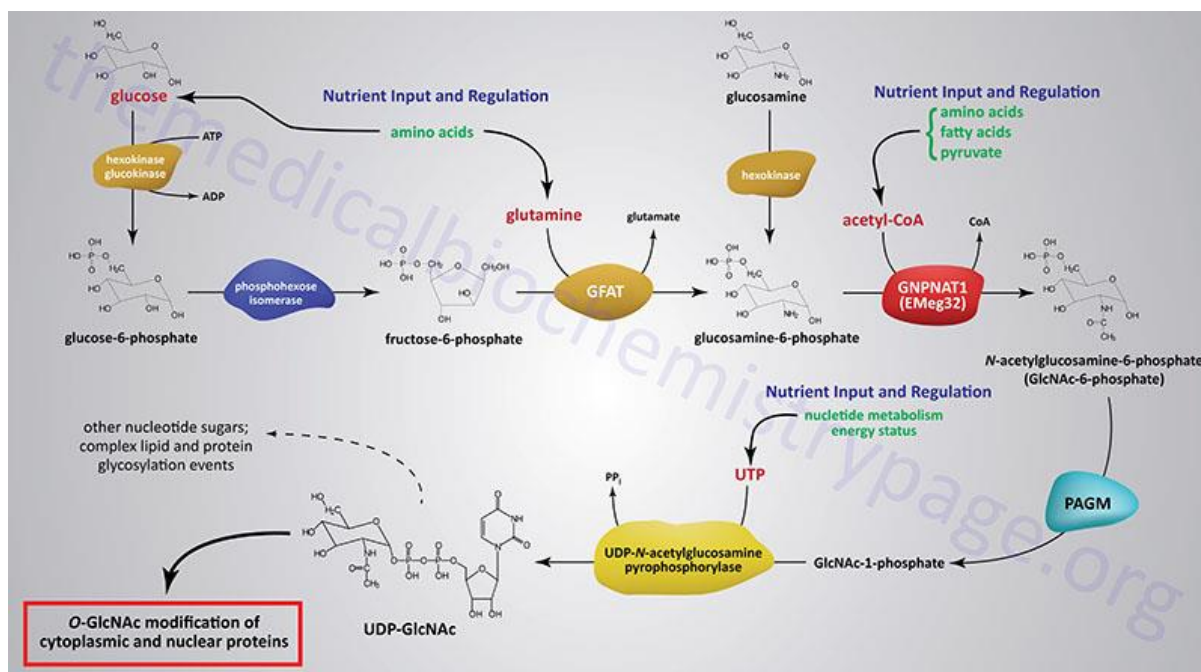
- Mitochondria are involved in the intrinsic pathway of apoptosis (programmed cell death) by releasing cytochrome c, which activates caspases that dismantle the cell.

Calcium Homeostasis:

- Mitochondria help regulate intracellular calcium levels, crucial for signal transduction, muscle contraction, and other cellular activities.

Reactive Oxygen Species (ROS) Production:

- Mitochondria are a source of ROS, which can act as signaling molecules but also cause oxidative damage if not properly regulated.



Lysosomes: Diverse Roles Due to Hydrolytic Activity of Enzymes

Lysosomes are membrane-bound organelles in eukaryotic cells that contain a wide variety of hydrolytic enzymes capable of breaking down different types of biomolecules. These enzymes include proteases, lipases, nucleases, and glycosidases, all of which function optimally at an acidic pH. Lysosomes are often referred to as the cell's "digestive system" because of their crucial role in degrading and recycling cellular waste, macromolecules, and foreign substances.

Structure of Lysosomes

- **Membrane:** A single phospholipid bilayer that separates the acidic interior of the lysosome from the cytoplasm.
- **Enzymes:** Contains over 50 different hydrolytic enzymes.
- **Acidic Environment:** Maintained by proton pumps (H⁺-ATPases) that actively transport hydrogen ions (protons) into the lysosome.

Diverse Roles of Lysosomes

1. Intracellular Digestion:

- **Autophagy:** Lysosomes degrade and recycle damaged or obsolete cellular components. During autophagy, portions of the cytoplasm and organelles are enclosed in autophagosomes, which then fuse with lysosomes, where their contents are digested.
- **Heterophagy:** Involves the digestion of extracellular material taken up by the cell via endocytosis or phagocytosis. Lysosomes fuse with endosomes or phagosomes, forming endolysosomes or phagolysosomes, where the ingested material is degraded.

2. Pathogen Destruction:

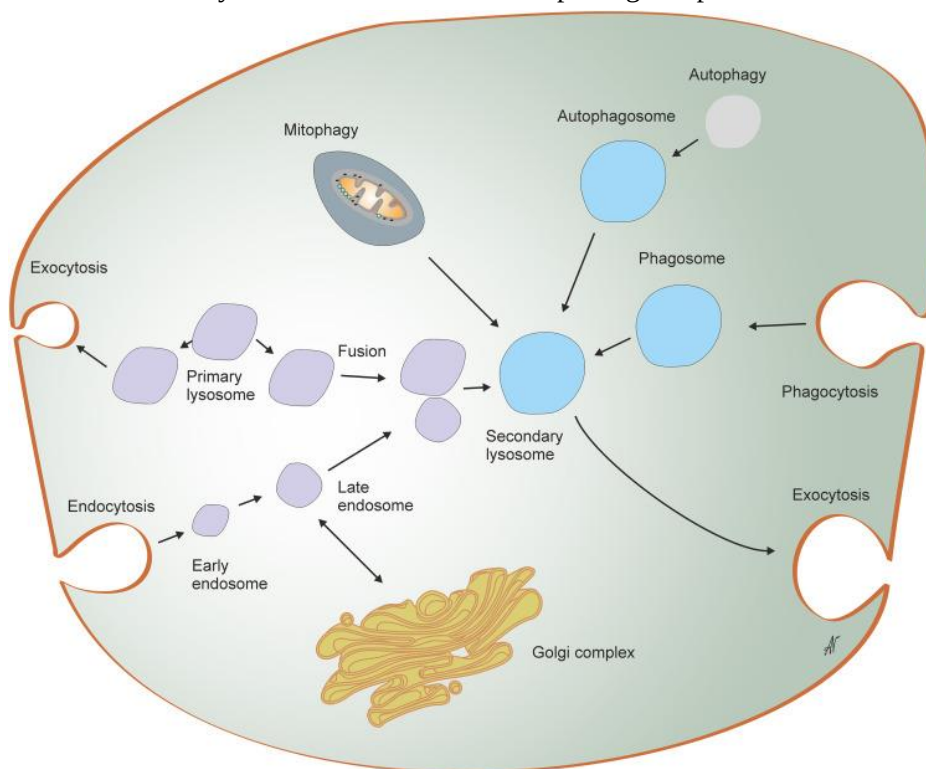
- **Phagocytosis:** Lysosomes play a crucial role in the immune response by destroying pathogens. Phagocytic cells, such as macrophages and neutrophils, engulf pathogens into phagosomes, which then fuse with lysosomes to form phagolysosomes where pathogens are digested by hydrolytic enzymes.

3. Apoptosis:

- Lysosomes can contribute to programmed cell death (apoptosis) by releasing their enzymes into the cytoplasm, leading to cellular breakdown and death.

4. Membrane Repair:

- Lysosomes are involved in repairing the plasma membrane. When the membrane is damaged, lysosomes can fuse with the membrane to patch up the lesion.



5. Metabolic Regulation:

- Lysosomes participate in nutrient sensing and metabolic regulation through the mTOR signaling pathway. They help regulate cellular metabolism by responding to the availability of nutrients and growth factors.

6. Secretion:

- Some specialized cells use

lysosome-related organelles for the regulated secretion of their contents. For example, in immune cells like cytotoxic T cells and natural killer (NK) cells, lysosomes contain cytotoxic molecules that are secreted to kill target cells.

Hydrolytic Enzymes in Lysosomes

- **Proteases:** Break down proteins into amino acids.
- **Lipases:** Degrade lipids into fatty acids and glycerol.
- **Nucleases:** Digest nucleic acids into nucleotides.
- **Glycosidases:** Hydrolyze carbohydrates into simple sugars.

Diseases Related to Lysosomal Dysfunction

Defects in lysosomal enzymes or in the transport of enzymes to lysosomes can lead to lysosomal storage diseases (LSDs), which are characterized by the accumulation of undigested substrates. Examples include:

- **Tay-Sachs Disease:** Caused by a deficiency in the enzyme hexosaminidase A, leading to the accumulation of GM2 ganglioside in nerve cells.
- **Gaucher Disease:** Caused by a deficiency in the enzyme glucocerebrosidase, leading to the accumulation of glucocerebroside.
- **Pompe Disease:** Caused by a deficiency in the enzyme acid alpha-glucosidase, leading to the accumulation of glycogen in cells.

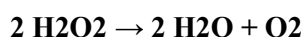
Peroxisomes: Metabolism of Hydrogen Peroxide

Peroxisomes are membrane-bound organelles found in eukaryotic cells that are involved in various metabolic processes, including the metabolism of hydrogen peroxide (H₂O₂). They contain several enzymes that participate in the breakdown of H₂O₂, a toxic byproduct of various cellular reactions, into water and oxygen. Peroxisomes play a crucial role in maintaining cellular redox balance and protecting cells from oxidative damage.

Enzymes Involved in Hydrogen Peroxide Metabolism

1. Catalase:

- Catalase is the key enzyme responsible for the breakdown of hydrogen peroxide into water and oxygen.
- It catalyzes the reaction:



- Catalase achieves this by facilitating the decomposition of hydrogen peroxide into water and molecular oxygen, effectively neutralizing its toxicity.
- Peroxisomes contain a high concentration of catalase, making them the primary site for hydrogen peroxide metabolism in the cell.

2. Peroxidases:

- Peroxisomes also contain peroxidases, such as peroxiredoxins, which can reduce hydrogen peroxide and other organic peroxides using reducing equivalents from molecules like NADPH.
- These peroxidases play a complementary role to catalase in detoxifying hydrogen peroxide and other reactive oxygen species (ROS) in the cell.

Importance of Hydrogen Peroxide Metabolism

1. Protection Against Oxidative Stress:

- Hydrogen peroxide is a reactive oxygen species (ROS) generated as a byproduct of various metabolic reactions, including cellular respiration.
- If not neutralized, hydrogen peroxide can damage cellular components such as DNA, proteins, and lipids through oxidative stress, leading to cell dysfunction and death.
- Peroxisomes help protect cells from oxidative damage by efficiently breaking down hydrogen peroxide into harmless water and oxygen.

2. Regulation of Cellular Redox Balance:

- Peroxisomes contribute to the regulation of cellular redox balance by controlling the levels of hydrogen peroxide and other ROS.
- They participate in maintaining an optimal cellular environment for metabolic reactions and signaling pathways by preventing ROS accumulation.

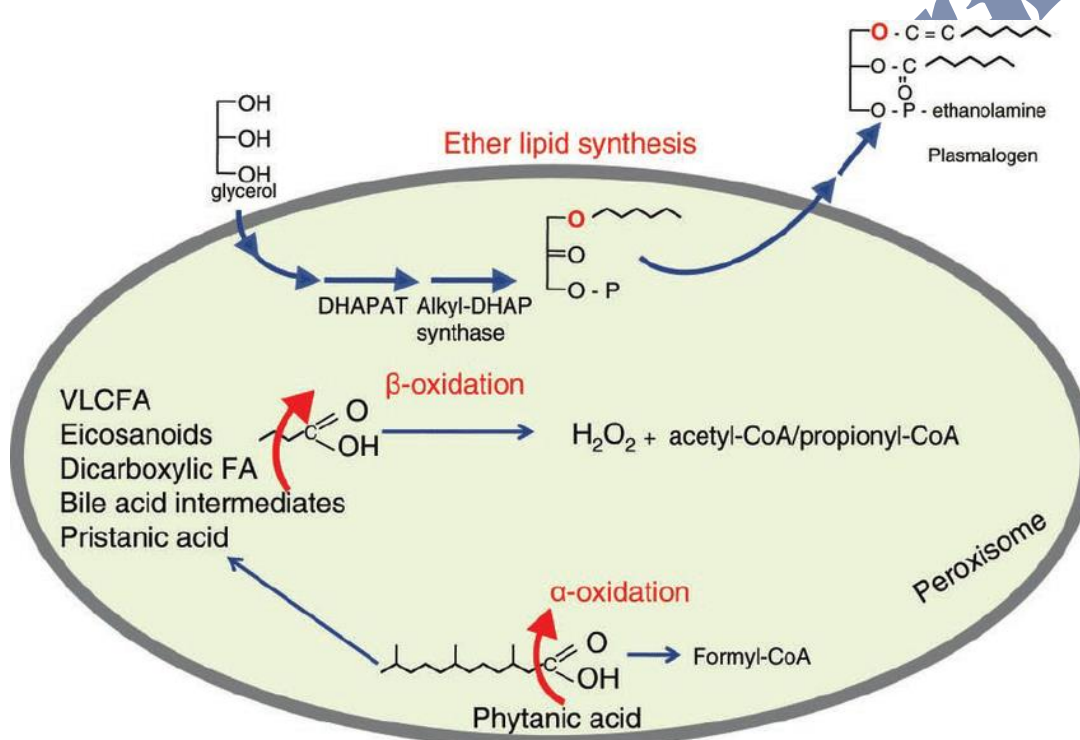
3. Cellular Signaling:

- Reactive oxygen species, including hydrogen peroxide, can also act as signaling molecules involved in various cellular processes, such as apoptosis, inflammation, and cell proliferation.

- Peroxisomes help regulate the levels of hydrogen peroxide, thereby influencing cellular signaling pathways and responses to stress and environmental stimuli.

Disorders Related to Peroxisomal Dysfunction

- Peroxisomal Biogenesis Disorders (PBDs):**
 - PBDs are a group of genetic disorders characterized by defects in peroxisome formation and function.
 - Examples include Zellweger syndrome, neonatal adrenoleukodystrophy, and infantile Refsum disease.
 - These disorders result in the accumulation of toxic substances and impaired metabolism of various molecules, leading to severe neurological, developmental, and metabolic abnormalities.



Glyoxysomes

Glyoxysomes are specialized organelles found in plants and some microorganisms, particularly in germinating seeds, where they play a crucial role in lipid metabolism. They are essentially specialized peroxisomes with additional enzymes required for the glyoxylate cycle, a variant of the Krebs cycle. The glyoxylate cycle enables plants to convert fatty acids into carbohydrates, which are essential for seedling growth before photosynthesis begins. Glyoxysomes contain enzymes such as isocitrate lyase and malate synthase, which are absent in typical peroxisomes but are necessary for converting fatty acids into sugars.

Plasma Membrane and Its Functions

The plasma membrane, also known as the cell membrane, is a phospholipid bilayer that surrounds the cytoplasm of a cell, separating it from the extracellular environment. The plasma membrane is a dynamic structure that performs several essential functions for the cell:

1. Cellular Barrier:

- The plasma membrane acts as a semi-permeable barrier, regulating the movement of substances into and out of the cell. It controls the passage of ions, nutrients, waste products, and signaling molecules.

2. Cellular Communication:

- The plasma membrane contains proteins, such as receptors and channels, that facilitate communication between the cell and its environment. These proteins allow the cell to detect and respond to external signals, such as hormones, neurotransmitters, and environmental cues.

3. Cell Adhesion:

- Specialized proteins in the plasma membrane, such as integrins and cadherins, mediate cell adhesion and interactions with neighboring cells and the extracellular matrix. Cell adhesion is crucial for maintaining tissue structure and integrity.

4. Cellular Recognition:

- Carbohydrate molecules, often attached to proteins and lipids on the extracellular surface of the plasma membrane, serve as markers for cell recognition and identification. These markers are involved in processes such as immune responses, tissue development, and cell-cell signaling.

5. Transport:

- The plasma membrane contains various protein channels, carriers, and pumps that regulate the movement of ions, nutrients, and other molecules across the membrane. This transport is essential for maintaining cellular homeostasis and supporting cellular processes such as nutrient uptake, ion exchange, and waste removal.

6. Cellular Signaling:

- The plasma membrane is involved in signal transduction, whereby extracellular signals are detected by receptors on the cell surface and transmitted to the cell interior, leading to cellular responses. This signaling regulates processes such as cell growth, differentiation, and survival.

7. Electrical Properties:

- The plasma membrane maintains a resting membrane potential, which is essential for the generation and propagation of electrical signals in excitable cells such as neurons and muscle cells.

The plasma membrane

The plasma membrane, also known as the cell membrane, is a dynamic and selectively permeable barrier that surrounds the cytoplasm of a cell, separating it from the extracellular environment. It is composed of a complex arrangement of lipids, proteins, and carbohydrates that form a fluid mosaic structure.

Chemical Composition of the Plasma Membrane

1. Phospholipids:

- Phospholipids are the primary structural components of the plasma membrane. They consist of a hydrophilic ("water-loving") phosphate head and two hydrophobic ("water-fearing") fatty acid tails.

- The arrangement of phospholipids in a bilayer forms the basic structure of the membrane, with the hydrophobic tails facing inward and the hydrophilic heads facing outward, towards the aqueous environments inside and outside the cell.

2. **Cholesterol:**

- Cholesterol molecules are interspersed within the phospholipid bilayer of the plasma membrane.
- Cholesterol helps maintain the fluidity and stability of the membrane by preventing phospholipids from packing too closely together or becoming too fluid.

3. **Proteins:**

- **Integral Proteins:** Embedded within the lipid bilayer, integral proteins traverse the membrane and may span from one side to the other. They can serve as channels, carriers, receptors, or enzymes.
- **Peripheral Proteins:** Found on the inner or outer surface of the membrane, peripheral proteins are not embedded within the lipid bilayer. They often interact with integral proteins or the cytoskeleton to help maintain the membrane's structure and function.

4. **Carbohydrates:**

- Carbohydrates are attached to proteins (glycoproteins) or lipids (glycolipids) on the extracellular surface of the plasma membrane.
- These carbohydrate chains, also known as glycocalyx, play roles in cell-cell recognition, adhesion, and communication.

Structure of the Plasma Membrane

• **Fluid Mosaic Model:**

- The plasma membrane is described by the fluid mosaic model, which depicts the membrane as a fluid, dynamic structure composed of a mosaic of lipids, proteins, and carbohydrates.
- The fluidity of the membrane allows for the lateral movement of phospholipids and proteins within the bilayer, enabling various cellular processes such as membrane trafficking, signal transduction, and cell-cell interactions.

• **Lipid Bilayer:**

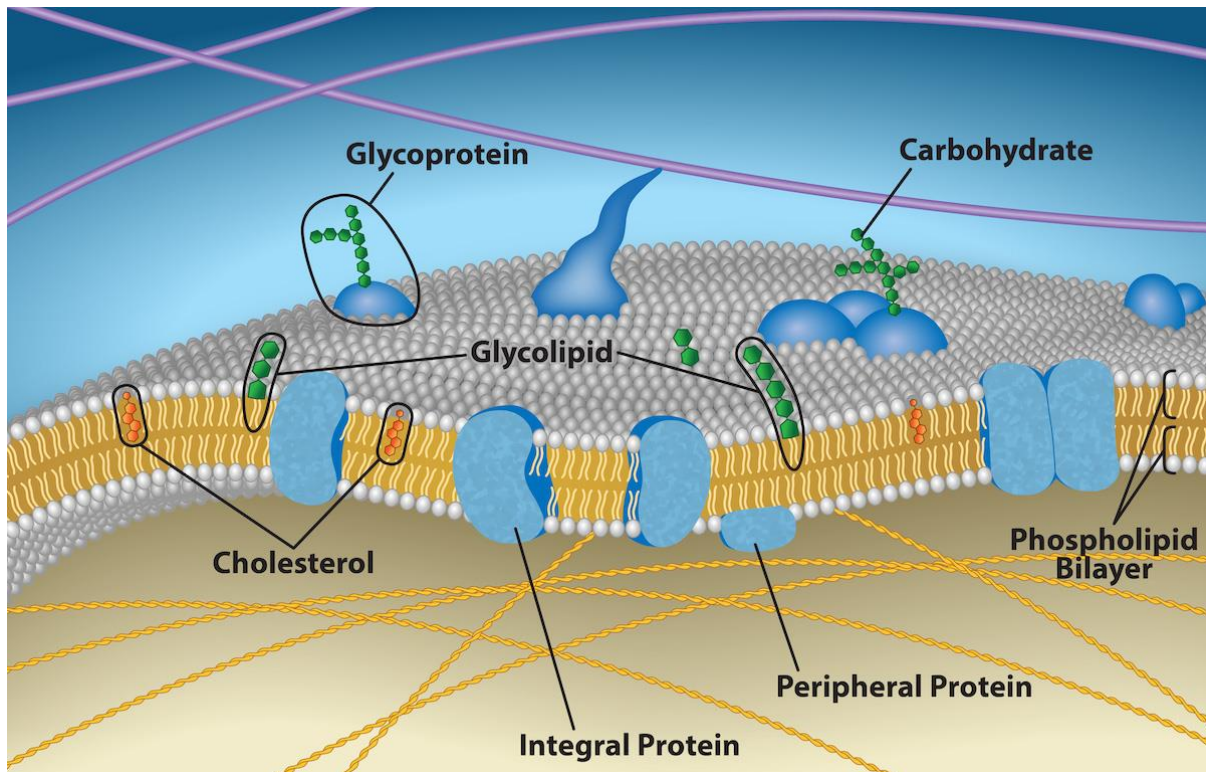
- The plasma membrane consists of two layers of phospholipids arranged with their hydrophobic tails facing inward and their hydrophilic heads facing outward.
- This lipid bilayer provides the basic structure of the membrane and serves as a barrier to the passage of hydrophilic molecules and ions.

• **Protein Distribution:**

- Integral proteins are embedded within the lipid bilayer, with some spanning the entire membrane (transmembrane proteins) and others extending partially into the hydrophobic core.
- Peripheral proteins are found on the inner or outer surface of the membrane and interact with integral proteins or the cytoskeleton.

• **Asymmetry:**

- The distribution of lipids and proteins across the plasma membrane is asymmetric, with different compositions on the inner and outer leaflets.
- This asymmetry is essential for various cellular processes, including membrane trafficking, cell signaling, and cell recognition.



Cell Permeability

Cell permeability refers to the ability of molecules to pass through the cell membrane. The plasma membrane is selectively permeable, meaning it allows certain substances to pass through while restricting others. The permeability of the membrane depends on factors such as the size, charge, and lipid solubility of the molecules.

Passive Transport: In passive transport, molecules move across the cell membrane without the expenditure of energy by the cell. This includes processes such as:

Diffusion: The movement of molecules from an area of higher concentration to an area of lower concentration until equilibrium is reached.

Facilitated Diffusion: The passive movement of molecules across the membrane facilitated by transport proteins, such as channels and carriers.

Osmosis: The diffusion of water across a selectively permeable membrane from an area of lower solute concentration to an area of higher solute concentration.

Active Transport: Active transport requires the expenditure of energy by the cell to move molecules against their concentration gradient, from an area of lower concentration to an area of higher concentration. This process is typically mediated by transport proteins called pumps, such as the sodium-potassium pump.

Endocytosis

Endocytosis is a cellular process by which cells engulf extracellular materials by forming vesicles derived from the plasma membrane. There are several types of endocytosis:

Phagocytosis: In phagocytosis, large particles, such as bacteria, cell debris, or other solid materials, are engulfed by the cell. Phagocytosis is primarily performed by specialized cells such as macrophages and neutrophils as part of the immune response.

Pinocytosis: Also known as "cell drinking," pinocytosis involves the non-selective uptake of small dissolved molecules and extracellular fluid into the cell. It is a continuous process that occurs in most cells to sample the extracellular environment and regulate the composition of the cell's internal environment.

Receptor-Mediated Endocytosis: This process involves the selective uptake of specific molecules bound to receptor proteins on the cell surface. Ligands bind to cell surface receptors, triggering the formation of clathrin-coated vesicles that are internalized by the cell.

Phagocytosis

Phagocytosis is a specialized form of endocytosis in which cells engulf large particles or microorganisms into vesicles called phagosomes. Phagocytosis is primarily performed by specialized cells called phagocytes, including macrophages, neutrophils, and dendritic cells. The process of phagocytosis involves several steps:

Recognition and Binding: The phagocyte recognizes and binds to the particle or microorganism through receptors on its surface.

Engulfment: The plasma membrane extends outward to surround the particle, forming a phagosome.

Phagosome Formation: The phagosome, containing the engulfed particle, is internalized within the cell.

Phagosome Maturation: The phagosome fuses with lysosomes, forming a phagolysosome. The lysosomal enzymes within the phagolysosome digest the particle, breaking it down into smaller components.

Digestion and Degradation: The particle is broken down into its constituent molecules, which can then be utilized by the cell for energy or other cellular processes.

Exocytosis: After digestion is complete, residual materials are expelled from the cell through exocytosis.

cytoskeleton

The cytoskeleton is a dynamic network of protein filaments found in the cytoplasm of eukaryotic cells. It provides structural support, maintains cell shape, facilitates cell movement, and is involved in various cellular processes such as cell division and intracellular transport. The cytoskeleton is composed of three main types of protein filaments: microfilaments (actin filaments), microtubules, and intermediate filaments.

Microfilaments (Actin Filaments)

- **Composition:** Microfilaments are thin, flexible filaments composed of actin protein subunits arranged in a double helix structure.
- **Function:**
 - Microfilaments play a crucial role in cell motility, including cell crawling, contraction, and cytokinesis (cell division).
 - They are involved in maintaining cell shape and providing mechanical support to the cell's plasma membrane.

- Microfilaments are also important in intracellular transport processes, such as the movement of organelles and vesicles along actin filaments.
- **Examples of Cellular Structures:** Microvilli in epithelial cells, contractile ring during cytokinesis, muscle contraction in muscle cells.

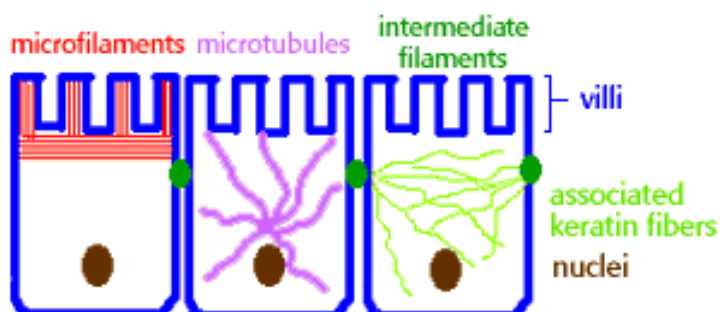
Microtubules

- **Composition:** Microtubules are hollow, cylindrical structures composed of tubulin protein subunits arranged in a protofilament structure.
- **Function:**
 - Microtubules serve as tracks for intracellular transport, facilitating the movement of organelles, vesicles, and molecular motors within the cell.
 - They are essential for maintaining cell shape and organization, as well as for positioning organelles within the cell.
 - Microtubules are involved in cell division, forming the mitotic spindle that separates chromosomes during mitosis and meiosis.
- **Examples of Cellular Structures:** Mitotic spindle during cell division, cilia and flagella for cell motility, centrioles and basal bodies.

Intermediate Filaments

- **Composition:** Intermediate filaments are fibrous proteins that form ropelike structures composed of various types of proteins, such as keratins, vimentins, and neurofilaments.
- **Function:**
 - Intermediate filaments provide mechanical strength and structural support to cells, particularly in tissues subjected to mechanical stress.
 - They contribute to the maintenance of cell shape and integrity, as well as the anchorage of organelles within the cytoplasm.
 - Intermediate filaments are important for cell-cell adhesion and tissue organization.
- **Examples of Cellular Structures:** Desmosomes, nuclear lamina, keratin filaments in epithelial cells.

Cytoskeletal components of intestinal epithelial cells



Comparison

Feature	Microfilaments (Actin Filaments)	Microtubules	Intermediate Filaments
Composition	Actin protein subunits	Tubulin protein subunits	Various fibrous proteins
Structure	Thin, flexible filaments	Hollow, cylindrical tubes	Ropelike structures

Function	Cell motility, shape maintenance, intracellular transport	Intracellular transport, cell division, shape maintenance	Mechanical strength, structural support, cell-cell adhesion
Examples	Microvilli, contractile ring, muscle contraction	Mitotic spindle, cilia, flagella	Desmosomes, nuclear lamina, keratin filaments

Replication

Definition: Replication is the process by which DNA molecules make exact copies of themselves.

Key Steps:

1. **Initiation:** DNA replication begins at specific sites called origins of replication, where the DNA double helix unwinds and separates into two strands.
2. **Elongation:** Enzymes called DNA polymerases add complementary nucleotides to each of the original DNA strands, synthesizing two new daughter strands.
3. **Termination:** Replication proceeds bidirectionally along the DNA strands until the entire molecule is copied. The process ends when replication forks from neighboring origins meet or reach the end of the DNA molecule.

Significance: DNA replication ensures that genetic information is faithfully transmitted from parent to daughter cells during cell division. It is essential for growth, development, and inheritance in all living organisms.

Transcription

Definition: Transcription is the process by which the information encoded in DNA is copied into RNA molecules.

Key Steps:

1. **Initiation:** RNA polymerase binds to a specific DNA sequence called the promoter region, marking the beginning of transcription.
2. **Elongation:** RNA polymerase moves along the DNA template strand, synthesizing a complementary RNA strand by adding ribonucleotides according to the base-pairing rules (A-U and G-C).
3. **Termination:** Transcription ends when RNA polymerase reaches a termination signal in the DNA sequence, causing RNA polymerase to dissociate from the DNA and release the newly synthesized RNA molecule.

Significance: Transcription is the first step in gene expression, where the genetic information stored in DNA is converted into functional RNA molecules such as messenger RNA (mRNA), transfer RNA (tRNA), and ribosomal RNA (rRNA). These RNA molecules play crucial roles in protein synthesis and various cellular processes.

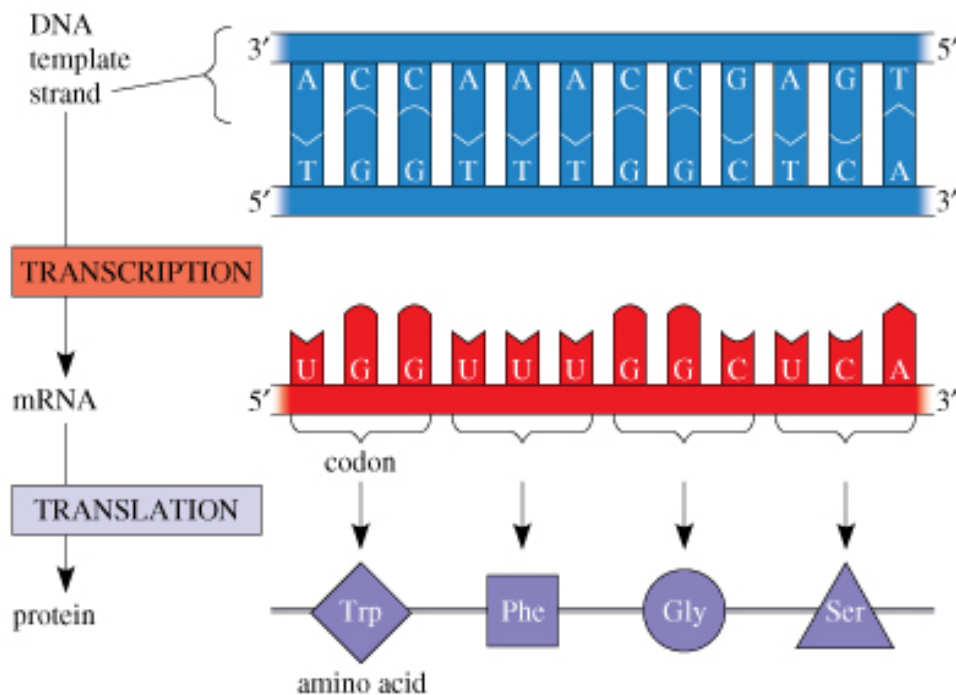
Translation

Definition: Translation is the process by which the information encoded in RNA molecules is used to synthesize proteins.

Key Steps:

1. **Initiation:** The small ribosomal subunit binds to the mRNA molecule at a specific sequence called the start codon (usually AUG), and the initiator tRNA carrying methionine binds to the start codon.
2. **Elongation:** The ribosome moves along the mRNA molecule in a 5' to 3' direction, catalyzing the formation of peptide bonds between amino acids carried by tRNA molecules in the A site and P site. The ribosome then translocates to the next codon, shifting the tRNA molecules to the P and E sites.
3. **Termination:** Translation ends when a stop codon (UAA, UAG, or UGA) is encountered on the mRNA molecule. Release factors bind to the stop codon, causing the ribosome to release the newly synthesized polypeptide chain and dissociate from the mRNA.

Significance: Translation is essential for protein synthesis, where the sequence of nucleotides in mRNA is translated into the sequence of amino acids in a protein. Proteins are the functional molecules that carry out most cellular processes, including enzymatic reactions, structural support, and cell signaling.



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